

FIXED INCOME CLEARING CORPORATION

Disclosure Framework for Covered Clearing Agencies and Financial Market Infrastructures

June 2026

ADVANCING FINANCIAL MARKETS. TOGETHER.™

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Responding Institution:	**Fixed Income Clearing Corporation ("FICC")**
Jurisdiction:	**State of New York, United States of America**
Authorities:	**U.S. Securities and Exchange Commission**

Except as noted in Section II, the information provided in this Disclosure Framework is accurate as of March 31, 2026; financial information and certain other data are provided as of the dates specified.

This Disclosure Framework can also be found on the DTCC website (www.dtcc.com).

For further information, please contact DisclosureFrameworks@dtcc.com.

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I. Executive Summary

The Committee on Payments and Market Infrastructures and the Technical Committee of the International Organization of Securities Commissions (collectively, "CPMI-IOSCO") recognize that financial market infrastructures ("FMIs"), which include payment systems, central securities depositories ("CSDs"), securities settlement systems, central counterparties ("CCPs") and trade repositories, each play a critical role in the financial system and the broader economy. FMIs facilitate clearing, settling and recording of monetary and other financial transactions, contributing to the goal of financial stability. CCPs, such as the Fixed Income Clearing Corporation ("FICC"), interpose themselves between counterparties to financial transactions. CPMI-IOSCO has recognized that, while properly managed FMIs bring great benefits to promoting market safety, they also have the potential to concentrate risk. Therefore, it is important that FMIs, such as FICC, have effective risk controls and adequate financial resources.

CPMI-IOSCO's 2012 report on the Principles for financial market infrastructures (the "FMI Principles") contains 24 FMI Principles covering the major types of risks faced by FMIs. One key objective of the FMI Principles is to encourage clear and comprehensive disclosure by FMIs through a public "Disclosure Framework" that explains how their businesses and operations reflect each of the applicable FMI Principles. The U.S. Securities and Exchange Commission ("SEC"), in Rule 17ad-22(e)(23), has adopted a similar approach to disclosure by covered clearing agencies.[1]

This Disclosure Framework covers FICC, a wholly owned subsidiary of The Depository Trust & Clearing Corporation ("DTCC"). FICC is comprised of two Divisions—the Government Securities Division ("GSD") and the Mortgage-Backed Securities Division ("MBSD"). GSD provides CCP services to its customers with respect to the U.S. government securities market, and MBSD provides such services to the U.S. mortgage-backed securities market. FICC is a clearing agency registered with, and under the supervision of, the SEC. In July 2012, FICC was designated as a systemically important financial market utility ("SIFMU") under Title VIII of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 ("Dodd-Frank"), and it is a "covered clearing agency" under the SEC's Standards for Covered Clearing Agencies.[2]

This Disclosure Framework is intended to provide relevant disclosure to FICC's stakeholders, including its Members and indirect users, on FICC's key services and the methods it uses to manage the risks to itself and others providing these services. In addition, this document facilitates FICC's compliance with CCAS 17ad-22(e)(23).

[1] 17CFR 240.17ad-22(e)(23).

[2] *See* Standards for Covered Clearing Agencies, Exchange Act Release No. 78961 (September 28, 2016), 81 FR 70786 (October 13, 2016) (hereinafter, "Covered Clearing Agency Standards" or "CCAS," and when referring to a specific rule, "CCAS 17ad-22(e)…).

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II. Summary of Material Changes since the Last Update of the Disclosure

This Disclosure Framework, which was comprehensively updated as of March 31, 2026, is published in adherence with FMI Principle 23 and CCAS 17ad-22(e)(23)(iv). The following is a quarterly summary of material changes since March 31, 2026:

Q2: April 1, 2026 – June 30, 2026

- None.

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III. General Background of FICC and Key Metrics

A. *General Description of FICC and Organization*

FICC operates under two Divisions—GSD and MBSD.[3] Each Division has its own rules and Members.[4] GSD provides trade comparison, netting, risk management, settlement and CCP services for the U.S. Government securities market, and MBSD provides the same services for the U.S. mortgage-backed securities market.

Regulatory, supervisory, and oversight framework

FICC was organized in 2003 as a business corporation under New York law and is a clearing agency registered with the SEC pursuant to the provisions of Section 17A of the Securities Exchange Act of 1934, as amended (the "Exchange Act"). As a registered clearing agency, FICC is subject to the requirements that are contained in the Exchange Act and in the SEC's regulations and rules thereunder. As noted above, these requirements include the Covered Clearing Agency Standards, which provide enhanced standards for entities that meet the definition of "covered clearing agency."[5] The Covered Clearing Agency Standards establish minimum requirements regarding how registered clearing agencies must maintain effective risk management procedures and controls as well as meet the statutory requirements under the Exchange Act on an ongoing basis. It is designed to enhance the regulatory framework for the supervision of clearing agencies.

In accordance with Dodd-Frank, FICC's designation as a SIFMU requires that it meet prescribed risk management standards and heightened oversight by the SEC in order to promote robust risk management and safety and soundness, reduce systemic risk, and support the stability of the broader financial system. FICC is also a "clearing corporation" within the meaning of Article 8 of the New York Uniform Commercial Code. These laws, regulations and rules are readily accessible to FICC's Members and the general public via the internet and through other public sources.

Organization

DTCC is the parent company of FICC. DTCC is a non-public holding company that owns three SIFMUs and related businesses. In addition to FICC, DTCC also owns The Depository Trust Company ("DTC"), the world's largest CSD and a registered clearing agency for the settlement of securities transactions for eligible securities and other financial assets, and National Securities Clearing Corporation ("NSCC"), a registered clearing agency and CCP that provides clearing, netting, settlement, risk management and CCP services for trades involving equities, corporate and municipal debt, exchange-traded funds and unit investment trusts in the U.S. DTC and NSCC are also SIFMUs.

DTCC, through its other subsidiaries, provides critical information, post-trade processing and transactional services, including through global trade repositories, to financial market participants in the U.S. and globally.

DTCC's common stock is owned by the financial institutions that are participants of its registered clearing agency/SIFMU subsidiaries. DTCC's governance arrangements—and those of its clearing agency

[3] GSD was previously known as the Government Securities Clearing Corporation ("GSCC"), which was established in 1986. MBSD was previously known as the MBS Clearing Corporation ("MBSCC"), which was established in 1979. FICC resulted from a merger of GSCC and MBSCC on January 1, 2003.

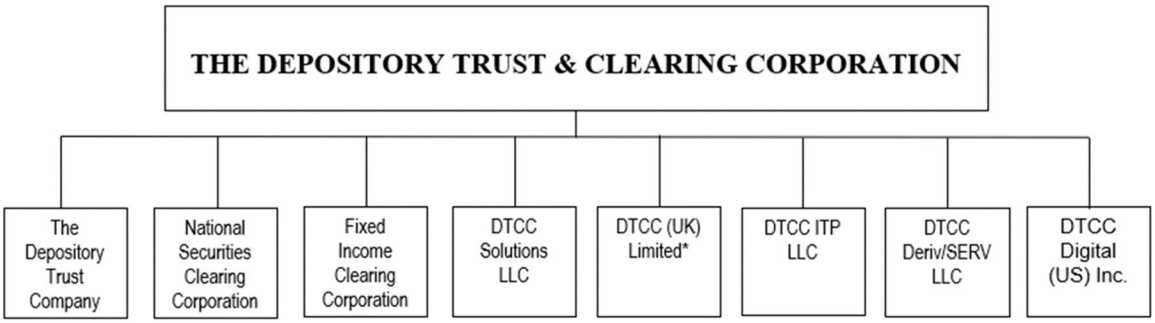
[4] GSD and MBSD have several membership types. For ease of description, when summarizing various GSD services and MBSD services in this Disclosure Framework, the term "Member," "GSD Member" or "MBSD Member," as applicable, are used throughout to denote a full-service participant that has access to GSD's guarantee/CCP services or MBSD's guarantee/CCP services, as applicable.

[5] *See supra* note 2. As a SIFMU, FICC is subject to the enhanced standards as a covered clearing agency.

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subsidiaries—are designed to promote the safety and efficiency of its clearing agency subsidiaries, support the stability of the broader financial system and promote the objectives of its participants. These governance arrangements are more fully described in response to Principle 2 (Governance) below.

DTCC's active direct subsidiaries are shown in the following chart:



* This entity directly holds DTCC's interest in non-US data repositories and other non-US businesses.

A description of the activities of DTCC's principal subsidiaries is available on the DTCC website.

B. *Key Services: System Design and Operation*

I. Government Securities Division

GSD offers a suite of services to support and facilitate the submission, comparison, risk management, netting and settlement of trades executed by its Members in the U.S. Government securities market. GSD acts as a CCP. In general, GSD novates eligible trades at the time of comparison of such trades. With respect to trades submitted in its tri-party service offerings, called "Sponsored GC" and "ACS Triparty", described in greater detail below, novation of a trade occurs when all of the requirements set forth in GSD Rule 3A, Section 7(b)(ii) and GSD Rule 8, Section 8(a)(ii), respectively, are met. GSD processes buy-sell transactions of U.S. Government securities (including Agencies) and repurchase agreement ("repo") transactions in U.S. Government securities (including Agencies) and mortgage-backed securities.

The following is a brief description of the core services and functions performed by GSD.

Trade Comparison/Real-Time Trade Matching

Both Divisions of FICC have a comparison system that enables real-time trade matching (RTTM® service). This interactive tool enables Members to automate the processing of their securities trades throughout the trading day. Using standardized international message formats, RTTM provides a common platform for collecting and matching trade data, enabling the parties to a securities trade to monitor and manage the status of their trades in real time. The result is an immediate confirmation for trade executions that is legal and binding.

A trade is deemed compared by GSD at the point in time at which GSD makes available to the Members on both sides of a transaction output (i.e., a report), indicating that their trade data have been compared. In general, upon comparison, GSD novates the trade, provided that the trade meets the requirements of the GSD Rules and was entered into in good faith. With respect to trades submitted in its tri-party service offerings, the Sponsored GC service and the ACS Triparty service, described in greater detail below, novation of a trade occurs when all of the requirements set forth in GSD Rule 3A, Section 7(b)(ii) and GSD Rule 8, Section 8(a)(ii), respectively, are met.

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Delivery-versus-payment activity—Netting and Settlement

GSD accepts buy/sell secondary market transactions in eligible securities for netting and settlement. In addition, GSD's Auction Takedown service enables Members to have their Treasury auction awards netted and novated with the rest of their GSD-eligible trading activity in the secondary markets, thereby reducing settlement risk and costs for Members' auction purchases. All auction awards are submitted to GSD by the Federal Reserve Bank of New York as fiscal agent for the U.S. Department of the Treasury on a locked-in basis, meaning that GSD automatically generates trade confirmations based on information supplied by the Federal Reserve Bank.

GSD also accepts delivery-versus payment repurchase transactions ("repos") for netting and settlement. Repos are financial instruments that enable firms to sell securities to obtain immediate funds for their own accounts, or for the benefit of their clients, and to simultaneously agree to repurchase the same (or similar) securities after a specified time at a given price, including interest calculated using a rate agreed upon at the time of execution. Repos can be executed directly between dealers, or on a blind-brokered basis through inter-dealer brokers. In dealer-to-dealer processing, dealers trade directly with one another. In blind brokering, repos are executed using inter-dealer brokers to ensure dealer anonymity.

Netting and settlement

GSD's netting system typically reduces the costs associated with securities transfers by reducing the number of securities movements required to settle transactions.

Once the trades described above for GSD Members have compared, they enter into GSD's Netting System. Through netting, GSD establishes a single net long or short position for each Member's daily trading activity in a given security. The Member's net position is the difference between all purchases (long) and all sales (short) in a given security. GSD replaces each net position with a settlement obligation for the scheduled settlement date whereby the Member settles with GSD as CCP.

Settlement occurs after GSD has informed its Members of their respective obligations (principally on the night of trade date in a T+1 market). Settlement occurs on the Federal Reserve's Fedwire® system or on the books of FICC's designated clearing bank. Members in turn instruct their clearing bank to transfer securities to GSD's clearing bank, The Bank of New York Mellon ("BNY Mellon"). FICC, acting as a CCP, receives the securities and turns the securities to the ultimate net buyers.[6]

On each business day, GSD makes available to each Member output that provides information that GSD deems sufficient to enable such Member to settle its net settlement positions on that business day.[7] Each Member, based on the information provided by GSD, then provides appropriate instructions to its clearing

[6] Because FICC is not a depository institution, it is not given direct access to the Fedwire service and therefore employs the services of BNY Mellon for this purpose. Prior to an entity activating its membership, GSD informs the Member of the clearing bank(s) that GSD will use to deliver eligible securities to Members and receive eligible securities from Members. Each Member must also notify GSD of the clearing bank(s) that the Member has designated to act on its behalf in the delivery of securities to GSD and the receipt of securities from GSD. Additionally, in GSD, with respect to the GCF Repo service, BNY Mellon is the agent for the settlement of GCF Repo transactions. GSD uses BNY Mellon, and GSD Members are required to use BNY Mellon with respect to the exchange of cash and collateral between cash lenders and repo dealers. As the Clearing Bank, BNY Mellon also values and allocates the collateral supporting the repos. As agent, the Clearing Bank does not require a membership category in FICC in its capacity as a clearing bank.

[7] The report includes the type of obligation (i.e., deliver or receive), name and reference number of the clearing bank, CUSIP number, settlement date, par value, final dollar value and other information descriptive of an eligible netting security.

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bank to deliver to GSD and/or to receive from GSD, on behalf of the Member, eligible netting securities against payment or receipt at the appropriate settlement value.

Obligations may be settled within a clearing bank (i.e., on the books of a clearing bank) when the deliver and attendant receive obligation are at the same clearing bank; otherwise, obligations are settled using the Fedwire service. All deliveries are made against payment.

Delivery-versus-payment activity is also processed through GSD's Sponsoring Member/Sponsored Member Service ("Sponsored Service"). The Sponsored Service permits Netting Members to sponsor certain institutional firms into GSD membership. A Sponsoring Member is permitted to submit to GSD for comparison, novation and netting certain types of eligible transactions between itself and its Sponsored Members or between its Sponsored Members and other third-party Netting Members.

Same-day settling activity

GSD novates and settles eligible same-day settling trades (i.e., the start legs of eligible repo transactions). These trades settle on a trade-for-trade basis with FICC.

Pair-off service

GSD provides an optional service for eligible Members whereby GSD systematically pairs-off such Members' failed securities settlement obligations during a specified time range in the afternoon. The failed securities settlement obligations could include (i) receive obligations and deliver obligations resulting from the previous night's net and (ii) obligations that were created intraday in order to settle a right of substitution or a same-day settling trade.

Tri-party activity

GSD provides a service known as the GCF Repo® service, which enables dealers to trade general collateral repos[8] based on rate, term and underlying product, throughout the day with inter-dealer broker Members ("brokers") on a blind basis. The service helps foster a highly liquid market for short-term financing.

Inter-dealer brokers (who are also members of GSD) are required to enter data on GCF Repo transactions for submission to GSD shortly after trade execution. Because the specific collateral is not known at the point of trade, the inter-dealer brokers submit all GCF Repo transactions to GSD using generic general collateral CUSIPs[9] that denote the underlying product. GCF Repo participants can trade in generic CUSIP numbers all day and then, after the netting process at the end of the day, allocate specific securities[10] to their net settlement obligations. GSD becomes the CCP for settlement purposes to each dealer party to a GCF Repo transaction and guarantees settlement of GCF Repo transactions upon receipt of trade data. GCF Repo transactions are settled on a tri-party basis,[11] which requires dealer Members to have an account with FICC's participating clearing bank, BNY Mellon.

[8] A general collateral repo is a repo in which the underlying securities collateral is nonspecific general collateral within an identified basket of eligible collateral. This is in contrast to a specific collateral repo where the underlying securities are specifically identified.

[9] Standardized generic CUSIP numbers established for GCF Repo processing are used to specify the acceptable type of underlying eligible collateral.

[10] The generic general collateral CUSIP defines the appropriate collateral that a dealer may use to satisfy the net settlement obligation in that CUSIP.

[11] A tri-party repo is one where a custodian bank acts as an intermediary between the two parties to the repo.

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GSD also has two tri-party service offerings. In its Sponsored Service, the tri-party service offering is called Sponsored GC and, in its Agent Clearing Service, the tri-party service offering is called ACS Triparty. These services offer Sponsoring Members and Agent Clearing Members, and their respective Indirect Participants, the ability to execute general collateral repos with each other and settle such repos on the tri-party repo platform of the relevant Clearing Agent Bank. Within the Sponsored GC tri-party service, Sponsoring Members may also participate in a Collateral-in-Lieu ("CIL") service offering, which uses a lien in favor of FICC on the purchased repo securities to reduce the margin and capital costs of providing cash providers with access to FICC.

The Centrally Cleared Institutional Triparty Service, or "CCIT" service, enables tri-party repurchase agreement transactions in GCF Repo Securities between Netting Members that participate in the GCF Repo Service and institutional counterparties (other than investment companies registered under the Investment Company Act of 1940, as amended). The institutional counterparties, called CCIT Members, are cash lenders in the transactions submitted to GSD. The CCIT service permits CCIT Members to benefit from FICC's novation, independent risk management and guaranty of completion of settlement of their eligible tri-party repo transactions with Netting Members.

Cross-Margining with Chicago Mercantile Exchange

FICC has established a cross-margining arrangement with the Chicago Mercantile Exchange (the "CME"). The FICC-CME cross-margining arrangement permits common members of FICC and CME (or affiliated members of FICC and CME) to reduce the aggregate margin requirements they must post to FICC and CME in respect of a portfolio of eligible FICC and CME cleared positions (e.g., cash Treasury positions and Treasury futures) on the condition that the members provide FICC and CME with the ability to look to the value of the entire portfolio in the event of the members' default. As a result of the cross-margining arrangement, FICC and CME are able to recognize risk offsets arising from the offsetting value of positions maintained by a common member (or a member and its affiliate) at FICC and CME when calculating the margin requirements for those positions. Each of FICC and CME separately calculate these risk offsets for the entire portfolio of eligible trades and give effect to the more conservative of the two calculations. The cross-margining agreement between FICC and CME covers certain cash U.S. Treasury securities cleared by FICC's GSD and various specified interest rate futures products cleared by CME. In September 2023, the SEC and the Commodity Futures Trading Commission ("CFTC") approved amendments to the cross-margining agreement, which is incorporated in FICC's rules and available on DTCC's website.[12] The amendments were implemented on January 22, 2024 and included (i) an expansion of the scope of CME products eligible for cross-margining, allowing for greater potential margin offsets; (ii) simplification of the overall margin calculation process which, based on portfolio-specific construction and market conditions, should generate margin savings in excess of those under the existing cross-margining agreement; and (iii) improvement in the efficiency and effectiveness of the default management process that FICC and CME would engage in if a common or affiliated member, as applicable, were to default.

Risk management

Risk management is the foundation for GSD's ability to guarantee settlement, as well as the means by which it protects itself and its Members from the risks inherent in the settlement process. Procedures are in place to ensure that Members comply with GSD's Rules. GSD maintains strict membership standards, including minimum financial requirements and Members are subject to ongoing review following admission. GSD's Clearing Fund addresses potential Member exposure through a number of risk-based component charges (as margin) calculated and assessed twice a day . GSD's key risk management practices and procedures, including its process for closing out the open positions of a defaulting Member and allocating any consequent losses, are discussed in more detail under Principles 3 through 7 and Principle 13.

[12] https://www.dtcc.com/~/media/Files/Downloads/legal/rules/ficc_cme_crossmargin_agreement.pdf

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In addition, if GSD incurred a loss in the liquidation of a Member that was not covered by the Member's Clearing Fund deposit, there is the possibility that GSD would receive funds to cover such loss (or part of the loss) from its cross-margining and cross-guaranty arrangements.

The Clearing Fund (which, in the aggregate, also operates as GSD's default fund) provides the collateralization required to cover a Member's exposure. The Clearing Fund consists of deposits posted by Members in the form of cash and eligible securities. To date, including through the 2008, 2011 and 2020 broker-dealer closeouts, FICC has never invoked its participant loss allocation procedures.

Additional services

In addition to the services described above, GSD offers a comparison-only service that is not guaranteed by GSD and for which it does not act as a CCP.

II. Mortgage-Backed Securities Division

MBSD processes (1) to-be-announced ("TBA") transactions, which are trades for which the actual identities of and/or the number of pools underlying each trade are not agreed to at the time of trade execution and (2) specified pool trades, which are trades for which all pool data is agreed upon by the Members at the time of execution. TBA transactions are comprised of (i) settlement balance order destined trades ("SBOD trades"), (ii) trade-for-trade transactions , (iii) stipulated trades, and (iv) TBA options trades.

MBSD's processing of these eligible transactions, which are pass-through mortgage-backed securities issued or guaranteed by Government National Mortgage Association ("Ginnie Mae"), Federal National Mortgage Association ("Fannie Mae") and Federal Home Loan Mortgage Corporation ("Freddie Mac"), consists of the following steps: trade matching, novation, TBA Netting, the Do Not Allocate ("DNA'") process, electronic pool notification allocation, pool comparison, Pool Netting, pool conversion and settlement.

MBSD's trade comparison and TBA Netting systems form the basis of all of its other services. All compared trades will be risk managed by MBSD, but the remainder of their life cycle differs according to their trade type as noted below:

- Compared SBOD trades proceed to MBSD's TBA Netting process, which is described in a subsequent section. Compared SBOD trades are guaranteed and legally novated by FICC at the time of comparison.

- Compared trade-for-trade transactions are not eligible for TBA Netting, but pools may be allocated before they can be settled, and those pools are eligible for MBSD's Pool Netting process. Compared trade-for-trade transactions are guaranteed and legally novated by FICC at the time of comparison.

- Compared stipulated trades are neither eligible for TBA Netting nor MBSD's Pool Netting, but pools must be allocated before they can be settled directly with FICC. Compared stipulated trades are guaranteed and legally novated by FICC at the time of comparison. The guarantee occurs at the generic level not the specific stipulation associated with the actual obligation.

- Compared specified pool trades are neither eligible for TBA Netting nor MBSD's Pool Netting, however, these trades settle directly with FICC. Compared specified pool trades are guaranteed and legally novated by FICC at the time of trade comparison. This guarantee occurs at the generic level not the specific pool associated with the actual obligation.

- Compared option trades are not eligible for any netting process. Option trades are not novated by FICC. Members also cannot allocate pools against option trades. Option trades must be bilaterally cancelled by both the buyer and the seller when the underlying TBA option is exercised or when it expires.

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Trade Comparison/Real-Time Trade Matching

As stated above in the GSD section, MBSD's trade comparison system is its RTTM service.

A trade is deemed compared by MBSD at the point in time at which MBSD makes available to the Members on both sides of a transaction output (i.e., a report) indicating that their trade data have been compared. All trades (except option trades) legally novate at the time of comparison.

Once trade-for-trade transactions and stipulated trades are matched and allocated through MBSD, settlement obligations are established between the Member and FICC, however, these trades do not enter the TBA Netting process. Once specified pool trades are matched through MBSD, settlement obligations are established between the Member and FICC. SBOD trades proceed to the TBA Netting process after they are matched. TBA Options are canceled by both Members at the time the underlying option is exercised or reaches its expiration date.

TBA Netting

MBSD performs the TBA Netting process four times per month, corresponding to each of the four primary settlement classes and dates established by Securities Industry Financial Markets Association ("SIFMA").[13]

Three business days prior to the SIFMA established settlement date (referred to as "72 Hour Day"), TBA Netting for the applicable class occurs. On this date, all compared SBOD trades within the class that has been designated for the TBA Netting process are netted with FICC as the counterparty. The net positions created by the TBA Netting process are referred to as the settlement balance order position ("SBO position"), which constitutes settlement obligations against which Members will (1) submit pool information ("Pool Instructs") for the Pool Netting process or (2) offset such SBO position with other SBO position or trade-for-trade transaction, as applicable, through the DNA process. The Pool Netting process and the DNA process are described in further detail below.

As noted above, trade-for-trade destined transactions are not netted at the TBA level, but like the SBO positions do constitute TBA settlement obligations against which Pool Instructs may be submitted. Specified pool trades and stipulated trades are also not netted at the TBA level, nor do such trades enter the Pool Netting.

DNA Process

The DNA process gives Members the ability to offset TBA obligations with other TBA obligations – meaning that, SBO positions and/or trade-for-trade transactions may be offset with other SBO positions and/or trade-for-trade transactions, as applicable, subject to certain restrictions. The purpose of this process is to reduce a Member's overall open TBA obligations by giving such Member the ability to exclude certain TBA obligations from the pool allocation process and securities settlement.

The DNA process is available two business days prior to the established settlement date of the TBA settlement obligations (known as "48 Hour Day") from 7:00 a.m.[14] through the business day before the contractual settlement date (known as "24 Hour Day") at 4:30 p.m. During this time, Members with two or more open TBA obligations with the same CUSIP number, and SIFMA designated settlement date may offset (i.e., "pair-off") such TBA obligations. To initiate the offset, Members are required to submit a request (referred to as a "DNA Request") to MBSD through the RTTM system. Upon FICC's validation of the DNA Request, a Member's TBA

[13] SIFMA publishes a calendar that specifies one settlement date per month for four different product classes (known as Classes A, B, C and D) that are used to categorize the various types of TBA securities. These product classes and the associated settlement dates are recognized by the industry, and they provide the foundation for MBSD's TBA Netting process.

[14] All times referred to herein are Eastern time.

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obligations are reduced and the Member will not be required to allocate pools against the TBA obligations related to this reduction.

Electronic Pool Notification Allocation

Electronic Pool Notification ("EPN") enables users to reduce risk and streamline their operations by providing an automated way for Members who have an obligation to deliver pools ("pool sellers") to transmit pool information efficiently and reliably to their counterparties ("pool buyers") in real time. The majority of pool allocations occur on 48 Hour Day. On this day, pool sellers must notify their pool buyers through the EPN system of the specific pools that such sellers intend to allocate in satisfaction of their SBO positions, trade-for-trade transactions and/or stipulated trades. Pool allocations occur for all TBA obligations, whether established on 72 Hour Day via the TBA Netting process or established upon comparison when the TBA trade-for-trade transaction or stipulated trade, as applicable, activity was submitted.

Pool allocations are not performed for specified pool trades because, as noted above, the pool that is to be delivered in connection with such trade is specified upon submission.

Pool Comparison

On 48 Hour Day, Members can submit Pool Instructs via the RTTM system for pool comparison, which is a prerequisite for Pool Netting. Pool sellers must notify their pool buyers through the EPN system of the pools that will be allocated in satisfaction of their TBA obligation. If a Member does not submit its Pool Instructs by the established deadline, FICC would determine and apply the Pool Instructs for such Member. This determination will be based on the allocated pools that a Member has submitted through the EPN Service.

Pool Netting

MBSD maintains two Pool Netting processes. The purpose of each process is to reduce the number of pool settlements by netting Pool Instructs stemming from TBA Netting Process and trade-for-trade activity to arrive at a single net position in a particular pool number for next-day delivery date. Although the initial Pool Netting runs daily, the majority of Pool Netting activity follows the SIFMA calendar. The initial Pool Netting process includes pool allocations that were timely submitted on 48 Hour Day by the deadline established by SIFMA. The second Pool Netting process is referred to as "Expanded Pool Netting." The Expanded Pool Netting process includes pool allocations that were not submitted by the deadline established for the initial Pool Netting process. The Expanded Pool Netting process occurs four times per month in accordance with the SIFMA designated settlement date.

At the start of each netting cycle, all Pool Instructs active in the system are evaluated for potential inclusion in Pool Netting.[15] Any compared Pool Instructs that are not eligible for Pool Netting will be converted into settlement obligations. Any uncompared Pool Instructs will be purged from the system. Members may elect to exclude their Pool Instructs from Pool Netting by submitting a customer delivery request ("CDR") through the EPN system. Pool Instructs in a CDR state are precluded from operational netting; however, the current face of all long CDR Pool Instructs are totaled to establish an aggregated long CDR amount in the same pool number, and the current face of all short CDR Pool Instructs are totaled to establish an aggregated short CDR amount in the same pool number.

All Pool Instructs selected for Pool Netting will be used to arrive at a single net position in each pool number that the Member must either deliver to or receive from FICC.

[15] In order to be included in this selection process, Pool Instructs must have (i) the same contractual settlement date, (ii) a delivery date equal to the next business date, and (iii) been compared.

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Settlement

MBSD employs the settlement services of BNY Mellon. Prior to an entity activating its membership, MBSD informs the Member of the clearing bank that MBSD will use to deliver eligible securities to Members and to receive eligible securities from Members. Each Member also informs MBSD of the clearing bank that the Member has designated to act on its behalf in the delivery of securities to MBSD and in the receipt of securities from MBSD.

On each business day, MBSD makes available to each Member output that provides information that MBSD deems sufficient to enable such Member to be able to settle its net settlement positions on that business day. Each Member, based on the information provided by MBSD, then provides appropriate instructions to its clearing bank to deliver to MBSD, and/or to receive from MBSD, on behalf of the Member, eligible netting securities against payment or receipt at the appropriate settlement value.

Obligations may be settled within a clearing bank (i.e., on the books of a clearing bank) when the deliver and attendant receive obligation are at the same clearing bank. Otherwise, obligations are settled using Fedwire. All deliveries are made against payment.

Risk management

Risk management is the foundation for MBSD's ability to guarantee settlement, as well as the means by which it protects itself and its Members from the risks inherent in the settlement process. Procedures are in place to ensure that Members comply with MBSD's Rules. MBSD maintains strict membership standards, including minimum financial requirements and Members are subject to ongoing review following admission. MBSD's Clearing Fund addresses potential Member exposure through a number of risk-based component charges (as margin) calculated and assessed daily MBSD's key risk management practices and procedures, including its process for closing out the open positions of a defaulting Member and allocating any consequent losses, are discussed in more detail under Principles 3 through 7 and Principle 13.

In addition, if MBSD incurred a loss in the liquidation of a Member that was not covered by the Member's Clearing Fund deposit, there is the possibility that MBSD would receive funds to cover such loss (or part of the loss) from its cross-guaranty arrangement. The Clearing Fund (which, in the aggregate, also operates as MBSD's default fund) provides the collateralization required to cover a Member's exposure. The Clearing Fund consists of deposits posted by Members in the form of cash and eligible securities. To date, including through the 2008, 2011 and 2020 broker-dealer closeouts, FICC has never invoked its Member loss allocation procedures.

Additional services

In addition to the core services described above, MBSD offers an EPN-only service that is not guaranteed by MBSD and for which it does not act as a CCP.[16]

C. *Key Metrics for FICC*

As a CCP, FICC provides specific quantitative disclosures with respect to certain FMI principles. These disclosures are intended to complement the narrative disclosures included in this Disclosure Framework. The FICC Quantitative Disclosures for Central Counterparties can be found on the DTCC website.

[16] MBSD Members are required to be EPN Members; however, one can be an EPN Member only and not a full-service Member with access to MBSD's guaranteed CCP services. The EPN services are described in the EPN Rules, which are available on the DTCC website.

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IV. Principle-by-Principle and Standard-by-Standard Summary Narrative Disclosure

Principle 1: Legal basis; CCAS 17ad-22(e)(1)

Principle 1: Legal Basis	**CCAS 17ad-22(e)(1)**
An FMI should have a well-founded, clear, transparent, and enforceable legal basis for each material aspect of its activities in all relevant jurisdictions. ***Key consideration 1***: The legal basis should provide a high degree of certainty for each material aspect of an FMI's activities in all relevant jurisdictions. ***Key consideration 2***: An FMI should have rules, procedures, and contracts that are clear, understandable, and consistent with relevant laws and regulations. ***Key consideration 3***: An FMI should be able to articulate the legal basis for its activities to relevant authorities, participants, and, where relevant, participants' customers, in a clear and understandable way. ***Key consideration 4***: An FMI should have rules, procedures, and contracts that are enforceable in all relevant jurisdictions. There should be a high degree of certainty that actions taken by the FMI under such rules and procedures will not be voided, reversed, or subject to stays. ***Key consideration 5***: An FMI conducting business in multiple jurisdictions should identify and mitigate the risks arising from any potential conflict of laws across jurisdictions.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (1) Provide for a well-founded, clear, transparent, and enforceable legal basis for each aspect of its activities in all relevant jurisdictions.

Legal basis for material aspects of FICC's activities in all relevant jurisdictions

With respect to GSD and MBSD (each, a "Division"), the material aspects of FICC's activities include:

(1) trade comparison and validation;

(2) trade guarantee, netting and novation of deliver and receive obligations;

(3) funds netting and position netting;

(4) funds settlement and finality;

(5) risk management and collateral arrangements;

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(6) default procedures and liquidity resources (including closing out a defaulting Member's open positions and closeout netting); and

(7) links, such as those with other CCPs.

In evaluating the legal basis for FICC's core activities, the United States is the critical venue for FICC's operations and the markets it serves and the location of collateral held by FICC. FICC is a New York corporation, and each Division's Rules and FICC's liquidity arrangements are expressly governed by New York law.[17]

All Members are required to execute membership agreements under which they agree to be bound by the applicable Division's Rules. Entities acting as settling banks on behalf of Members execute agreements covering such arrangements and are required, under the applicable Division's Rules, to be "Funds-Only Settling Bank Members" with respect to GSD or "Cash Settling Bank Members" with respect to MBSD; as such, they execute a membership agreement under which they also agree to be bound by the relevant Division's Rules that are applicable to Funds-Only Settling Bank Members or Cash Settling Bank Members, respectively (which include the requirements for the funds-only settlement/cash settlement process at the relevant Division). Sponsored Members and other limited purpose members are required to execute membership agreements under which they agree to be bound by the particular Rules that are applicable to such participants. All such agreements are expressly governed by New York law. Each Division's Rules are public and can be found on the DTCC website.

Federal law, principally the Exchange Act and Title VIII of Dodd-Frank, governs the activities of FICC as a registered clearing agency, SRO and SIFMU.

FICC ensures that its legal basis provides a high degree of legal certainty for each material aspect of its activities in all relevant jurisdictions:

1. *By structuring each Division's activities and Rules in accordance with the laws of the relevant jurisdictions.*

With respect to each Division, FICC's activities and its Rules are structured in accordance with the laws of New York and the United States. The principal laws comprising the legal framework under which FICC operates include: (1) the Exchange Act, particularly Sections 17A and 19, (2) the New York Business Corporation Law (the "New York BCL"), (3) the New York Uniform Commercial Code (the "New York UCC"), particularly Articles 8 and 9, (4) general New York contract law, (5) the Securities Act of 1933, as amended, (6) the Federal Deposit Insurance Act, as amended (the "FDIA"), (7) the U.S. Bankruptcy Code (the "Bankruptcy Code"), (8) the Federal Deposit Insurance Corporation Improvement Act of 1991, as amended ("FDICIA"), (9) the Securities Investor Protection Act of 1970, as amended ("SIPA"), and (10) Dodd-Frank, particularly Title II, regarding the orderly liquidation authority ("OLA"), and Title VIII, regarding payment, clearing and settlement supervision.

The ability of FICC to enforce each Division's Rules to accomplish its core clearance and settlement and risk management functions has been repeatedly confirmed through analogous court decisions with respect to its clearing agency affiliates, NSCC and DTC.[18] Moreover, courts have routinely held that state-law challenges to the existence or operation of clearing agency-related SEC-approved programs are federally

[17] FICC is also qualified to do business in New Jersey and Florida, where it maintains certain operations.

[18] *See generally* Pet Quarters, Inc. et al. v. Depository Trust and Clearing Corporation et al., 559 F.3d 772 (8th Cir. 2009); Whistler Investments, Inc. et al. v. The Depository Trust and Clearing Corporation et al., 539 F.3d 1159 (9th Cir. 2008); Nanopierce Technologies, Inc. et al. v. The Depository Trust and Clearing Corporation et al., 168 P.3d 73 (Nev. 2007).

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preempted because they conflict with congressional directives as set forth in Section 17A of the Exchange Act.[19]

2. Through the Proposed Rule Change and Advance Notice Processes.

As a clearing agency registered with the SEC, the Exchange Act provides a framework under which each Division's Rules are adopted and enforced. Each Division's Rules are filed with and reviewed by the SEC.[20] As a clearing agency registered under Section 17A of the Exchange Act, a self-regulatory organization subject to Section 19 of the Exchange Act, and a SIFMU under Title VIII of Dodd-Frank, FICC is required to follow: (1) a specified process[21] whenever it proposes a new rule or a change or amendment to either Division's Rules (a "Proposed Rule Change," and the process, the "Proposed Rule Change Process") and (2) a specified process[22] whenever it proposes to make a change to either Division's rules, procedures or operations that could materially affect the nature or level of risks presented by FICC (a "Material Change," and the process, the "Advance Notice Process").

Under the Proposed Rule Change Process, generally, before a Proposed Rule Change may take effect, (1) the change and an explanatory statement, completed pursuant to the SEC Form 19b-4, must be filed with the SEC and posted by FICC on DTCC's website, (2) notice of the filing and the substantive terms or description of the change must be published by the SEC in the *Federal Register* for public review and comment, and (3) the SEC must approve the change (or the change must otherwise be permitted to take effect[23]). The SEC is required to disapprove a Proposed Rule Change if it does not find that the change is consistent with the requirements of the Exchange Act and the rules and regulations thereunder that are applicable to FICC.

Similar submission, disclosure and publication requirements apply to the Advance Notice Process, where, generally, FICC must provide 60 days advance notice to the SEC (an "Advance Notice") before a Material Change may take effect, describing the nature of the change, its expected effects on risks to FICC, its Members or the market and how FICC plans to manage any identified risks. A copy of the Advance Notice must also be: (1) provided by FICC to the Board of Governors of the Federal Reserve System (the "FRB"), (2) posted by FICC on DTCC's website and (3) published by the SEC in the *Federal Register* for public review and comment. The SEC must consult with the FRB in regard to an Advance Notice of a Material Change, and the SEC may object to the change if it determines that the change would be inconsistent with the objectives and principles for risk management standards described in Section 805(b) of Dodd-Frank or the rules and regulations thereunder that are applicable to FICC.

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[19] *See* Whistler Investments, 539 F.3d at 1167-68 (affirming the district court's dismissal of all claims on the grounds of preemption by Section 17A of the Exchange Act); Pet Quarters, 559 F.3d at 780-82; Nanopierce Technologies, 168 P.3d at 76 (concluding that "because the state law on which [plaintiffs] base their claims poses an obstacle to [DTCC, DTC and NSCC's] accomplishment of congressional objectives as explicitly stated in and gleaned from the [Exchange Act's] framework, and because [DTCC, DTC and NSCC's] compliance with both state and federal requirements concerning the securities transactions at issue in this case is impossible, section 17A of the [Exchange Act] preempts [plaintiffs'] claims.").

[20] Each Division's Rules, as originally in effect at the time of FICC's registration as a clearing agency, were filed with and reviewed by the SEC as part of the registration process. Subsequent changes in each Division's Rules have been similarly filed with and reviewed by the SEC.

[21] This process is set forth in Section 19(b) of the Exchange Act and Exchange Act Rule 19b-4.

[22] This process is set forth in Section 806(e) of Dodd-Frank and Exchange Act Rule 19b-4.

[23] In certain limited circumstances, including fee changes, Proposed Rule Changes may become effective upon filing. Proposed Rule Changes may also become effective summarily if it appears to the SEC that such action is necessary for the protection of investors, the maintenance of fair and orderly markets or the safeguarding of securities or funds. However, any Proposed Rule Change that becomes effective upon filing or summarily is subject to SEC review and the right of the SEC to take action thereafter.

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In more limited circumstances, FICC may also obtain legal certainty regarding certain activities by requesting an exemption from legal or regulatory requirements from the SEC pursuant to the SEC's authority under Section 17A and/or Section 36 of the Exchange Act or by requesting a "no-action" letter from SEC staff concluding that SEC staff would not recommend that the SEC take enforcement action against FICC based on the facts and representations described in its "no-action" request.

3. *By requiring or otherwise obtaining legal opinions, analyses or advice.*

FICC requires applicants for membership to provide a legal opinion concerning the membership agreement (which incorporates the applicable Division's Rules). This opinion provides FICC with the necessary comfort to confirm that the membership agreement will be binding and enforceable on the applicant when it becomes a Member.

If an applicant is organized under the laws of a jurisdiction outside of the United States, FICC relies on a legal opinion with respect to the laws of the non-U.S. jurisdiction that specifically addresses issues such as FICC's ability to enforce the applicable Division's Rules (including its netting, guaranty, novation of delivery and receive obligations, and default management rules) under the applicable insolvency regime of the applicant's home jurisdiction, and the enforceability of the choice of New York law to govern the membership agreement and the applicable Division's Rules.

FICC also obtains legal analyses or advice as it deems appropriate in connection with new services, changes in law and other matters. For example, where applicable, an annual review is conducted with respect to each jurisdiction in which its participants are located to confirm the continued enforceability of its legal opinions and the advice it has received regarding relevant local licensing requirements.

FICC's role as CCP, trade guaranty and netting

FICC's authority to clear and settle securities transactions and act as a CCP is provided through: (1) its powers as a New York corporation under the New York BCL; (2) its registration as a clearing agency under the Exchange Act, which provides for the registration and regulation of clearing agencies and also prescribes standards for their operation and governance; (3) the New York UCC, which contains provisions for the operation of clearing corporations (including registered clearing agencies such as FICC); and (4) general New York contract law, which provides the legal basis under which FICC guarantees trades and becomes a CCP through novation. The timing of FICC's guarantee and the timing of novation are clearly set forth in each Division's Rules.

As a general matter, U.S. law supports netting arrangements relating to securities transactions. In particular, the definition of "clearing agency" in the Exchange Act clearly recognizes that one of the roles of a clearing agency is to "reduce the number of settlements of securities transactions."[24] FDICIA supports "netting contracts" (which include the rules of a clearing organization), providing for the netting of payment obligations and payment entitlements between and among clearing organizations (including registered clearing agencies such as FICC) and their members (including liquidation or closeout values relating to such obligations or entitlements). "Payment" includes both cash payments and noncash deliveries. Each Division's Rules (including the netting and limited cross guaranty agreements that FICC has with other registered clearing agencies[25]) are a "netting contract" within the meaning of FDICIA.

FDICIA provides that (1) provisions of any security agreement or arrangement or other credit enhancement related to one or more netting contracts between any two members of a clearing organization (including the clearing organization itself) shall be enforceable in accordance with their terms (with limited exceptions); (2) notwithstanding any other provisions of state or federal law (with limited exceptions), the payment

[24] 15 U.S.C. 78c(a)(23)(A).

[25] These arrangements are described under Principles 9 (Money settlements), 13 (Participant-default rules and procedures) and 20 (FMI links).

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entitlements and obligations of members of a clearing organization (including the clearing organization itself) shall be terminated, liquidated, accelerated and netted in accordance with and subject to the conditions of any applicable netting contract; (3) the only obligation or entitlement of a member of a clearing organization to make or receive payment under a single netting contract to or from other members of a clearing organization (including the clearing organization itself) shall be equal to its net obligation or entitlement under the netting contract; and (4) no stay, injunction, avoidance, moratorium or similar proceeding or order, whether issued or granted by a court, administrative agency or otherwise, shall limit or delay application of otherwise enforceable netting contracts.

The provisions of FDICIA applicable to clearing organization netting therefore override any conflicting provisions of state or federal law, including the Bankruptcy Code, except to the extent otherwise expressly provided. The netting provisions of FDICIA were designed to reduce systemic risk to the financial markets. In addition, amendments to the FDIA, FDICIA, the Bankruptcy Code and SIPA in 2005 and 2006 included provisions that validate master netting agreements in respect of securities, commodities, forward, swap and repurchase transactions. The foregoing provisions also support the cross-margining arrangement between FICC and the CME.

FICC's legal basis supports the finality of transfers of financial instruments and funds, and the effectiveness of its risk management and default management rules and procedures (including the closeout of positions of an insolvent Member).

Each Division's Rules provide (1) the timing of, and requirements for, settlement, and (2) procedures that will be followed in the event of a failure of a Member to perform its obligations or the insolvency of a Member. As a general matter, U.S. law, including general New York contract law, the New York UCC, the New York BCL, the Exchange Act, the FDIA, FDICIA, the Bankruptcy Code, SIPA and Title II of Dodd-Frank, supports the settlement of securities transactions in accordance with each Division's Rules, and the ability of FICC to effectuate its risk management and default management rules and procedures (including closing out the open positions of an insolvent Member).

Settlement Finality

For GSD CCP services, other than the GCF Repo service, the CCIT service, the Sponsored GC service and the ACS Triparty service, each Division's Rules provide that all securities settlements are to be effected on a delivery versus payment basis either on the Fedwire service or on the books of FICC's clearing bank pursuant to FICC's agreement with its clearing bank. Settlement in connection with the GCF Repo service and the CCIT service occurs on the books of the clearing bank and is final when the transfer of securities and related funds occurs per the agreement between FICC and the clearing bank. Settlement with respect to tri-party trades cleared through the Sponsored GC service and ACS Triparty service occurs on the books of the relevant Clearing Agent Bank. The legal basis is provided by general New York contract law, the New York UCC and the Exchange Act, and is protected in the event of the insolvency of a Member (with limited exceptions) by relevant provisions of the FDIA, FDICIA, the Bankruptcy Code, SIPA and Title II of Dodd-Frank. Finality regarding the Fedwire service is governed by the FRS operating circular. The legal basis for settlement finality is described more fully in Principle 8.

GSD funds-only settlement (GSD Rule 13) and MBSD cash settlement (MBSD Rule 11) are described more fully in Principle 9 (Money settlements). Cash debits and credits are paid out via the National Settlement Service ("NSS"), a payment system operated by the Federal Reserve System ("FRS"), through the use of cash settling banks appointed by Members, with DTC assuming the role of FICC's settlement agent in NSS. The individual debits and credits of each Member using the same settling bank are totaled. Once the net debits/credits are approved at the settling bank level, the Federal Reserve debits or credits each settling bank. Each settling bank then debits or credits the account of each Member for which the bank settles. Funds transfers become final at the time that funds are moved through NSS.[26]

[26] NSS is governed by Federal Reserve Bank Operating Circular 12 (Multilateral Settlement).

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Clearing Fund payments are currently paid by wire transfer through the Fedwire® service, a payment system operated by the FRS.[27] Payment to the receiving party (FICC) through Fedwire is final and irrevocable upon the crediting of the receiving party's account, or when the payment order is sent to the receiving party, whichever is earlier. Payment orders generally are processed promptly following the applicable Federal Reserve Bank's receipt of a transfer message.[28]

Risk Management and Default Procedures.

Each Division's Rules also provide risk management tools: GSD Rule 4 and MBSD Rule 4, as applicable, provide for the Clearing Fund (margin) formula, collection of Clearing Fund (margin) from Members, the form and manner in which such funds or collateral is provided, and the means by which losses are to be allocated. With respect to activity clearing through the Collateral-in-Lieu offering of the Sponsored GC service, in lieu of collecting margin, in most cases, FICC is granted a lien on the purchased repo securities that are subject to the Sponsored GC CIL Trade. The lien is documented in a "CIL Custodial Agreement Supplement" between the Clearing Agent Bank, FICC, the Netting Member or Indirect Participant that is the counterparty to the repo transaction ("GC Funds Borrower") and the Sponsored Member (referred to as the "GC Funds Lender").

In addition, GSD Rules 3 and 3A and MBSD Rule 3, as applicable, provide FICC with the ability to obtain adequate assurances of a Member's ability to perform its operational and financial obligations. Additionally, the cross-guaranty arrangement described under Principle 13 is structured as a "netting contract" under FDICIA.

The foregoing provisions, including the validity of FICC's lien and collateral protections, are enforceable under general New York contract law, New York UCC and the Exchange Act and are protected in the event of the insolvency of a Member (with limited exceptions) by relevant provisions of the FDIA, FDICIA, the Bankruptcy Code, SIPA and Title II of Dodd-Frank. In the case of deposits of Clearing Fund cash, title is transferred to FICC by the transfer of the funds to accounts of FICC.

In the case of a Member default or insolvency, FICC may cease to act for the Member, and in that event, proceed to close out the open positions of the Member,[29] and once that is completed, determine a net gain or loss on the closeout of those positions.[30] GSD Rule 21 (Restrictions on Access to Services), GSD Rule 22 (Insolvency of a Member) and GSD Rule 22A (Procedures for when the Corporation Ceases to Act), and MBSD Rule 14 (Restrictions on Access to Services), MBSD Rule 16 (Insolvency of a Member) and MBSD Rule 17 (Procedures for when the Corporation Ceases to Act), set out the circumstances under which FICC may cease to act for a Member, and the types of actions it may take.

As described more fully below, clearing agency closeout provisions (including access to and liquidation of collateral) are supported, with limited exceptions, under FDICIA, the Bankruptcy Code, SIPA, the FDIA and Title II of Dodd-Frank.

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[27] Funds transfers through Fedwire are governed by Subpart B of Regulation J, which incorporates the provisions of Article 4A of the UCC, and Federal Reserve Bank Operating Circular 6 "Funds Transfers Through the Fedwire® Funds Service."

[28] *See FRB, The Fedwire® Funds Service Disclosure, December 5, 2025, at 52, available at https://www.frbservices.org/binaries/content/assets/crsocms/financial-services/wires/funds-service-disclosure.pdf.*

[29] With respect to such defaulting Member, under GSD Rule 22A and MBSD Rule 17, FICC will generally exit from its system those transactions that have not reached the point where FICC's trade guaranty attaches, unless the Board of Directors determines otherwise.

[30] The process by which FICC closes out transactions of a defaulting Member is described in greater detail under Principle 13 (Participant-default rules and procedures).

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(a) FDICIA

As noted above, FDICIA supports the effectiveness of "netting contracts," which include the rules of a clearing organization, providing for the netting of payment obligations and payment entitlements between and among clearing organizations and their members (including liquidation or closeout values relating to such obligations or entitlements). "Payment" includes both cash payments and noncash deliveries. Each Division's Rules are a "netting contract" within the meaning of FDICIA.

Further, Section 404(h) of FDICIA provides that the provisions of any security agreement or arrangement or other credit enhancement related to one or more netting contracts between any two members of a clearing organization (including the clearing organization itself) shall be enforceable in accordance with their terms (with limited exceptions), and shall not be stayed, avoided or otherwise limited by any state or federal law (with limited exceptions).

FDICIA also provides that (1) Section 404 of FDICIA shall be given effect notwithstanding that a member of a clearing organization is a failed member, and (2) no stay, injunction, avoidance, moratorium or similar proceeding or order, whether issued or granted by a court, administrative agency or otherwise, shall limit or delay application of otherwise enforceable netting contracts in accordance with Section 404 of FDICIA.

(b) The Bankruptcy Code, SIPA, the FDIA and Title II of Dodd-Frank

The insolvency regime applicable to a FICC Member will generally depend on the jurisdiction in which the Member is organized, its form of organization and its regulatory status (among other factors). The U.S. insolvency regimes to which Members may be subject include Chapter 11 of the Bankruptcy Code (reorganization), Chapter 7 of the Bankruptcy Code (liquidation), SIPA (with respect to members of the Securities Investor Protection Corporation ("SIPC")), the receivership provisions of the FDIA (with respect to insured depository institutions) and Title II of Dodd-Frank regarding OLA (with respect to covered financial companies).[31]

The insolvency of a FICC Member that is not a member of SIPC, an insured depository institution or a covered financial company would typically be handled under Chapter 11 of the Bankruptcy Code or subchapter III or IV of Chapter 7 of the Bankruptcy Code. Although the automatic stay, prohibitions on *ipso facto* provisions and avoidance powers of the bankruptcy trustee are generally applied with respect to cases conducted under Chapters 11 and 7 of the Bankruptcy Code, the Bankruptcy Code contains various exceptions and safe harbors to those provisions that support the finality of securities transactions processed through securities clearing agencies and a clearing agency's closeout of the insolvent member's open positions.[32] The Bankruptcy Code also provides similar exceptions and safe harbors that apply to master netting agreements.[33]

[31] As discussed below, legal risk relating to the insolvency regime applicable to a non-U.S. Member is identified, analyzed and mitigated through the use of legal opinions, analyses and/or advice.

[32] Examples of such exceptions and safe harbors include: Sec. 362(b)(6) (exceptions from the automatic stay for the exercise of contractual rights under a security agreement related to a securities contract and to net out amounts owed under one or more securities contracts); Sec. 546(e) (limitations on avoidance powers for certain transfers in connection with a securities contract); Sec. 555 (protection for the exercise of a contractual right to liquidate, terminate or accelerate a securities contract).

[33] Examples of such exceptions and safe harbors include: Sec. 362(b)(27) (exceptions from the automatic stay for the exercise of contractual rights under a security agreement related to a master netting agreement and to net out amounts owed under a master netting agreement, to the extent such rights would be exercisable in relation to the individual contracts subject to the master netting agreement); Sec 546(j) (limitations on avoidance powers for certain transfers in connection with a master netting agreement); Sec. 561 (protection for the exercise of a contractual right to liquidate, terminate or accelerate one or more

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The insolvency of a FICC Member that is a member of SIPC, and whose customers would be entitled to advances from the SIPC fund created under SIPA to protect customers, would be handled under SIPA. Although SIPC proceedings generally involve (1) a protective decree of a federal district court that, among other things, freezes a member's assets, and (2) the application of the Bankruptcy Code's automatic stay and avoidance powers, there are exceptions and safe harbors in SIPA, as well as provisions typically included in such protective decrees, that support the finality of securities transactions processed through securities clearing agencies and their ability to close out the open positions of the insolvent member.[34]

The insolvency of a FICC Member that is an insured depository institution would be handled under the receivership provisions of the FDIA. Although the Federal Deposit Insurance Corporation ("FDIC") generally has avoidance powers in cases under the FDIA, there are exceptions and safe harbors in the FDIA that support the finality of securities transactions processed through securities clearing agencies.[35]

The FDIA does provide for up to a one-business-day stay of closeout actions while the FDIC determines whether to transfer all the insured depository institution's qualified financial contracts ("QFCs"), including securities contracts to a successor institution. If the FDIC does not effectuate such a transfer, the FDIA protects the rights of parties to QFCs to terminate, liquidate or accelerate such contracts, net amounts owing thereunder and exercise rights under security agreements related to such contracts. Moreover, the FDIA requires the FDIC to transfer all or none of the QFCs between the institution and a given counterparty and all related collateral, and thus prohibits the FDIC from effectuating a transfer that would disrupt a bilateral netting set.

In the event the FDIC is appointed as receiver for a Member, FICC expects to be in discussions with the FDIC regarding FICC's acceptance of such a successor institution as a substitute Member.

The provisions of Title II of Dodd-Frank regarding OLA provide for the appointment of the FDIC as receiver for certain systemically important entities (defined in Title II of Dodd-Frank as "covered financial companies"). Although Title II provides the FDIC with certain avoidance powers in such cases, there are exceptions and safe harbors under Title II which are similar to those under the FDIA that support the finality

contracts under a master netting agreement, to the extent such right would be exercisable in relation to the individual contracts).

[34] Examples of these exceptions and safe harbors include 15 U.S.C. §78eee(b)(2)(C), which provides that the automatic stay shall not apply to "any contractual rights [including rights set forth in a rule or bylaw of a securities clearing agency] of a creditor to liquidate, terminate, or accelerate a securities contract, commodity contract, forward contract, repurchase agreement, swap agreement, or master netting agreement, as those terms are defined in [the Bankruptcy Code], to offset or net termination values, payment amounts, or other transfer obligations arising under or in connection with one or more of such contracts or agreements, or to foreclose on any cash collateral pledged by the debtor, whether or not with respect to one or more of such contracts or agreements." Moreover, protective decrees often recite many of the stay exceptions and safe harbors found in the Bankruptcy Code and SIPA and also contain additional stay exceptions and safe harbors not contained in the Bankruptcy Code and SIPA, including those designed to enable clearing agencies to timely effectuate a closeout or settlement.

[35] Examples of these exceptions and safe harbors include 12 U.S.C. §1821(e)(8)(C), which provides that, notwithstanding any federal or state law relating to the avoidance of preferential or fraudulent transfers, the FDIC, whether acting as such or as conservator or receiver of an insured depository institution, "may not avoid any transfer of money or other property in connection with any qualified financial contract [including a securities contract] with an insured depository institution" (except in cases involving an intent to hinder, delay or defraud).

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of securities transactions processed through securities clearing agencies.[36]

As under the FDIA, Title II of Dodd-Frank provides for up to a one-business-day stay of closeout actions while the FDIC determines whether to transfer all the covered financial company's QFCs to a successor institution. If the FDIC does not effectuate such a transfer, Title II protects the rights of parties to QFCs to terminate, liquidate or accelerate such contracts, net amounts owing thereunder and exercise rights under security agreements related to such contracts. Moreover, as under the FDIA, Title II of Dodd-Frank requires the FDIC to transfer all or none of the QFCs between the covered financial company and a given counterparty and all related collateral, and thus prohibits the FDIC from effectuating a transfer that would disrupt a bilateral netting set.

In the event the FDIC is appointed as receiver for a Member under Title II of Dodd-Frank, FICC expects to be in discussions with the FDIC regarding FICC's acceptance of such a successor institution as a substitute Member.

The exceptions and safe harbors contained in FDICIA, the Bankruptcy Code, SIPA, the FDIA and Title II of Dodd-Frank that support the finality of securities transactions and the closeout of the insolvent Member's open positions provide FICC with a high degree of certainty as to the effectiveness of its risk management and default management rules and procedures.

Transparency of each Division's Rules

FICC makes the following resources, among others, available to the public on the DTCC website: (1) each Division's Rules; (2) submitted Proposed Rule Change and Advance Notice filings; (3) white papers and other reports addressing initiatives under consideration or in process, or other issues of import to stakeholders, including the membership; (4) Important Notices that address issues of import to each Division's membership (including notice of when FICC ceases to act for a Member and information as to how it will handle pending transactions of the defaulting Member); and (5) this Disclosure Framework.

The Proposed Rule Change Process and Advance Notice Process, as described above, require filings to be made publicly available – which is done both by the SEC by publication in the *Federal Register*,[37] and by FICC, by posting on the DTCC website. Interested parties are provided an opportunity to comment publicly on such proposals, including the ability to raise any concerns. Any such concerns may be addressed or responded to as part of the SEC's review.

The SEC is required to disapprove a Proposed Rule Change if it does not find that the change is consistent with the requirements of the Exchange Act and the rules and regulations thereunder that are applicable to FICC. Similarly, the SEC may object to a Material Change that is the subject of an Advance Notice if it determines that the change is inconsistent with the objectives and principles for risk management standards described in Section 805(b) of Dodd-Frank or the rules and regulations thereunder that are applicable to FICC.

[36] Examples of these exceptions and safe harbors include 12 U.S.C. § 5390(c)(8)(C), which provides that, notwithstanding any federal or state law relating to the avoidance of preferential or fraudulent transfers, the FDIC, whether acting as such or as receiver of a covered financial company, "may not avoid any transfer of money or other property in connection with any qualified financial contract [including a securities contract] with a covered financial company" (except in cases involving an intent to hinder, delay or defraud).

[37] The SEC also makes Proposed Rule Change and Advance Notice filings publicly available on its website.

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Degree of certainty for rules and procedures; conflict of laws issues

FICC achieves a high level of confidence that each Division's Rules and material contracts are enforceable in all relevant jurisdictions:

1. *By obtaining legal opinions, analyses and advice and other legal comfort.*

Each Division's Rules and, generally, FICC's material contracts are governed by the law of New York. As described above, U.S. and New York law provide a clear and comprehensive framework for the enforceability of each Division's Rules and FICC's material contracts.

FICC identifies, analyzes and mitigates legal risks arising from potential conflict-of-law issues in a variety of ways. FICC's operations are based in the United States, and each Division's Members are predominantly U.S.-domiciled entities. With respect to membership of non-U.S. entities, GSD Rule 2A and MBSD Rule 2A, as applicable, set forth FICC's policy with respect to the admission of such Members. Among the requirements for such applicants is the provision of a legal opinion of counsel qualified in the applicant's home jurisdiction, to the effect that the choice of New York law to govern the membership agreement and the applicable Division's Rules and the submission to jurisdiction of the federal and state courts located in New York will be enforceable against the Member in its home jurisdiction. These opinions facilitate analysis of any legal risk that may arise as a result of the applicant's participation in FICC. FICC can mitigate conflicts of laws risks or other legal risks that may be identified as a result of such opinions and analysis by, among other things, imposing conditions on membership or additional requirements and/or requiring special representations from an applicant.[38]

FICC also obtains legal opinions, analyses and/or advice, as it deems necessary or appropriate, to confirm, among other things, that each Division's Rules and FICC's material contracts are consistent with relevant laws and regulations prior to becoming effective. FICC will also obtain legal opinions, analyses and/or advice in the future, as it deems necessary or appropriate, to address new services, changes in law or other matters. For example, where applicable, an annual review is conducted with respect to each jurisdiction in which its participants are located to confirm the continued enforceability of its legal opinions, and the advice it has received regarding relevant local licensing requirements.

2. *Through the Proposed Rule Change and Advance Notice Processes.*

As noted above, generally before a Proposed Rule Change may take effect, the SEC must approve the change. In order to do so, it must determine that the change is consistent with the requirements of the Exchange Act and the rules and regulations thereunder that are applicable to FICC. Similarly, before a Material Change takes effect, the SEC, following consultation with the Federal Reserve, must not object to the change. It may object to the change if it determines that the change is inconsistent with the objectives and principles for risk management standards described in Section 805(b) of Dodd-Frank or the rules and regulations thereunder that are applicable to FICC. These processes, together with the opportunity for public review and comment provided thereby, provide a clear record and statutory basis for the enforceability of each Division's Rules.

[38] For example, GSD Rule 4 (Clearing Fund and Loss Allocation) and MBSD Rule 4 (Clearing Fund and Loss Allocation) provide that FICC reserves the right to require the entity to deposit additional amounts to the Clearing Fund in an instance where FICC, in its sole discretion, believes the entity presents legal risk.

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Principle 2: Governance; CCAS 17ad-22(e)(2)

Principle 2: Governance	**CCAS 17ad-22(e)(2)**
An FMI should have governance arrangements that are clear and transparent, promote the safety and efficiency of the FMI, and support the stability of the broader financial system, other relevant public interest considerations, and the objectives of relevant stakeholders. ***Key consideration 1***: An FMI should have objectives that place a high priority on the safety and efficiency of the FMI and explicitly support financial stability and other relevant public interest considerations. ***Key consideration 2***: An FMI should have documented governance arrangements that provide clear and direct lines of responsibility and accountability. These arrangements should be disclosed to owners, relevant authorities, participants, and, at a more general level, the public. ***Key consideration 3***: The roles and responsibilities of an FMI's board of directors (or equivalent) should be clearly specified, and there should be documented procedures for its functioning, including procedures to identify, address, and manage member conflicts of interest. The board should review both its overall performance and the performance of its individual board members regularly. ***Key consideration 4***: The board should contain suitable members with the appropriate skills and incentives to fulfill its multiple roles. This typically requires the inclusion of non-executive board member(s). ***Key consideration 5***: The roles and responsibilities of management should be clearly specified. An FMI's management should have the appropriate experience, a mix of skills, and the integrity necessary to discharge their responsibilities for the operation and risk management of the FMI. ***Key consideration 6***: The board should establish a clear, documented risk-management framework that includes the FMI's risk-tolerance policy, assigns responsibilities and accountability for risk decisions, and addresses decision making in crises and emergencies. Governance arrangements should ensure that the risk-management and internal control functions have sufficient authority,	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (2) Provide for governance arrangements that: (i) Are clear and transparent; (ii) Clearly prioritize the safety and efficiency of the covered clearing agency; (iii) Support the public interest requirements in Section 17A of the Act (15 U.S.C. 78q-1) applicable to clearing agencies, and the objectives of owners and participants; (iv) Establish that the board of directors and senior management have appropriate experience and skills to discharge their duties and responsibilities; (v) Specify clear and direct lines of responsibility; and (vi) Consider the interests of participants' customers, securities issuers and holders, and other relevant stakeholders of the covered clearing agency.

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independence, resources, and access to the board. ***Key consideration 7***: The board should ensure that the FMI's design, rules, overall strategy, and major decisions reflect appropriately the legitimate interests of its direct and indirect participants and other relevant stakeholders. Major decisions should be clearly disclosed to relevant stakeholders and, where there is a broad market impact, the public.	

Ownership and Board structure

FICC is a wholly owned subsidiary of DTCC. DTCC is user owned and governed pursuant to a Shareholders Agreement.[39] The DTCC common shareholders include banks, broker-dealers, and other companies in the financial services industry that are participants of one or more of DTCC's clearing agency subsidiaries, including FICC. DTCC common shares are allocated to participants in accordance with a formula based on their relative usage of the services of the three clearing agencies.

Individuals elected to the DTCC Board of Directors are also elected to the Boards of Directors of FICC and of its affiliates, DTC and NSCC, and these boards generally operate together with the DTCC Board (collectively, the "Board"). DTCC (and FICC) is a New York business corporation. As such, DTCC's business is subject to the oversight of the Board, and managed on a day-to-day basis by DTCC's senior management.

The Board is currently composed of (i) directors who represent clearing agency participants, including broker/dealers, custodian and clearing banks and investment institutions ("participant directors"); (ii) non-participant directors; (iii) directors who are designated by DTCC's Series A and Series B preferred shareholders (ICE and the Financial Industry Regulatory Authority, respectively); (iv) DTCC's Non-Executive Chairman; and (v) a member of DTCC senior executive management (the President and Chief Executive Officer). In addition to the Series A and Series B preferred stock, DTCC has issued additional series of preferred stock, none of which have rights to appoint directors. The non-participant directors are not employed by any user of DTCC's services. Non-participant directors bring additional skills and expertise to the Board, mitigate potential conflicts of interest among participant directors and introduce different perspectives. Collectively, the participant directors, non-participant directors, the Non-Executive Chairman, the management director and representatives of the Series A and B preferred shareholders provide a diverse representation of DTCC's stakeholders and ensure that the broad interest of the public is represented. All directors, except those designated by the preferred shareholders, are elected annually to one-year terms.

As noted in the Board of Directors Mission Statement and Charter (available on the DTCC website), the Board is responsible for providing direction to and overseeing the conduct of the affairs of the corporation in the interests of the corporation, its shareholders and other stakeholders including investors, issuers and participants in and regulators of the financial markets that DTCC serves. The Board plays a key role in policy development and establishment of corporate objectives, financial management and operational planning.

[39] The Fifth Amended and Restated Shareholders Agreement, dated March 20, 2024 (the "Shareholders Agreement"), among DTCC, DTC, NSCC, FICC, and the common shareholders, Series A preferred shareholders and Series B preferred shareholders of DTCC.

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The names and backgrounds of the members of the Board and DTCC senior management are available on the DTCC website.

The Board's Mission Statement provides that:

> The Board will discharge its oversight responsibilities and exercise its authority in a manner, consistent with applicable legal and regulatory provisions and with regulatory expectations of a systemically important financial market infrastructure, that:
>
> - Promotes the safe, sound and efficient operation of DTCC and its subsidiaries, including the clearance and settlement activities conducted by its registered clearing agency subsidiaries;
>
> - Safeguards the securities and funds in the custody of its registered clearing agency subsidiaries, the SIFMUs and protects investors and the public interest;
>
> - Fosters the safe, sound and efficient operation of services provided by DTCC and its subsidiaries supporting the global system for processing transactions in financial instruments and related activities;
>
> - Fosters cooperation and coordination between parties engaged in the clearance and settlement of securities transactions;
>
> - Seeks to develop the services and businesses of DTCC and its subsidiaries in a manner promoting further safety, soundness and efficiency broadly in the global system for processing transactions in financial instruments and related activities; and
>
> - Leverages DTCC's role as a leader in financial services with respect to risk management and systemic risk management, promoting sound practices in governance and in transparency to its membership and user community and in its role as a systemic component of the financial market infrastructure supporting the operation of orderly and efficient markets in the interest of the investing public as required under Section 17A of the Securities Exchange Act of 1934.

Collectively, these objectives are designed, particularly with respect to the activities of DTC, FICC and NSCC, to promote the prompt and accurate clearance and settlement of securities transactions and the safeguarding of securities and funds in the custody or control of DTC, FICC and NSCC or for which they are responsible.

In furtherance of supporting the relevant public interest considerations, DTCC consults with individual participants, members of the Board, standing advisory councils, industry associations, regulators and others. Individual product managers are actively engaged with relevant industry groups and associations in partnering to identify ways that DTCC can continue to develop operational, technological and risk mitigation solutions for the capital markets industry.

DTCC management prioritizes resiliency, security and efficiency as evidenced through reviews of businesses and initiatives by enterprise-wide management committees, including the Executive Committee, the Management Risk Committee ("MRC"), the Investment Management Committee ("IMC"), and the IT Governance Committee, organized to focus on such issues. Initiatives are also reviewed at the Board level by the relevant Board committee, as appropriate. For example, risk initiatives are reviewed by the MRC and escalated as appropriate to the Board Risk Committee. The Compensation and Human Resources Committee, in consultation with the Board Risk Committee Chair, seeks to ensure that compensation policies do not encourage excessive risk-taking and meet evolving standards on compensation practices.

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DTCC works to foster a culture of openness and dialogue at the management and Board levels with its regulators (sometimes referred to as the “supervisors”).

Governance structure and arrangements

The governance arrangements, which also identify the roles and lines of responsibility and accountability within DTCC, are described in by-laws, the Board of Directors Mission Statement and Charter, the Procedure for the Nomination, Evaluation and Appointment of Candidates to the DTCC, DTC, FICC, NSCC Boards of Directors, and each of the Board Committee Charters, all of which are publicly available on the DTCC website. The Board Charter includes provisions relating to board composition, meetings, election of directors, board committees, duties and responsibilities of directors, and director qualification standards. With respect to management, the roles and responsibilities of management, including reporting lines, are provided in job descriptions.

The Board currently has designated seven (7) standing Committees, and each director currently serves on at least two Board committees. The committees are briefly described below:

Audit Committee: The Board has established an Audit Committee to assist the Board in overseeing (1) the integrity of DTCC’s financial statements and financial reporting; (2) the overall effectiveness of DTCC’s internal control environment; (3) the effectiveness of DTCC’s process for monitoring compliance with applicable laws, regulations and the Code of Conduct; (4) the performance and coverage of the internal audit function; (5) the external auditor’s independence, performance and coverage; (6) functional areas within the Committee’s jurisdiction to ensure proper communication of any issues or risks material to the clearing agencies; and (7) legal, compliance and regulatory risks as they relate to DTCC or its subsidiaries or any clearing agency’s provision of clearance and settlement services as an industry utility. The roles and responsibilities of the Audit Committee are outlined in the Audit Committee Charter.

Compensation and Human Resources Committee: The Board has established a Compensation and Human Resources Committee to assist it in overseeing DTCC's human capital management programs to ensure that they are aligned with DTCC’s strategic goals and objectives and to continually improve business performance. Programs included in the scope of the Compensation and Human Resource Committee’s responsibilities include those related to talent acquisition, retention, executive leadership development, performance management, succession planning, total rewards, diversity and the fostering of a high-performance corporate culture. The Compensation and Human Resources Committee further advises on the extent to which human capital management programs are periodically reviewed and revised so as to be in alignment with industry and regulatory best practices. The roles and responsibilities of the Compensation and Human Resources Committee are outlined in the Compensation and Human Resources Committee Charter.

Enterprise Services Committee: The Board has established an Enterprise Services Committee to assist the Board in overseeing all aspects of the Company’s operational performance relating to the Company’s Enterprise Services, including but not limited to Custody Clearing and Settlement Operations, Asset Servicing Operations, Enterprise Client Operations, Enterprise Digital Solutions, Enterprise Data Solutions, Governance and Reporting, and Enterprise Resiliency Office.

Executive Committee: The Board has established an Executive Committee to exercise powers of the Board in the event that an emergency or other time-sensitive matter arises, and it is not practicable to assemble the entire Board. The roles and responsibilities of the Executive Committee are outlined in the Executive Committee Charter.

Governance Committee: The Board has established a Governance Committee to assist the Board in (1) identifying, screening and reviewing individuals qualified to serve as directors and recommending to the Board candidates for nomination for election at the annual meeting of shareholders or to fill Board vacancies; (2) developing, recommending to the Board and overseeing implementation of DTCC's corporate governance policies and procedures and the Board Code of Ethics; and (3) reviewing on a regular basis the overall corporate governance framework of DTCC and recommending enhancements and/or

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changes when necessary. The roles and responsibilities of the Governance Committee are outlined in the Governance Committee Charter.

Risk Committee: The Board has established a Board Risk Committee ("BRC") to assist the Board in fulfilling its responsibilities for oversight of DTCC's risk management activities focusing on the following critical aspects: (1) oversight of risk management systems and processes designed to identify and manage credit, market, liquidity and operational risks to DTCC and its subsidiaries; and (2) due to DTCC's unique capabilities and position, oversight of DTCC's efforts to mitigate certain "systemic risks" that may undermine the stable operation of the financial system. To provide additional dedicated focus and align oversight of specialized risk categories with the domain expertise of other non-standing committees that have been established, or may be established from time to time, by the Board, certain risks will be overseen by these other Board committees. While these risks will not be directly overseen by the BRC, the BRC will, nonetheless, coordinate risk oversight with these other Board committees as appropriate to achieve a comprehensive and holistic oversight of the organization's risk-related matters. The Board retains the authority to review matters brought to the BRC and request immediate escalation to the full Board should the Board deem appropriate. The roles and responsibilities of the BRC and its governance arrangements, which have been designed to prioritize the clearing agencies' safety and efficiency in support of the public interest and the prompt and accurate clearance and settlement of securities transactions, as well as the objectives of clearing agency members and participants, as required under the Exchange Act are outlined in the Risk Committee Charter.

Technology & Cyber Committee: The Board has established a *Technology & Cyber Committee to* assist the Board in overseeing all aspects of the Company's technology and cyber program, procedures, performance and risks relating to the Company's information technology and cybersecurity infrastructure.

Advisory Councils

In order to best be able to consider the interests of participants' customers, securities issuers and holders, and other relevant stakeholders of the covered clearing agency including investors, issuers, participants, participants' customers, other persons that access the national system for clearance and settlement indirectly, and regulators of the financial markets in which FICC serves, FICC has established certain advisory councils comprised of representatives from clearing agency participants, their customers, and other stakeholders with appropriate seniority and expertise in the relevant subject matter. Additionally, as required by SEC Rule 17ad-25(j), FICC solicits council representatives' views on material developments in the clearing agencies' risk management and operations and escalates these viewpoints to the Board or the appropriate Board committee for its consideration.

Review of Board performance and conflicts of interest

The Governance Committee of the Board reviews and addresses director conflicts of interest. Directors are subject to the DTCC Board Code of Ethics and Conflicts of Interest Policy (the "Code"). The Code , which is available on the DTCC website, describes how conflicts of interest are identified and how they are to be addressed. The Governance Committee is responsible for reviewing and approving any changes to the Code.

The Board conducts an annual self-evaluation of its performance and the performance of its committees. The Governance Committee is responsible for coordinating and providing oversight of the annual self-evaluation, including determining the methodology for the evaluation, and overseeing its execution. The results of the self-evaluations are reviewed by the respective committee, the Governance Committee and the Board.

Director nomination process

As a registered clearing agency, FICC is subject to Section 17(A) of the Exchange Act, which provides that the rules of a clearing agency must assure a "fair representation of its shareholders and participants in the

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selection of its directors and administration of its affairs…." DTCC's shareholders are the participants and members of its three clearing agency subsidiaries, including FICC's Members, pursuant to GSD Rule 49 and MBSD Rule 39. This "fair representation" requirement is currently satisfied by allocating shares, pursuant to the Shareholders Agreement, with attendant voting rights to elect directors, in accordance with a formula based on usage of the services of the three clearing agencies.

Each year, typically in December, a nominations solicitation letter along with a copy of the Selection Process and Character and Fitness Standards for Board of Director Candidates is sent to the participants of DTC, FICC and NSCC, and is also provided via Important Notice on DTCC's website. This begins the process by which participants of the clearing agency subsidiaries may nominate individuals for election as participant directors. The Governance Committee reviews the individuals nominated and recommends for Board approval the full director slate to be presented for election at DTCC's Annual Shareholders' Meeting Using cumulative voting, DTCC's common shareholders vote on the slate at such meeting.

The Selection Process and Character and Fitness Standards for Board of Director Candidates set forth the skill sets that are necessary for Board members. The skill sets are reviewed annually by the Governance Committee, and changes, if any, are recommended to the Board for approval.

Each director is required to complete a skills and experience questionnaire. In accordance with its Committee Charter, on a yearly basis, the Governance Committee reviews the skills and experience of the current directors and determines, among other things, whether there are any gaps in skills or experiences that would be desirable to have filled by a new director nominee. The Board Charter contains a list of qualifications and areas of expertise and experience that Board members should have.

Individuals are nominated for election as directors based on their ability to represent DTCC's diverse base of participants, and DTCC's governance is specifically structured to help achieve this objective. The non-participant Board members are individuals with specialized knowledge of financial services, but who bring an independent perspective since they are not employed by firms that use clearing agency services. Collectively, the Board composition reflects an appropriate balance of participant and non-participant directors. Board members serve on a variety of Board committees with responsibility to oversee aspects of DTCC and its clearing agency subsidiaries' operations.

The Board plays an integral role in the oversight of the firm, ensuring DTCC services (including those provided by FICC) continue to meet the evolving needs of participants.

Management experience, skills, integrity and performance

DTCC has talent management programs and succession planning programs that are reviewed, updated and presented to the Compensation and Human Resources Committee annually. On an annual basis, the full Board is updated on the company's succession planning process and on the status of development plans for key individuals. For very senior level positions, multiple members of the Executive Committee and selected Board Members may be involved in the interview and selection process. For the most senior levels of the organization, a search committee of the Board may be formed to conduct the process.

Performance appraisals are prepared on an annual basis for personnel, including Managing Directors.

On an annual basis, the CEO reviews the performance of the Executive Committee members and discusses the performance with the members of the Compensation and Human Resources Committee.

The performance and compensation of the CEO and Non-Executive Chairman are reviewed by the Compensation and Human Resources Committee and presented to the full Board for approval.

A Risk Assessment of the Executive Committee (and selected additional Managing Directors) is conducted on an annual basis by the Group Chief Risk Officer, the Chief Compliance Officer and the General Auditor

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and reviewed with the Compensation and Human Resources Committee and the Chairs of the Risk and Audit Committees.

The Board of Directors measures management's performance through DTCC's enterprise performance management framework (the "EPM Framework") which measures performance across corporate goals aligned to four strategic pillars, corporate goals, enterprise health metrics, and fundamentals. Metrics are periodically reviewed throughout each year through (i) the review of status reports prepared by management, and (ii) the Board Committees that have oversight responsibilities for particular activities. In addition, the Board's Compensation and Human Resources Committee receives periodic updates throughout the year on the achievement of the established EPM Framework for the year. As provided in its Charter, the Compensation and Human Resources Committee is also responsible for the following:

- In consultation with the Board Risk Committee Chair, assuring that corporate goals and objectives are aligned with the interests of DTCC's stakeholders and do not encourage executive officers to take undue risks;

- Reviewing and assessing, in consultation with the BRC Chair, whether the compensation program:

    - Promotes appropriate risk identification and management approaches,

    - Discourages executive officers from taking unnecessary or excessive risks,

    - Includes a regular assessment of key performance indicators to measure and compensate executive officers, and

    - Maintains a high level of responsiveness to concerns and recommendations raised by officers;

- Obtaining input from the Group Chief Risk Officer, General Auditor and Chief Compliance Officer regarding executives' performance in relation to risk management/behaviors and considering that input when determining compensation; and

- Ensuring that critical matters and material systemic risk concerns are escalated to the full Board.

Personnel integrity is a core DTCC value. DTCC maintains Codes of Conduct that set forth the foundational principles that govern DTCC's business. All personnel must comply with the applicable Code of Conduct and are expected to have a clear understanding of and commitment to the high ethical standards outlined in the Code of Conduct.

Risk management framework

DTCC maintains a risk management framework for comprehensively managing legal, credit, liquidity, operational, general business, investment, custody, and other risks. The goal of the Corporate Risk Framework is to define DTCC's risk management program and provide guidelines to manage key risks across the organization in a comprehensive, consistent and effective manner, enabling DTCC to achieve its strategic business objectives while remaining consistent with its risk tolerances. While risk types are diverse, their effective management requires a core set of common principles and common processes for identifying, assessing, measuring, monitoring, mitigating and reporting risk. The Board has ultimate responsibility for the effectiveness of the framework's implementation. In carrying out this responsibility, applicable guidelines make clear that the Board is not expected to itself implement the framework or to carry out the day-to-day management of risks. Rather, the Board, in discharging its ultimate responsibilities over risk management matters works closely with management and may assign tasks to board committees. DTCC's risk management framework is documented through the DTCC Corporate Risk Management Policy, which includes the several Risk Tolerance Statements in effect from time to time. Each Risk Tolerance Statement pertains to a specified risk within the "risk family framework" established to guide DTCC's discussion of risk. The DTCC Corporate Risk Management Policy is reviewed, updated (as

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appropriate), and approved by the MRC, at least annually, and is reviewed and approved by the Board Risk Committee annually.

The DTCC Corporate Risk Management Policy is the overarching policy from which all risk management policies and procedures within the primary purview of the Group Chief Risk Office, among other areas, derive. The Group Chief Risk Office consists of various areas, each of which has its own risk management policies designed to address that area's specific domain of expertise. These policies are reviewed and updated (as appropriate) at least annually.

Oversight of the Group Chief Risk Office is the responsibility of the BRC. The Group Chief Risk Officer has a direct reporting line to the Chair of the BRC and reports, administratively, to the DTCC President and Chief Executive Officer. The BRC has responsibility to review and approve the compensation recommendation of the Group Chief Risk Officer. This specialized Board oversight over the Group Chief Risk Office is designed to provide the Group Chief Risk Office with independence from the operating business areas of DTCC.

The risk management framework is more fully described under Principle 3 (Framework for the comprehensive management of risks).

In addition, DTCC has established policies and procedures that address decision making in emergencies regarding the Clearing Agencies, including FICC.

Internal Control Functions (Internal Audit and Compliance)

The mission of the Internal Audit Department ("IAD") is to provide independent and objective assurance services to assist the enterprise in maintaining effective risk management and control practices. IAD supports the Board in its oversight of the governance, risk and control framework and seeks to achieve and maintain the highest level of professional standards, while helping the company achieve its objectives. The Audit Committee is responsible for overseeing the internal audit function. IAD is governed by policies reviewed and approved by the Audit Committee and the Board.

IAD's assurance services include evaluating operations and internal controls to validate that the enterprise's assets and member assets held by DTCC and its subsidiaries (including FICC) are safeguarded, and that policies, standards, and procedures are being followed. Additionally, IAD helps management proactively identify risks that require controls or risk mitigation strategies for new businesses, products, and services, and to properly maintain controls in existing and expanding businesses.

The scope of IAD's activities is subject to the Audit Committee's review and approval, but is not otherwise restricted. The General Auditor directs all internal audit activities in support of its mission to support the Board and to enhance the control environment.

IAD is expected to be independent in the execution of its activities. As noted above, independence is achieved through organizational placement and reporting lines. To help maintain independence, staff assignments are rotated periodically as is practicable. In addition, the General Auditor, among others, periodically meets privately with the Audit Committee, without members of management or the external accounting firm present. The Audit Committee Chair conducts the General Auditor's annual performance assessment and the Audit Committee reviews and approves the compensation recommendation for the General Auditor.

The Compliance Department is a control function dedicated to protecting DTCC and preserving the safety and soundness of the operations of DTCC and its subsidiaries (including FICC) in the global financial community through compliance with applicable laws and regulations and by providing advice, training and education to business units and management. Compliance works with other control functions and with management to make sure that potential regulatory risks and liabilities are monitored, identified, escalated and appropriately addressed. The Compliance Department's main responsibilities are the development and

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maintenance of DTCC's compliance program ("Compliance Program"). The Audit Committee is responsible for overseeing the design and management of the Compliance Program, including policies and procedures reasonably designed to ensure that compliance issues are resolved effectively and expeditiously by the Chief Compliance Officer and senior management. The Chief Compliance Officer assists the Audit Committee in the performance of its duties with regards to oversight of legal and regulatory compliance and Code of Conduct matters.

The Chief Compliance Officer, among others, periodically meets privately with the Audit Committee, without members of management or the external accounting firm present. The Audit Committee is responsible for reviewing and endorsing management's appointment, reassignment or dismissal of a Chief Compliance Officer, and presenting that appointment, reassignment or dismissal to the Board for its approval and the Audit Committee reviews and approves the compensation recommendation for the Chief Compliance Officer.

Disclosure

Major decisions of the Board that are not confidential are disclosed to relevant stakeholders, as appropriate. To the extent such decision is reflected in a change to the Rules of FICC or constitutes a material change to the nature or level of risk presented by FICC, disclosure is made though public regulatory filings with the appropriate regulators, as applicable, and often in Important Notices, which are published on the DTCC website. Decisions and information with a broader impact or of more general interest are often disclosed through press releases, articles, white papers, Important Notices and other materials posted to the DTCC website, including DTCC's Annual Reports and the financial statements of the clearing agency subsidiaries, as applicable. Section VI of this Disclosure Framework provides a list of information and documents that is publicly available on the DTCC website.

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Principle 3: Framework for the comprehensive management of risks; CCAS 17ad-22(e)(3)

Principle 3: Framework for the comprehensive management of risks	**CCAS 17ad-22(e)(3)**
An FMI should have a sound risk-management framework for comprehensively managing legal, credit, liquidity, operational, and other risks. ***Key consideration 1***: An FMI should have risk-management policies, procedures, and systems that enable it to identify, measure, monitor, and manage the range of risks that arise in or are borne by the FMI. Risk-management frameworks should be subject to periodic review. ***Key consideration 2***: An FMI should provide incentives to participants and, where relevant, their customers to manage and contain the risks they pose to the FMI. ***Key consideration 3***: An FMI should regularly review the material risks it bears from and poses to other entities (such as other FMIs, settlement banks, liquidity providers, and service providers) as a result of interdependencies and develop appropriate risk-management tools to address these risks. ***Key consideration 4:*** An FMI should identify scenarios that may potentially prevent it from being able to provide its critical operations and services as a going concern and assess the effectiveness of a full range of options for recovery or orderly wind-down. An FMI should prepare appropriate plans for its recovery or orderly wind-down based on the results of that assessment. Where applicable, an FMI should also provide relevant authorities with the information needed for purposes of resolution planning.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (3) Maintain a sound risk management framework for comprehensively managing legal, credit, liquidity, operational, general business, investment, custody, and other risks that arise in or are borne by the covered clearing agency, which: (i) Includes risk management policies, procedures, and systems designed to identify, measure, monitor, and manage the range of risks that arise in or are borne by the covered clearing agency, that are subject to review on a specified periodic basis and approved by the board of directors annually; (ii) Includes plans for the recovery and orderly wind-down of the covered clearing agency necessitated by credit losses, liquidity shortfalls, losses from general business risk, or any other losses; (iii) Provides risk management and internal audit personnel with sufficient authority, resources, independence from management, and access to the board of directors; (iv) Provides risk management and internal audit personnel with a direct reporting line to, and oversight by, a risk management committee and an independent audit committee of the board of directors, respectively; and (v) Provides for an independent audit committee.

Risk Management Framework

DTCC maintains a risk management framework for comprehensively managing legal, credit, liquidity, operational, general business, investment, custody, and other risks that arise in or are borne by FICC. The framework includes policies, procedures, and systems that are designed to identify, measure, monitor, and manage such risks to FICC.

The risk management framework is outlined in DTCC's Corporate Risk Management Policy, attached to which are the set of Risk Tolerance Statements described below. In addition, FICC has filed and the SEC has approved a number of risk management documents: (i) the Clearing Agency Risk Management Framework, (ii) the Clearing Agency Stress Testing Framework; (iii) the Clearing Agency Model Risk Management Framework; (iv) the Clearing Agency Price and Margin Input Data Framework; (v) the Clearing Agency Liquidity Risk Management Framework; (vi) the Clearing Agency Operational Risk Management

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Framework; (vii) the Clearing Agency Investment Policy; (viii) the Clearing Agency Policy on Capital Requirements; (ix) the Clearing Agency Capital Replenishment Plan; and (x) Clearing Agency Framework for Certain Requirements on Governance and Conflicts of Interest (collectively, the "Clearing Agency Risk Frameworks").

The DTCC Corporate Risk Management Policy, the Risk Tolerance Statements, and the Clearing Agency Risk Frameworks, as well as the various applicable supporting policies and procedures thereunder, are reviewed by their respective owners and approved by the MRC at least annually. They are also approved by the BRC at least annually.

FICC effectively manages legal, credit, liquidity, operational, general business, investment, custody, and other risks that arise in or are borne by it, including systemic risk to the broader marketplace, *in part*, by monitoring its activities against the Risk Tolerance Statements and the risk tolerance thresholds therein contained. The Risk Tolerance Statements align to the Risk Families identified below (the " *Risk Families represent the Level 1 (L1) categories of the Risk Taxonomy*)):[40]

- Clearance & Settlement Financial Risk
- Corporate Financial Risk[41]
- Data Management Risk
- Model Risk
- Legal & Regulatory Compliance Risk
- Processing & Operations Risk
- Information Technology Risk
- Cybersecurity & Information Protection Risk
- Human Capital Risk
- Business Continuity Risk
- Third Party Risk
- Strategic Risk

Risk Tolerance Statements outline the risk mitigation objectives and controls for managing FICC's Risk Families. Each Risk Tolerance Statement, among other things: (i) identifies the Risk Families to which it applies; (ii) identifies the key indicators (or metrics) that categorize the tolerance thresholds (from "within tolerance" to "exceeds tolerance"); and (iii) identifies the governance and escalation protocols applicable to any risk metric breach that would result in FICC falling outside of the accepted tolerance level.

The effective management of FICC's diverse risks requires a core set of common principles and common processes for identifying, assessing, measuring, monitoring, mitigating, and reporting risk. The DTCC Corporate Risk Management Policy and the Risk Tolerance Statements define the approach through which risk tolerances are established, communicated, and monitored. These documents, in combination with the Clearing Agency Risk Frameworks, have been designed to provide guidelines for managing FICC's risk in a comprehensive, consistent, and effective manner. Adherence to the DTCC Corporate Risk Management Policy, the Risk Tolerance Statements, and the Clearing Agency Risk Frameworks supports FICC achieving its business objectives while remaining within its established risk tolerance levels.

[40] These risks and risk families are also referred to as the Key Clearing Agency Risks in the Clearing Agency Risk Management Framework.

[41] DTCC/FICC defines this Key Risk to include those risks addressed by Principle 15 (General business risk) and Principle 16 (Custody and investment risk) below.

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The Three Lines of Defense.

All DTCC personnel, including all personnel supporting FICC's businesses, are responsible for identifying, assessing, measuring, monitoring, mitigating, and reporting risk. DTCC personnel are expected and encouraged to be proactive, not reactive, in managing risks. DTCC operates under a core set of common risk management principles and processes where risk management methodologies are integrated into all significant operations, including FICC's operations, through an approach involving the three lines of defense described below.

- The First Line of Defense: The first line of defense is comprised of the various FICC business lines and supporting functional units including Product Management, Global Business Operations, Information Technology, and other areas critical to FICC's daily operations and functioning. Their mandate is to manage risk proactively on a day-to-day basis.

- The Second Line of Defense: The second line of defense is comprised of DTCC's control functions, including the Legal Department, the Privacy Office, Compliance, and those areas that fall within the Group Chief Risk Office, as outlined below. Their mandate is to provide advice and guidance to the first line of defense for adhering to established risk standards and/or to monitor compliance with such established risk standards.

- The Third Line of Defense: The third line of defense is the Internal Audit Department ("IAD"). IAD's mission is to challenge the adequacy of DTCC's control environment to promote the organizations resiliency and security in safeguarding the global financial markets. IAD provides independent and objective assurance to assist in DTCC's maintenance of effective risk management and control practices.

Policies, Procedures and Systems

DTCC maintains a comprehensive set of policies, procedures, and systems designed to identify, measure, monitor, and manage FICC's Risk Families. The set of policies and procedures includes the DTCC Corporate Risk Management Policy, the Clearing Agency Risk Frameworks, and the underlying policies and procedures that support their execution.

Comprehensive Management of Risks

The risks presented by FICC's business lines and supporting functional units are monitored through a process of assessing inherent risk (that is, risk in the absence of any mitigating controls) and then evaluating such inherent risk against the strength of the then-existing controls (and, as appropriate, new and additional controls). Assessments are prepared and documented at least annually, and are supported by quarterly metrics, which consolidate pertinent operational risk and control data to support an overall assessment of the applicable business line's or functional unit's inherent risk and residual risk. The data collected in these metrics and risk assessments supports planning and helps guide decision-making. Additional actions are performed to assess and manage operational risk, including the collection of internal incident data, external event data, metrics, key risk indicators, monitoring issues, and preparing issue reports. Additionally, FICC's performance relative to the Risk Tolerance Statements is assessed quarterly and is shared with senior management, the BRC, and MRC for further discussion. FICC's management of its operational risks, including business continuity and technology system arrangements, is discussed more fully in Principle 17 (Operational risk) below.

FICC's monitoring and management of credit and market risks are addressed in Principles 4 and 6, respectively. FICC's mitigation of collateral risk is addressed in Principle 5 (Collateral). FICC's management of its liquidity risks is described in Principle 7 (Liquidity risk). Finally, FICC's management of custody and investment risks is discussed in Principle 16 (Custody and investment risk).

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DTCC employs a structured approach for the implementation of new initiatives, which includes defining the initiative being proposed (e.g., description, estimated investment cost), evaluating and approving initiatives and, among other things, conducting a comprehensive risk assessment of new initiatives that are in scope. Such new initiatives are subject to necessary governance and oversight to enable FICC to bring initiatives to market in a timely and efficient manner, while helping ensure, where relevant, that these initiatives are designed in a way that appropriately mitigates risk to FICC, its Members and the financial markets.

The Group Chief Risk Office and IAD

The Group Chief Risk Office. The Group Chief Risk Office is the dedicated, independent control function specifically charged with monitoring and managing FICC's risks. The Group Chief Risk Office's responsibilities include: (i) day-to-day management of FICC's credit, market, liquidity, operational and systemic risks; (ii) establishing the processes by which FICC's business lines and other functional units report on risk, actively monitoring the Risk Tolerance Statements and risk assessments described above, and regularly reporting its findings to the MRC and the BRC; and (iii) promptly escalating risk tolerance threshold breaches and/or material risk exposures in accordance with the Risk Tolerance Statements.

The Group Chief Risk Office is divided into two core risk areas: (i) Operational & Technology Risk, and (ii) Financial Risk Management ("FRM").

The head of the Group Chief Risk Office is the Group Chief Risk Officer. The Group Chief Risk Officer has a direct reporting line to the Chair of the BRC and is responsible for assisting the BRC, as needed, in the performance of its duties. The BRC is responsible for approving the Group Chief Risk Office's annual budget and for monitoring the Group Chief Risk Office's performance.

The Internal Audit Department. As the third line of defense, IAD is independent from all other FICC business lines, support areas, and control functions. The head of IAD is the General Auditor. The General Auditor reports directly to the Chair of the Audit Committee and is responsible for assisting the Audit Committee in the performance of its duties. The Audit Committee is responsible for reviewing and approving IAD's annual budget, as well as for annually reviewing and approving IAD's structure, staffing, and resources. IAD evaluates FICC's operations and internal controls to validate that FICC is providing services in a safe and sound manner, consistent with applicable regulatory requirements and guidance, that its Members' assets are safeguarded, and that policies, and procedures are being followed. The General Auditor's annual report on the control environment, and annual analysis of significant industry trends in risks and control is reviewed by the Audit Committee.

Management and Board Committees and Board Oversight

Management oversight. While responsibility for proactive, day-to-day management of risk lies with business line and functional unit managers and their staff, senior management oversight is achieved through the following management committees:

- The Executive Committee: The Executive Committee provides enterprise-wide strategic direction for, and oversight of DTCC's (including FICC's) mission, vision and values as well as its businesses, technology and operations, human capital, financial health, and risk management. Among its responsibilities, the Executive Committee periodically reviews and assesses overall performance, including but not limited to, goals, metrics, targets (including the enterprise performance management framework), and budget, and approves recommendations for change as necessary. The Executive Committee provides oversight to and governance for the other management committees described below.

- The Management Risk Committee: The MRC is primarily responsible for implementing DTCC's risk management framework by overseeing the management of credit, market, liquidity, operational, and systemic risks in accordance with the DTCC Corporate Risk Management Policy and the Risk Tolerance Statements. Among its responsibilities, the MRC: (i) reviews and approves at least

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annually those policies and procedures that are critical to FICC's management of clearing agency risks, including, for example, credit, market, liquidity, operational, and systemic risks; (ii) where applicable, approves applicants for FICC membership, and oversees other FICC Rules and participation matters: (iii) reviews reporting of Risk Families and related risk tolerance thresholds for FICC, and escalates, as appropriate, those matters requiring Board level consideration to the BRC or other Board committees.

- Investment Management Committee: The IMC was formed to drive a more rigorous and disciplined approach to how DTCC manages business planning and allocates resources for the annual IT budget. The IMC is responsible for overseeing, monitoring, and approving DTCC's portfolio of investments and initiatives, ensuring that information technology and financial resources are deployed to the organization's highest priorities in alignment with established governance protocols for new initiatives.

- IT Governance Committee ("ITGC"): The ITGC (i) provides oversight of DTCC's IT department and IT strategic objectives, (ii) assesses performance and progress against that strategy, and (iii) provides oversight of technology capabilities, including but not limited to capacity management and performance of IT infrastructure. ITGC also (i) reviews IT process risk and controls (PRC), control testing, and risk assessments; (ii) reviews critical matters and concerns related to IT's existing and/or new initiatives and services, as appropriate; and (iii) regularly provides a consolidated and FICC-level view, as appropriate, of critical IT governance metrics to DTCC's Technology and Cyber Committee for awareness and escalates related matters as appropriate.

Board oversight. The Board is responsible for providing direction to and overseeing the conduct of the affairs of DTCC in the interests of DTCC, its shareholders, and other stakeholders - including investors, issuers, and participants in the financial markets served by DTCC and FICC. The Board discharges its oversight responsibilities and exercises its authority in a manner that, among other things, is designed to promote the safe, sound, and efficient operation of FICC and the services it provides, and that promotes safety, soundness, and efficiency broadly in the global system for processing financial transactions. The Board fulfills its role either directly or by delegation of certain responsibilities to committees of the Board. The Board Committees and their responsibilities are described in Principle 2 (Governance) above.

Information and Incentives for Member Management of Risk

Members are provided with a significant amount of information, and incentives, to enable them to understand, monitor and manage the risks they pose to FICC, as further described under Principle 23 (Disclosure of Rules, Key Procedures, and Market Data). First, the respective Rules of GSD and MBSD, which are publicly available, establish Membership responsibilities, including Members' daily responsibilities to meet Clearing Fund requirements and complete settlement. Changes to the GSD Rules and MBSD Rules are subject to the Proposed Rule Change and Advance Notice processes (collectively, "Filing Requirements"), as described in Principle 1 (Legal basis) above. Pursuant to the Filing Requirements, proposed changes to the respective Division's Rules are published in the *Federal Register* and subject to public review and comment.

On a daily basis FICC provides Members a number of reports showing their activity in all phases of the settlement cycle; this includes transaction messages and reports, Clearing Fund requirements, and settlement reports/accounting summaries.

To assist firms in understanding FICC's products, services and applications, FICC provides a number of tools, including user guides, as well as training courses through DTCC's learning portal. Topics covered include not only functional and operational aspects of securities clearing through FICC, but also risk management practices and methodologies. In addition, FICC has provided Members with the FICC Client Calculator, which is a tool that simulates real and hypothetical portfolios through the same engine that FICC uses on a daily basis to calculate the VaR component of the Required Fund Deposit. This tool allows Members to monitor and manage the impact that their trading activities may have on their Required Fund Deposit.

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Each GSD and MBSD Member is required to meet its Clearing Fund requirements which are based on the Member's outstanding positions as well as its intraday trading and settlement activity. Clearing Fund reports help Members manage their market risk on an ongoing basis. The resizing of a Member's margin requirement (at least twice a day for GSD Members and once per day for MBSD Members) and its direct correlation with the amount of risk created by a Member's trade activity provide incentives for Members to carefully monitor and manage the risks that they pose to FICC. Members are incentivized to monitor their Clearing Fund requirements continuously, as these requirements have a direct impact on the amount of collateral and liquidity at their disposal, as further described in Principle 5 (Collateral) and Principle 7 (Liquidity risk).

FICC also has the ability to require additional Clearing Fund deposits or other adequate assurances (such as limitations on their activity), as a risk mitigant from those firms it considers may pose a risk to GSD or MBSD, as applicable, or their memberships.

Further, FICC has a number of disciplinary tools at its disposal under its Rules. These include monetary fines, reporting certain incidents to regulators of affected Members, and the potential to limit access to one or more services or limit activities, up to and including termination of membership should a Member fail to meet its obligations to FICC.

Material interdependency risks

Given its central role in the U.S. securities markets, FICC could potentially pose a number of risks to other entities, or the broader financial system, while it is also exposed to risks that may originate from other entities. In certain instances, FICC is engaged in multiple and differing relationships with a single third party. For example, a third party may be a FICC Member, as well as a settling bank for other FICC Members, as well as a liquidity provider to FICC. In such cases, an impact on that single third party entity would pose greater risk to FICC, and as a result the broader financial system, than an impact on another entity with only a single relationship with FICC.

These interdependent relationships require special consideration and oversight because of their potential for causing "systemic risk" to the broader financial system. "Systemic risk" in the financial markets environment relevant to FICC is defined as the risk that the effect of an adverse event or series of events within the broadly defined financial services industry, including the financial industries critical infrastructures, caused by members or inflicted through external channels, is transmitted across the industry, markets, products and/or structures. This transmission, in turn, may lead to severe impairment, disruption, or degradation of the effectiveness and/or efficiency of the intermediation function, which may also impair the unencumbered functioning of critical infrastructures such as FICC. DTCC's Emerging and Systemic Risk team ("ESR") within Operational Risk leads the efforts to analyze and address systemic risk issues, leveraging existing structures and resources. ESR ultimately sits within the Group Chief Risk Office, and as such, is a "second line of defense" function.

Accordingly, in addition to how FICC monitors third parties by virtue of any one of their relationships as set forth below, these entities are also monitored holistically by ESR.

As regards to risks that may be posed by Members, FICC has established clear and objective membership requirements for each category of membership, which provide fair and open access for firms seeking membership, while maintaining prudent risk management standards that enable FICC to manage the material risks resulting from membership activities. As described more fully under Principle 4 (Credit risk), FICC monitors GSD Members and MBSD Members on an ongoing basis and assesses credit risk through the use of the CRRM which is designed to focus the level of scrutiny and ongoing monitoring in a manner consistent with a Member's internal risk rating.

In a similar manner, FICC has established credit standards for entities providing settling bank services to its Members, and investment and custody services. While settling banks are selected by, and act on behalf of, Members, entities providing such services are participants of GSD and MBSD and, as such,

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are held to strict operational standards, and subject to fines and other disciplinary measures (in their capacity as Members or Settling Bank Only Members) should they fail to abide by such requirements.

FICC's material service providers are subject to a comprehensive third-party review and vetting process. FICC assesses its material service provider risks periodically, throughout the lifecycle of a third party service provider relationship. This includes protocols for identifying, assessing, reporting, and mitigating third party risk and potential exposures. Mitigants may take the form of contractual protections, or additional or backup providers where deemed appropriate and feasible. For example, FICC utilizes price feeds from multiple third-parties as an additional risk mitigant, and further employs price review and data scrubbing procedures to minimize the risk of using inaccurate prices. In addition, in compliance with SEC Rule 17ad-25(i), FICC has established policies and procedures to manage risks from relationships with service providers for core services (CSPs). FICC's third party risk management processes are also discussed under Principle 17 (Operational risk) below.

As described in the General Background section of this Disclosure Framework, because FICC is not a depository institution, it is not given direct access to the Fedwire system and therefore employs the clearing bank services of BNY Mellon.[42] GSD Rule 12 and MBSD Rule 9 provide that FICC shall notify each GSD Member and MBSD Member, as applicable (prior to the entity activating its membership) of the clearing bank or banks that FICC will use to deliver eligible securities to Members and to receive eligible securities from Members, and by product, the types of securities that each such clearing bank will so deliver and receive. In turn, each Member (prior to activating its membership) must notify FICC of the clearing bank or banks that the Member has designated to act on its behalf in the delivery and receipt of securities to or from GSD or MBSD, as applicable. As stated above, obligations may be settled on the books of a clearing bank when the deliver and attendant receive obligations are at the same clearing bank. Otherwise, obligations are settled using the Fedwire system. All deliveries are made against full payment. Participation in GSD's GCF Repo and CCIT service requires dealer GSD Members to have an account with BNY Mellon. It should be noted that for non-GCF Repo and non-CCIT activity, members are not limited to BNY Mellon. The clearing banks have an integral role in the settlement of FICC eligible transactions.

FICC uses the central bank model with a tiered settlement arrangement for GSD's and MBSD's funds-only settlement process and settlement occurs in central bank funds.[43] The funds-only settlement process is a cash pass-through process (i.e., those members who are in a debit position submit payments that are then used to pay Members in a credit position). The settlement ultimately occurs via the Federal Reserve's NSS, with DTC acting as a settling agent, and funds transfers are final when effected on the NSS. Similarly, Members are required to engage a settling bank that meets FICC's settling bank criteria to effect funds-only settlement on the central bank's NSS on behalf of the Members. FICC confirms that fund transfers have been effected twice each business day with respect to GSD and once each business day with respect to MBSD. With respect to securities settlement, the GSD Rules and the MBSD Rules clearly set forth its obligations with respect to securities settlement. Every securities delivery, whether to or from GSD and MBSD, as applicable, is made against payment.

As described in the General Background section of this Disclosure Framework, in the CME-FICC cross-margining arrangement, FICC and CME are able to recognize risk offsets arising from the offsetting value of positions maintained by a common member (or a member and its affiliate) at FICC and CME when calculating the margin requirements for those positions. Each of FICC and CME separately calculate these risk offsets for the entire portfolio of eligible trades and give effect to the more conservative of the two calculations. The SEC and the CFTC approved the Restated Agreement in September 2023 and any

[42] Historically, JPMorgan Chase ("JPM") provided clearing bank services to FICC and its Members. In September 2018, JPM exited the business and discontinued these services. As a result, BNY Mellon is currently the only bank that provides clearing bank services to FICC.

[43] The GSD Rules refer to this process as the "funds-only settlement" process, and the MBSD Rule refers to this process as the "cash settlement" process.

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changes to the Restated Agreement require a rule filing with the SEC, subjecting the arrangement to an SEC approval process.

Scenarios that may prevent an FMI from providing core operations and services

FICC identifies scenarios that may potentially prevent it from providing its core operations and services, including through the development of recovery and orderly wind-down plans. FICC leverages its existing stress testing methodologies, as described in Principle 4, to identify scenario assumptions and inputs for scenarios that may potentially prevent FICC from being able to provide its core services due to uncovered credit loss and uncovered liquidity shortfalls. As part of this process, FICC has simulated single and multiple Member failures coinciding with significant market price changes to assess financial resource adequacy and impact to FICC's ability to continue to provide core operations and services. FICC maintains financial resources—the Clearing Fund (which in the aggregate serves as the default fund for loss mutualization purposes) and certain of its own resources—designed to enable it to cover potential losses resulting from the failure of the Member or Member family to which it has the largest credit exposure (a "Cover One" standard). These tests, in general, show that such resources are sufficient to handle defaults larger than a Cover One standard.

In addition, DTCC maintains business continuity plans that address the loss of people, premises and/or technology as an additional element to ensure the continuing operation of critical services. This is discussed in Principle 17 (Operational Risk).

Recovery or orderly wind-down plans

FICC maintains a Recovery and Wind-down ("R&W") Plan to be used by the Board and management in the event FICC encounters scenarios that could potentially prevent it from being able to provide its core services as a going concern. The R&W Plan identifies (i) the recovery tools available to address the risks of (a) uncovered losses or liquidity shortfalls resulting from the default of one or more Members, and (b) losses arising from non-default events such as, for example, damage to its physical assets, a cyber-attack, or custody or investment losses, and (ii) the strategy for implementation of such tools. The R&W Plan also establishes the strategy and framework for the orderly wind-down of FICC and the transfer of its business in the remote event the implementation of the available recovery tools does not successfully return FICC to financial viability.

The R&W Plan is structured as a roadmap and identifies the tools that FICC may use to effectuate a recovery from such events. It provides, among other matters, (i) a description of FICC's services, including the criteria used to determine which services are considered core as well as the staffing roles and Core Service Providers necessary to support them; (ii) a description of the governance around the overall recovery and wind-down program, including the requirement to inform the SEC as soon as practicable when the covered clearing agency is considering implementing a recovery or orderly wind-down; (iii) unique scenarios that may potential prevent the covered clearing agency from being able to provide its core services; (iv) a discussion of tools available to FICC to mitigate credit/market and liquidity risks, including recovery indicators and triggers, and the governance around management of a stress event along a "Crisis Continuum" timeline; (v) a discussion of potential non-default losses and the resources available to address such losses, including recovery triggers and tools to mitigate such losses; and (vi) the framework and approach for the orderly wind-down and transfer of FICC's business in a manner designed to provide continuity of access to FICC's core services; and (vii) procedures for testing the covered clearing agency's ability to implement the recovery and orderly wind-down plans at least every 12 months with stakeholders and Participants, where practicable.

Key factors considered in developing the R&W Plan and the types of tools available to FICC were its governance structure and the nature of the markets within which FICC operates. As a result, many of the tools available to FICC are FICC's existing, business-as-usual risk management and default management tools, which would continue to be applied in scenarios of increasing stress. In addition to business-as-usual tools, the R&W Plan describes FICC's other principal recovery tools, which include (i) identifying, monitoring

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and managing general business risk and holding sufficient liquid net assets funded by equity ("LNA") to cover potential general business losses pursuant to the Clearing Agency Policy on Capital Requirements, filed with and approved by the SEC ("Capital Policy"), (ii) maintaining the Clearing Agency Capital Replenishment Plan ("Replenishment Plan"), also filed and approved by the SEC, as a viable plan for the replenishment of capital should FICC's equity fall close to or below the amount being held pursuant to the Capital Policy, and (iii) the process for the allocation of losses among FICC and the membership, as provided in Rule 4 of the respective Division's Rules.

GSD Rule 22D and MBSD Rule 17B (Wind-down of the Corporation) and other rules, including GSD Rule 50, and MBSD Rule 40 (Market Disruption and Force Majeure) help facilitate the implementation of the R&W Plan, particularly FICC's strategy for winding down and transferring its business, and provide the membership with transparency and certainty with respect to these matters.

The development and maintenance of the R&W Plan is facilitated by DTCC's Office of Recovery & Resolution Planning ("R&R Team"). The R&R Team is responsible for the development and ongoing maintenance of both the R&W Plans and the overall recovery and wind-down planning process. The Executive Committee is responsible for overseeing the Recovery & Resolution Planning Program. The Board reviews and approves the R&W Plan at least every 12 months or following material changes to the SIFMU's operations that would significantly affect the viability or execution of the Plan.

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Principle 4: Credit risk; CCAS 17ad-22(e)(4)

Principle 4: Credit risk	**CCAS 17ad-22(e)(4)**
An FMI should effectively measure, monitor, and manage its credit exposure to participants and those arising from its payment, clearing, and settlement processes. An FMI should maintain sufficient financial resources to cover its credit exposure to each participant fully with a high degree of confidence. In addition, a CCP that is involved in activities with a more-complex risk profile or that is systemically important in multiple jurisdictions should maintain additional financial resources sufficient to cover a wide range of potential stress scenarios that should include, but not be limited to, the default of the two largest participants and their affiliates that would potentially cause the largest aggregate credit exposures to the CCP in extreme but plausible market conditions. All other CCPs should maintain, at a minimum, total financial resources sufficient to cover the default of the one participant and its affiliates that would potentially cause the largest aggregate credit exposures to the CCP in extreme but plausible market conditions. ***Key consideration 1***: An FMI should establish a robust framework to manage its credit exposures to its participants and the credit risks arising from its payment, clearing, and settlement processes. Credit exposure may arise from current exposures, potential future exposures, or both. ***Key consideration 2***: An FMI should identify sources of credit risk, routinely measure and monitor credit exposures, and use appropriate risk-management tools to control these risks. ***Key consideration 3***: A payment system or SSS should cover its current and, where they exist, potential future exposures to each participant fully with a high degree of confidence using collateral and other equivalent financial resources (see Principle 5 on collateral). In the case of a DNS payment system or DNS SSS in which there is no settlement guarantee but where its participants face credit exposures arising from its payment, clearing, and settlement processes, such an FMI should maintain, at a minimum, sufficient resources to cover the exposures of the two participants and their affiliates that would create the largest aggregate credit exposure in the system.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (4) Effectively identify, measure, monitor, and manage its credit exposures to participants and those arising from its payment, clearing, and settlement processes, including by: (i) Maintaining sufficient financial resources to cover its credit exposure to each participant fully with a high degree of confidence; (ii) To the extent not already maintained pursuant to paragraph (e)(4)(i) of this section, for a covered clearing agency providing central counterparty services that is either systemically important in multiple jurisdictions or a clearing agency involved in activities with a more complex risk profile, maintaining additional financial resources at the minimum to enable it to cover a wide range of foreseeable stress scenarios that include, but are not limited to, the default of the two participant families that would potentially cause the largest aggregate credit exposure for the covered clearing agency in extreme but plausible market conditions; (iii) To the extent not already maintained pursuant to paragraph (e)(4)(i) of this section, for a covered clearing agency not subject to paragraph (e)(4)(ii) of this section, maintaining additional financial resources at the minimum to enable it to cover a wide range of foreseeable stress scenarios that include, but are not limited to, the default of the participant family that would potentially cause the largest aggregate credit exposure for the covered clearing agency in extreme but plausible market conditions; (iv) Including prefunded financial resources, exclusive of assessments for additional guaranty fund contributions or other resources that are not prefunded, when calculating the financial resources available to meet the standards under paragraphs (e)(4)(i) through (iii) of this section, as applicable; (v) Maintaining the financial resources required under paragraphs (e)(4)(ii) and (iii) of this section, as applicable, in combined or separately maintained clearing or guaranty funds;

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[Not applicable, as FICC is not a payment system or SSS.] ***Key consideration 4***: A CCP should cover its current and potential future exposures to each participant fully with a high degree of confidence using margin and other prefunded financial resources (see Principle 5 on collateral and Principle 6 on margin). In addition, a CCP that is involved in activities with a more-complex risk profile or that is systemically important in multiple jurisdictions should maintain additional financial resources to cover a wide range of potential stress scenarios that should include, but not be limited to, the default of the two participants and their affiliates that would potentially cause the largest aggregate credit exposure for the CCP in extreme but plausible market conditions. All other CCPs should maintain additional financial resources sufficient to cover a wide range of potential stress scenarios that should include, but not be limited to, the default of the participant and its affiliates that would potentially cause the largest aggregate credit exposure for the CCP in extreme but plausible market conditions. In all cases, a CCP should document its supporting rationale for, and should have appropriate governance arrangements relating to, the amount of total financial resources it maintains. ***Key consideration 5***: A CCP should determine the amount and regularly test the sufficiency of its total financial resources available in the event of a default or multiple defaults in extreme but plausible market conditions through rigorous stress testing. A CCP should have clear procedures to report the results of its stress tests to appropriate decision makers at the CCP and to use these results to evaluate the adequacy of and adjust its total financial resources. Stress tests should be performed daily using standard and predetermined parameters and assumptions. On at least a monthly basis, a CCP should perform a comprehensive and thorough analysis of stress testing scenarios, models, and underlying parameters and assumptions used to ensure they are appropriate for determining the CCP's required level of default protection in light of current and evolving market conditions. A CCP should perform this analysis of stress testing more frequently when the products cleared or markets served display high volatility, become less liquid, or when the size or concentration of positions held by a CCP's participants increases significantly. A full validation of a CCP's risk-management model should be performed at least annually.	(vi) Testing the sufficiency of its total financial resources available to meet the minimum financial resource requirements under paragraphs (e)(4)(i) through (iii) of this section, as applicable, by: (A) Conducting stress testing of its total financial resources once each day using standard predetermined parameters and assumptions; (B) Conducting a comprehensive analysis on at least a monthly basis of the existing stress testing scenarios, models, and underlying parameters and assumptions, and considering modifications to ensure they are appropriate for determining the covered clearing agency's required level of default protection in light of current and evolving market conditions; (C) Conducting a comprehensive analysis of stress testing scenarios, models, and underlying parameters and assumptions more frequently than monthly when the products cleared or markets served display high volatility or become less liquid, or when the size or concentration of positions held by the covered clearing agency's participants increases significantly; and (D) Reporting the results of its analyses under paragraphs (e)(4)(vi)(B) and (C) of this section to appropriate decision makers at the covered clearing agency, including but not limited to, its risk management committee or board of directors, and using these results to evaluate the adequacy of and adjust its margin methodology, model parameters, models used to generate clearing or guaranty fund requirements, and any other relevant aspects of its credit risk management framework, in supporting compliance with the minimum financial resources requirements set forth in paragraphs (e)(4)(i) through (iii) of this section; (vii) Performing a model validation for its credit risk models not less than annually or more frequently as may be contemplated by the covered clearing agency's risk management framework established pursuant to paragraph (e)(3) of this section; (viii) Addressing allocation of credit losses the covered clearing agency may face if its collateral and other resources are insufficient to fully cover its credit exposures, including the repayment of any funds the covered clearing agency may borrow from liquidity providers; and

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Key consideration 6: In conducting stress testing, a CCP should consider the effect of a wide range of relevant stress scenarios in terms of both defaulters' positions and possible price changes in liquidation periods. Scenarios should include relevant peak historic price volatilities, shifts in other market factors such as price determinants and yield curves, multiple defaults over various time horizons, simultaneous pressures in funding and asset markets, and a spectrum of forward-looking stress scenarios in a variety of extreme but plausible market conditions. ***Key consideration 7***: An FMI should establish explicit rules and procedures that address fully any credit losses it may face as a result of any individual or combined default among its participants with respect to any of their obligations to the FMI. These rules and procedures should address how potentially uncovered credit losses would be allocated, including the repayment of any funds an FMI may borrow from liquidity providers. These rules and procedures should also indicate the FMI's process to replenish any financial resources that the FMI may employ during a stress event, so that the FMI can continue to operate in a safe and sound manner.	(ix) Describing the covered clearing agency's process to replenish any financial resources it may use following a default or other event in which use of such resources is contemplated.

Framework for management of credit risk

The measurement, monitoring and management of credit risk are part of the overall risk management framework described under Principle 3 (Framework for the comprehensive management of risks). As noted in that section, FICC manages its credit exposures through strict membership admission criteria, ongoing Member surveillance and regular margining.

FICC's credit risk vis-à-vis Members consists of the risk that a Member defaults in the performance of its obligations to FICC. FICC addresses this risk through its membership requirements, ongoing monitoring and surveillance and margining methodologies.

Membership requirements. FICC has established participation criteria and requirements relating to financial resources, creditworthiness and operational capability. Rule 2, Rule 2A, Rule 3, Rule 3A (for GSD's Sponsored Service) and Rule 8 (for GSD's Agent Clearing Service) of the GSD Rules and Rule 2, Rule 2A and Rule 3 of the MBSD Rules set forth the specific requirements for full-service membership in each Division. Members must meet minimum financial and other criteria for membership, including minimum capital requirements. All applicants for membership must provide FICC with certain information, which is reviewed by FRM and other relevant departments as necessary to confirm, among other matters, that the applicant has (1) sufficient financial ability to make anticipated contributions to the Clearing Fund and to meet obligations to the respective Division; (2) either an established business history, or personnel with sufficient operational background and experience to ensure the ability of the firm to conduct such a business and (3) appropriate settling and clearing bank arrangements. FICC employs a robust membership process that entails a thorough review prior to approval by the Board Risk Committee or through its delegated authority to the MRC and the Counterparty Credit Risk ("CCR") team. Details regarding the application process are set forth in response to Principle 18 (Access and participation requirements).

Each Division has a limited purpose membership category for Settling Banks. Each Member is required to appoint a settling bank to perform their funds-only/cash settlement obligations. The review and ongoing

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monitoring of Settling Banks is discussed more fully under Principle 9 (Money settlement). All Members are required to designate a clearing bank for purposes of delivering securities to, and receiving securities from, FICC in satisfaction of settlement obligations.

Ongoing monitoring and surveillance. FICC monitors its credit exposures with respect to the risk that a Member defaults through the ongoing surveillance of its Members' financial strength and default risk. On an ongoing basis, Members are required to provide financial and other information to FICC, as outlined in GSD Rule 3 and MBSD Rule 3, to ensure that they meet the membership standards on an ongoing basis. In addition, FRM reviews publicly available information such as earnings releases, equity prices, and news as part of its Member surveillance. FICC utilizes a credit risk rating model (referred to as the credit risk rating matrix, or "CRRM", as further described below) to evaluate and rate the credit risk of FICC's U.S. bank, foreign bank, and U.S. broker-dealer Members, and rate such Members based upon qualitative and quantitative information. These ratings are used to set surveillance levels and may impact Clearing Fund requirements. All Members are subject to a credit review at least every 12 to 24 months.

Clearing Fund. Each FICC Division collects Clearing Fund deposits from its Members using a risk-based margin methodology. These amounts operate, individually, as the Member's margin, and the aggregate of all such Members' deposits is referred to collectively as the Clearing Fund, which operates as each Division's default fund. This risk-based methodology enables FICC to identify the risks posed by a Member's unsettled portfolio and to quickly adjust and collect additional deposits as needed to cover those risks. Each Member's required Clearing Fund deposit ("Required Fund Deposit") is calculated at least twice daily for GSD Members and at least once daily for MBSD Members pursuant to a formula set forth in GSD Rule 4 and the GSD Margin Component Schedule, and MBSD Rule 4, respectively, each of which is designed to provide sufficient funds to cover FICC's exposure to the Member fully with a high degree of confidence. In addition, FICC maintains financial resources sufficient to cover a wide range of potential stress scenarios (discussed more fully below); including the default of the Member and its affiliated Members that would cause the largest aggregate credit exposure to the respective Division in extreme, but plausible, market conditions. As noted above, these resources include each Division's Clearing Fund, which operate as FICC's default fund. The Clearing Fund formula, discussed more fully under Principle 6 (Margin), takes account of a variety of risk factors through the application of a number of components.

Measurement, monitoring and management of credit risk

FICC Members are monitored on a daily, monthly/quarterly and periodic basis. FICC monitors Members' credit risk through review and analysis of financial and operational information and through market, news and regulatory monitoring. Members' ongoing reporting and information requirements are set forth in GSD Rule 3 and MBSD Rule 3.

The CRRM is produced systematically from financial metrics contained in the financial reports. The rating is expressed on a defined scale that categorizes Members according to their relative creditworthiness. The model incorporates both quantitative factors and qualitative factors. The assigned risk analyst considers whether additional qualitative factors (e.g. regulatory history, type of audit opinion issued and material management changes) warrant a manual override of the model-generated rating. The resulting rating helps determine the level of financial review that will be performed on that Member and may impact the Member's Clearing Fund requirement. Members with a weaker internal credit rating are automatically placed on the respective Division's Watch List. Members on the Watch List may be subject to additional monitoring than those Members with a stronger credit rating.

Members can be added to the Watch List for a variety of reasons, including concerns about regulatory issues, changes to senior management at the Member firm, etc. A Member can be added to the Watch List regardless of the rating generated for the firm by the CRRM. Members not subject to monitoring through the CRRM may be added to the Watch List, as FICC deems necessary and appropriate.

GSD calculates a Clearing Fund requirement for each GSD Member twice a day, and MBSD calculates the Clearing Fund requirement for each MBSD Member once a day. The calculation is based upon each

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Member's unsettled and pending transactions. The formula uses a risk-based methodology and includes a number of components as described below. In addition, each Division may impose, as applicable, a premium charge when a Member's VaR charge exceeds a defined capital amount.[44] Any shortfalls in a Member's required deposit must be paid each business day by 9:30 a.m. and 2:45 p.m. with respect to GSD and 9:30 a.m. with respect to MBSD. In addition, FICC may call for additional Clearing Fund on an intraday basis, as needed. Any such amounts must be satisfied within the timeframe specified by GSD and MBSD.

In order to mitigate counterparty exposure to each Member, each Division calculates the VaR of the unsettled obligations of each Member to a 99 percent confidence interval with a three-day liquidation/hedge horizon and a ten-year historical look-back period, as the basis of its Clearing Fund requirement, if it includes sufficient stressed market scenarios. If, however, FICC determines that the historical look-back period does not contain adequate shocks, an additional period of stressed market conditions may be added to the ten-year historical data. Each Division has additional margin charges for each Member's total Clearing Fund requirement.

With respect to GSD, in addition to VaR, an additional component that impacts the Member's Required Fund Deposit is the Blackout Period Exposure Adjustment. The Blackout Period Exposure Adjustment is an additional charge or reduction to the Required Fund Deposit to address risks to the Corporation that may arise due to potential overvaluation of mortgage-backed securities pledged to collateralize Sponsored GC, ASC Triparty and GCF Repo Transactions during the Blackout Period. Other charges may be applicable to reflect additional exposure for reasons such as late submission of requisite financial statements. Activity cleared through the CIL service offering within Sponsored GC ("Sponsored GC CIL Trades") is generally not subject to a Clearing Fund requirement and, instead, each Sponsoring Member utilizing this tri-party service is required to collect a haircut and grant FICC a lien on the purchased repo securities subject to the Sponsored GC CIL Trades. In the limited circumstances when FICC does collect margin with respect to Sponsored GC CIL Trades, the margin components generally mirror the components applied to Sponsored GC activity.

Separate and apart from collecting Clearing Fund, marking to market is conducted to reflect the difference between the contract value of a trade and the current market value of the security. Twice each business day for GSD and once each business day for MBSD, each Member must pay (or is entitled to collect) an aggregate funds-only settlement amount across all CUSIPs in which it has outstanding positions. For GSD, the main components of this amount include, among other payments, a mark-to-market amount for every net settlement position, a mark-to-market amount for every forward net settlement position, fail marks for obligations that were scheduled to settle and have not yet settled, coupon payments and other adjustments. For MBSD, the cash settlement process that occurs pursuant to MBSD Rule 11 consists of a mark-to-market amount, principal and interest payments, Treasury Market Practices Group (TMPG) fail charges, billing and other adjustments. Each Division's funds-only settlement process is a cash pass-through process; i.e., those Members that are in a debit position submit payments that are then used to pay Members in a credit position.

The Divisions employ backtesting and stress-testing to determine the adequacy of their Clearing Fund requirements. The backtests compare the Clearing Fund requirements with actual price changes in a portfolio. The portfolio values are calculated by using the actual positions in a Member's portfolio and the observed security price changes over the assumed liquidation/hedge period of three days. Each Division employs stress-testing to determine whether there will be adequate coverage levels for potential losses in a portfolio incurred during extreme, but plausible, market conditions. The scenarios used in the stress tests fall into two broad categories—historical scenarios and hypothetical or "what-if" scenarios. FICC maintains

[44] For GSD, the VaR Charge is compared to a Member's "Netting Member Capital," a defined term in the GSD Rules that means Net Capital, net assets or equity capital as applicable, to a Netting Member based on its type of regulation. For MBSD, the VaR Charge is compared to a Member's "Excess Capital," a defined term in the MBSD Rules that means Excess Net Capital, net assets, or equity capital as applicable to a Clearing Member based on its type of regulation.

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policies and procedures, in accordance with the CCAS, that address the process for conducting these tests and calculating the resulting metrics. Backtesting and stress-testing are also discussed under Principle 6 (Margin) and Principle 7 (Liquidity), respectively.

For example, FICC, and its affiliates, DTC and NSCC, have adopted the Clearing Agency Stress Testing Framework ("Stress Testing Framework"), which sets forth the manner in which FICC tests the sufficiency of its prefunded financial resources in accordance with applicable legal requirements, including CCAS 17ad-22(e)(4) and addresses related matters. The Stress Testing Framework has been filed with and approved by the SEC,[45] and is reviewed and approved by the Board on an annual basis.

Both Divisions also may impose a Backtesting Charge on Members based on their end-of-day, and with respect to GSD based on their intraday, backtesting performance. The Backtesting Charge is assessed for those Members whose portfolios experience overall margin backtesting deficiencies over the prior 12-month period. The Backtesting Charge is calculated by each Division to mitigate exposures to the Division caused by settlement risks that may not be adequately captured by the Division's portfolio volatility model. As a further mitigant of credit risk, under GSD Rule 3 and MBSD Rule 3, the Divisions may seek additional assurances of financial responsibility (as well as operational capability) from a Member as they deem necessary or advisable. This may include, for example, increased Clearing Fund deposits.

Coverage of current and potential future exposures to Members and financial resources

The Clearing Fund formula is designed to calculate individual Member Clearing Fund (margin) requirements at a 99 percent confidence level, assuming a three-day liquidation/hedge period and a ten-year historical look-back period, if it includes sufficient stressed market scenarios. If, however, FICC determines that the historical look-back period does not contain adequate shocks, an additional period of stressed market conditions may be added to the ten-year historical data. The adequacy of these amounts is measured through backtesting.

FICC maintains additional prefunded financial resources sufficient to cover a wide range of potential stress scenarios, including the default of the Member and its affiliated Members that would cause the largest aggregate credit exposure to FICC in extreme but plausible market conditions (referred to as a "Cover One" standard).

GSD clears and settles securities issued or guaranteed by the United States, U.S. government agencies or instrumentalities, or U.S. government-sponsored corporations. MBSD clears and settles mortgage-backed pass-through securities issued or guaranteed by Ginnie Mae, Freddie Mac and Fannie Mae. MBSD trades and positions are guaranteed at the TBA level. FICC does not consider these securities to have a more complex risk profile of the type contemplated by Key Consideration 4 of Principle 4, or its CCAS analogue. Moreover, FICC is not systemically important in multiple jurisdictions. Accordingly, each Division maintains financial resources in amounts required, at a minimum, to meet a Cover One standard; this is consistent, and in accordance, with the requirements of CCAS 17ad-22 (e)(4)(ii) and (iii).

Additional financial resources

With respect to each Division, if a defaulting Member has any outstanding obligations to FICC arising from its default, FICC would first apply the amounts on deposit to the Clearing Fund and Eligible Clearing Fund Securities pledged from the defaulting Member (along with other resources of, or attributable to, the defaulting Member (or certain permitted margin affiliates) that FICC may access under each Division's Rules, including under available clearing agency cross-guaranty agreements and clearing agency cross-margining agreements).

[45] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

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To address potential tail losses, FICC maintains additional prefunded resources. These consist of (i) the Clearing Fund of each Division, which, in the aggregate, serves as each Division's default fund, and (ii) an amount, referred to as FICC 's "Corporate Contribution", equal to 50 percent of its General Business Risk Capital Requirement, as described in more detail under Principles 13 (Participant-default rules and procedures) and 15 (General business risk). As regards to the composition of the Clearing Fund at each Division, Clearing Fund deposits are cash, but a portion of a Member's Required Fund Deposit may be evidenced by open account indebtedness, secured by certain eligible U.S. government and agency securities, as described in more detail under Principle 5 (Collateral). Not less than 40 percent of a Member's Required Fund Deposit must be in cash. With respect to the Corporate Contribution, FICC maintains sufficient liquid net assets to fund such amount. All of the prefunded financial resources that FICC thus relies upon to cover losses are liquid and readily available to meet its obligations.

Stress-testing

As described above, FICC employs stress-testing to evaluate whether it will have adequate financial resources to cover potential losses during extreme, but plausible, market conditions. The Stress Testing Framework describes, among other matters, FICC's stress-testing methodologies.

FICC employs daily stress-testing to determine whether it will have sufficient prefunded financial resources to cover potential losses in a wide range of scenarios including, at a minimum, on a Cover One standard, during extreme, but plausible, market conditions. The scenarios used to conduct these stress tests fall into two broad categories: historical scenarios and hypothetical scenarios. The historical scenario set includes at least 50 stress scenarios selected from a look-back period of over 10 years. In addition, FICC supplements the scenarios with historical stresses outside the look-back period, such as the volatile bear market in US Treasuries in 1994. The hypothetical scenarios are constructed according to potential market conditions. Current positions in a portfolio are used in the stress tests. Stress-testing results and the appropriateness of the underlying assumptions and parameters are used for both historical and hypothetical scenarios, are reviewed on a monthly basis and as appropriate more frequently than monthly when the products cleared or the markets served by FICC display high volatility or become less liquid, or when the size or concentration of positions held by Members increases significantly. Stress-testing results are reviewed at least monthly by the internal Enterprise Stress Test Committee ("ESTC"), a group comprised primarily of risk management, business and control function senior management.

The ESTC has responsibility for the review, oversight, escalation, and governance of credit stress-testing related activities at FICC. This committee meets at least monthly and its responsibilities include (but are not limited to) reviewing stress-testing policies, procedures, methodology, and test results; approving stress scenarios; and ensuring stress-testing activities meet DTCC's risk tolerance requirements applicable to FICC. Updates to stress scenarios are escalated for review by the MRC on a quarterly basis, or more frequently if appropriate.

FICC maintains written policies and procedures that describe its process related to the calculation and monitoring of the following stress-test metrics: (i) maximum Member-level daily historical and hypothetical stress test results run daily and reported on a monthly basis; (ii) maximum family-level daily historical and hypothetical stress test results run daily and reported on a monthly basis; and (iii) as a record of the financial resources necessary to meet a Cover One standard, the maximum observed family-level historical stress deficiency relative to the total aggregate Clearing Fund requirement.

Stress tests are conducted daily. Exceptions or substantial changes in stress-test results are reviewed by FRM management in order to identify the causes and formulate responses, as needed. The results of this review and related metrics are reported and discussed monthly with the MRC and shared monthly with the BRC and FICC's supervisors.

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FICC and its affiliates DTC and NSCC have adopted the Clearing Agency Model Risk Management Framework ("Model Risk Framework"),[46] which provides that models utilized for the Clearing Fund formula, as well as stress testing models, be validated prior to their implementation and not less often than annually thereafter. In conducting a model validation, FICC would verify that the model is performing (or continuing to perform) as expected in accordance with its design objectives and business purpose, and identify any deficiencies that would call into question the continuing validity of any such model.

Model validation

Model Risk Management("MRM") maintains the FICC model inventory of all methods, systems that meet FICC's definition of a model and validates all models and material changes thereof, in FICC model inventory. MRM is responsible for oversight of all risk models. In accordance with the Model Risk Framework, all new models (including material changes to existing models) are subject to independent validation and approval or provisional approval by MRM prior to their use in production. The margin stress-testing and CRRM models are also reviewed at least annually by MRM. The results of these reviews are provided to the MRC, the BRC and FICC's supervisors. If a model issue is detected, deficiencies would be escalated by MRM and additional mitigation strategies would be discussed with the MRC and BRC.

Allocation of credit losses and replenishment of resources

If, after closing out and liquidating a defaulting Member's positions, FICC were to suffer a loss, that loss would first be satisfied by the amounts on deposit to the Clearing Fund and Eligible Clearing Fund Securities pledged from the defaulting Member (along with any other resources of, or attributable to, the defaulting Member that FICC may access under each Division's Rules). If these amounts are insufficient to fully remediate the loss, then FICC would apply the applicable Division's loss waterfall provisions (contained in GSD Rule 4 and MBSD Rule 4, respectively), which allocate and assess losses among FICC and each Division's membership as applicable. This process is described under Principle 13 (Participant-default rules and procedures). The replenishment of any prefunded resources that may be applied is also discussed in Principle 13.

[46] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

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Principle 5: Collateral; CCAS 17ad-22(e)(5)

Principle 5: Collateral	**CCAS 17ad-22(e)(5)**
An FMI that requires collateral to manage its or its participants' credit exposure should accept collateral with low credit, liquidity, and market risks. An FMI should also set and enforce appropriately conservative haircuts and concentration limits. ***Key consideration 1***: An FMI should generally limit the assets it (routinely) accepts as collateral to those with low credit, liquidity, and market risks. ***Key consideration 2***: An FMI should establish prudent valuation practices and develop haircuts that are regularly tested and take into account stressed market conditions. ***Key consideration 3***: In order to reduce the need for procyclical adjustments, an FMI should establish stable and conservative haircuts that are calibrated to include periods of stressed market conditions, to the extent practicable and prudent. ***Key consideration 4***: An FMI should avoid concentrated holdings of certain assets where this would significantly impair the ability to liquidate such assets quickly without significant adverse price effects. ***Key consideration 5***: An FMI that accepts cross-border collateral should mitigate the risks associated with its use and ensure that the collateral can be used in a timely manner. ***Key consideration*** 6: An FMI should use a collateral management system that is well-designed and operationally flexible.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (5) Limit the assets it accepts as collateral to those with low credit, liquidity, and market risks, and set and enforce appropriately conservative haircuts and concentration limits if the covered clearing agency requires collateral to manage its or its participants' credit exposure; and require a review of the sufficiency of its collateral haircuts and concentration limits to be performed not less than annually.

Collateral eligibility

GSD and MBSD accept cash (in U.S. dollars) and certain eligible securities as Clearing Fund collateral. Due to the nature of their businesses, GSD and MBSD limit the non-cash securities currently accepted to collateralize a Member's open account Clearing Fund indebtedness to U.S. Government Treasury securities, U.S. Agency securities guaranteed by the U.S. Government, and certain U.S. Agency/Government Sponsored Enterprise mortgage-backed pass-through securities (collectively, "Eligible Clearing Fund Securities"). Not less than 40 percent of a Member's required deposit must be in cash or a combination of cash and U.S. Treasury securities. The composition of cash and eligible securities is reported to Management Risk Committee at least quarterly.

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Under GSD Rule 4, the same eligibility criteria applies to collateral deposited on behalf of Segregated Indirect Participants to meet Segregated Customer Margin requirements.[47]

GSD and MBSD do not accept cross-border collateral.

GSD and MBSD maintain a collateral management system ("Clearing Fund Management System" or "CFM") that validates that posted collateral meets security eligibility standards and values aggregate collateral to determine sufficiency of posted amounts.

Collateral haircuts and valuation

GSD and MBSD apply haircuts to Eligible Clearing Fund Securities that reflect the potential of such assets' value to decline both in normal and in stressed markets. The schedule of haircuts is publicly available on the DTCC.com website and includes additional concentration haircuts for mortgage-backed pass-through securities posted in amounts above specified levels.

Because GSD and MBSD limit the collateral they accept to U.S. Government Treasury securities, U.S. Agency securities guaranteed by the U.S. Government and certain U.S. Agency/Government Sponsored Enterprise mortgage-backed pass-through securities, they face limited exposure to wrong-way risk with respect to their collateral.

Haircuts and concentration limits are backtested daily, consolidated in monthly reporting, and a comprehensive review is completed no less than quarterly. If FRM determines that the risk of a particular security may have increased significantly, ad hoc reviews would be conducted. The sufficiency of haircuts is evaluated through use of backtests and stress tests. This review is conducted initially by management within FRM, and any recommended changes are escalated to the Management Risk Committee. Before making adjustments to the haircuts they apply to such collateral, GSD and MBSD measure the potential impact of such adjustments. However, given the limited nature and high quality of acceptable collateral, adjustment of haircuts is infrequent.

Collateral securities are re-priced every night, from pricing sources utilized by FRM's Securities Valuation unit. Pricing updates may be performed instantaneously and will update collateral valuations within GSD's and MBSD's collateral management system accordingly.

FICC, and its affiliates, DTC and NSCC, have adopted a Clearing Agency Price and Margin Input Data Framework ("PMID Framework"), which sets forth the manner in which FICC identifies, measures, monitors, and manages the risks related to (i) the pricing of securities processed or otherwise held by it and (ii) inputs to FICC's margin methodologies, in accordance with applicable legal requirements, including CCAS 17ad-22(e)(4)(i) and Rule 17ad-22(e)(6)(iv), and addresses related matters. The PMID Framework has been filed with and approved by the SEC,[48] and is reviewed and approved by the Board on an annual basis.

Eligible Clearing Fund Securities are held on deposit at a custodian bank for GSD and MBSD. With respect to each Division, securities pledged by Members to secure their Clearing Fund open account indebtedness are available for immediate access by GSD and MBSD, as applicable, and may be re-pledged by GSD and MBSD, respectively, if there is a need for liquidity to facilitate completion of settlement.[49]

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[47] The conditions adopted in Note H of Exchange Act Rule 15c3-3a require that Segregated Customer Margin be met with either cash, U.S. Treasury securities or "qualified customer securities" (as defined in Note H(c) of Rule 15c3-3a and which are generally the same as Eligible Clearing Fund Securities) to qualify for a debit in the Rule 15c3-3a formula.

[48] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register

[49] GSD Rule 4, Section 5 and MBSD Rule 4, Section 5.

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Collateral management

Clearing Fund management is performed by the Treasury Operations staff, which is part of the Finance Department. This group consists of dedicated staff in multiple locations to facilitate the management of GSD's and MBSD's Clearing Fund collateral using the CFM system in order to ensure smooth operations even during times of market stress.

Under GSD's and MBSD's Rules, Members are required to make Clearing Fund deposits within specified timeframes (generally, Required Fund Deposits must be satisfied by 9:30 a.m. and 2:45 p.m. for GSD Members and 9:30 a.m. for MBSD Members or, for intraday calls, within a timeframe specified by FICC), and are permitted to request the return of excess amounts and/or substitute collateral, subject to the review of Clearing Fund Management for collateral sufficiency, prior to specified cutoff times. The CFM system is designed to interface with several internal and external systems in the daily management of the Clearing Fund collateral. FRM's system calculates Clearing Fund requirements and provides real-time requirements feeds to CFM. Account Administration's master file stores Members' wire instructions, which CFM reads to create delivery instructions for the return of any permitted excess deposits. The CFM system validates Clearing Fund parameter rules to ensure minimum cash levels are maintained, validates security eligibility, and applies haircuts on collateral and values aggregate collateral to determine the overall sufficiency of deposited amounts. The CFM system is utilized by Members to make online requests to withdraw collateral or have the system evaluate collateral substitution values.

Treasury Operations is also responsible for the investment of Clearing Fund cash. The Clearing Agency Investment Policy governs FICC's investments of cash, including the custody and overnight investment of Clearing Fund cash. The management and investment of cash is described under Principle 16 (Custody and investment risks).

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Principle 6: Margin; CCAS 17ad-22(e)(6)

Principle 6: Margin	**CCAS 17ad-22(e)(6)**
A CCP should cover its credit exposures to its participants for all products through an effective margin system that is risk-based and regularly reviewed. ***Key consideration 1***: A CCP should have a margin system that establishes margin levels commensurate with the risks and particular attributes of each product, portfolio, and market it serves. ***Key consideration 2***: A CCP should have a reliable source of timely price data for its margin system. A CCP should also have procedures and sound valuation models for addressing circumstances in which pricing data are not readily available or reliable. ***Key consideration 3***: A CCP should adopt initial margin models and parameters that are risk-based and generate margin requirements sufficient to cover its potential future exposure to participants in the interval between the last margin collection and the close out of positions following a participant default. Initial margin should meet an established single-tailed confidence level of at least 99 percent with respect to the estimated distribution of future exposure. For a CCP that calculates margin at the portfolio level, this requirement applies to each portfolio's distribution of future exposure. For a CCP that calculates margin at more-granular levels, such as at the sub portfolio level or by product, the requirement must be met for the corresponding distributions of future exposure. The model should (a) use a conservative estimate of the time horizons for the effective hedging or close out of the particular types of products cleared by the CCP (including in stressed market conditions), (b) have an appropriate method for measuring credit exposure that accounts for relevant product risk factors and portfolio effects across products, and (c) to the extent practicable and prudent, limit the need for destabilising, procyclical changes. ***Key consideration 4***: A CCP should mark participant positions to market and collect variation margin at least daily to limit the build-up of current exposures. A CCP should have the authority and operational capacity to make intraday margin calls and payments, both scheduled and unscheduled, to participants.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (6) Cover, if the covered clearing agency provides central counterparty services, its credit exposures to its participants by establishing a risk-based margin system that, at a minimum: (i) Considers, and produces margin levels commensurate with, the risks and particular attributes of each relevant product, portfolio, and market; and, if the covered clearing agency provides central counterparty services for U.S. Treasury securities, calculates, collects, and holds margin amounts from a direct participant for its proprietary positions in Treasury securities separately and independently from margin calculated and collected from that direct participant in connection with U.S. Treasury securities transactions by an indirect participant that relies on the services provided by the direct participant to access the covered clearing agency's payment, clearing, or settlement facilities; (ii) (A) Marks participant positions to market and collects margin (including variation margin or equivalent charges if relevant) at least daily; (B) Monitors intraday exposures on an ongoing basis; (C) Includes the authority and operational capacity to make intraday margin calls, as frequently as circumstances warrant, including the following circumstances: (1) When risk thresholds specified by the covered clearing agency are breached; or (2) When the products cleared or markets served display elevated volatility; and (D) Documents when the covered clearing agency determines not to make an intraday call pursuant to its written policies and procedures required under paragraph (e)(6)(ii)(C) of this section; (iii) Calculates margin sufficient to cover its potential future exposure to participants in the interval between the last margin collection and

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Key consideration 5: In calculating margin requirements, a CCP may allow offsets or reductions in required margin across products that it clears or between products that it and another CCP clear, if the risk of one product is significantly and reliably correlated with the risk of the other product. Where two or more CCPs are authorised to offer cross-margining, they must have appropriate safeguards and harmonised overall risk-management systems. ***Key consideration 6***: A CCP should analyse and monitor its model performance and overall margin coverage by conducting rigorous daily backtesting and at least monthly, and more-frequent where appropriate, sensitivity analysis. A CCP should regularly conduct an assessment of the theoretical and empirical properties of its margin model for all products it clears. In conducting sensitivity analysis of the model's coverage, a CCP should take into account a wide range of parameters and assumptions that reflect possible market conditions, including the most-volatile periods that have been experienced by the markets it serves and extreme changes in the correlations between prices. ***Key consideration 7***: A CCP should regularly review and validate its margin system.	the close out of positions following a participant default; (iv) (A) Uses reliable sources of timely price data and other substantive inputs; (B) Uses procedures (and, with respect to price data, sound valuation models) for addressing circumstances in which price data or other substantive inputs are not readily available or reliable, to ensure that the covered clearing agency can continue to meet its obligations under this section; and (C) Such procedures under paragraph (e)(6)(iv)(B) of this section must include either: (1) The use of price data or substantive inputs from an alternate source; or (2) If it does not use an alternate source, the use of a risk-based margin system that does not rely on substantive inputs that are unavailable or unreliable; (v) Uses an appropriate method for measuring credit exposure that accounts for relevant product risk factors and portfolio effects across products; (vi) Is monitored by management on an ongoing basis and is regularly reviewed, tested, and verified by: (A) Conducting backtests of its margin model at least once each day using standard predetermined parameters and assumptions; (B) Conducting a sensitivity analysis of its margin model and a review of its parameters and assumptions for backtesting on at least a monthly basis, and considering modifications to ensure the backtesting practices are appropriate for determining the adequacy of the covered clearing agency's margin resources; (C) Conducting a sensitivity analysis of its margin model and a review of its parameters and assumptions for backtesting more frequently than monthly during periods of time when the products cleared or markets served display high volatility or become less liquid, or when the size or concentration of positions held by the covered clearing agency's participants increases or decreases significantly; and (D) Reporting the results of its analyses under paragraphs (e)(6)(vi)(B) and (C) of this section to appropriate decision makers at the covered clearing agency, including but not limited

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	to, its risk management committee or board of directors, and using these results to evaluate the adequacy of and adjust its margin methodology, model parameters, and any other relevant aspects of its credit risk management framework; and (vii) Requires a model validation for the covered clearing agency's margin system and related models to be performed not less than annually, or more frequently as may be contemplated by the covered clearing agency's risk management framework established pursuant to paragraph (e)(3) of this section.

Overview of margin methodology

Each Division calculates and collects Clearing Fund and, at GSD Segregated Customer Margin, from its Members using a risk-based margin methodology that includes a sensitivity-based VaR approach. These amounts (a Member's "Required Fund Deposit" or, for a GSD Member with Segregated Indirect Participants, its "Segregated Customer Margin") operate as the Member's margin, and the aggregate of all such Members' Required Fund Deposits is, collectively, the Clearing Fund of each Division, which operates as FICC's default fund with respect to each Division. The sensitivity approach enables FICC to identify the risks posed by a GSD Member's or MBSD Member's unsettled portfolio and to quickly adjust and collect additional deposits as needed to cover those risks. Each GSD Member's Required Fund Deposit and Segregated Customer Margin are calculated at least twice daily, and each MBSD Member's Required Fund Deposit is calculated at least once daily, pursuant to a formula that is outlined in Rule 4 of each Division's Rules. Each Division's formula is designed to provide sufficient funds to cover this risk of loss. The formula accounts for a variety of risk factors through the application of a number of components, each described in Rule 4 of MBSD's Rules and in the Margin Component Schedule in the GSD Rules.

A Member's Required Fund Deposit (and Segregated Customer Margin) will vary daily based on its proprietary and indirect participant trading activity and financial status. The calculated requirement is based upon the total unsettled (fails) and pending (future settling) transactions of the Member and its indirect participants. The amount due, payable via the Fedwire system, is the difference between the Member's calculated Required Fund Deposit (and Segregated Customer Margin) and the amount currently on deposit. This amount is due each day by 9:30 a.m. and 2:45 p.m. for GSD Members and by 9:30 a.m. for MBSD Members, regardless of the time zone the Member operates in. FICC may call for additional margin on an intraday basis, as needed. Any such amounts must be satisfied within the timeframe specified by GSD and MBSD. Failure to submit payment by the deadline may subject the Member to a warning letter or a fine, according to the schedule provided in each Division's Rules. Such fines may be reported to the SEC. Repeated lateness may result in additional penalties or disciplinary action, including termination of membership pursuant to each Division's Rules, as applicable.

The GSD Rules are designed to provide for the separate and independent calculation, collection and holding of margin deposited by a Netting Member to support its proprietary transactions and margin deposited by a Netting Member to support the transactions of an indirect participant (a Sponsored Member or Executing Firm Customer). This separate margining is accomplished by requiring Netting Members to establish separate Accounts where proprietary activity and indirect participant activity are recorded (under GSD Rule 2B), and then calculating and collecting margin for those Accounts through separate Margin Portfolios and Deposit IDs (under GSD Rule 4).

Further, the GSD Rules also permit Netting Members to elect for margin for certain indirect participants ("Segregated Indirect Participants") to be calculated on a gross basis and legally segregated from the

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margin deposited for the Netting Member's proprietary transactions as well as transactions of other indirect participants.

Segregated Customer Margin requirements must be met exclusively by the delivery of cash and Eligible Clearing Fund Securities of the Segregated Indirect Participant (subject to limited prefunding as permitted by Note H and provided for in Section 3 of GSD Rule 2B). Under Section 3 of GSD Rule 2B, Sponsoring Members and Agent Clearing Members represent that the deposits made for Segregated Customer Margin requirements are customer funds and securities.

GSD Rule 4 provides for the segregation of such Segregated Customer Margin in a manner that satisfies the conditions for a broker-dealer to record a debit in the customer or PAB reserve formula set forth in Note H to Rule 15c3-3a. These conditions are principally set forth in Section 1a of GSD Rule 4 and provide, among other things, that (1) Segregated Customer Margin is calculated separately for each Segregated Indirect Participant and the Netting Member delivers that amount of margin for each Segregated Indirect Participant on a gross basis; and (2) Segregated Customer Margin is held in a separate account that is segregated from any other account of FICC and is, among other things, used exclusively to clear, settle, novate, and margin U.S. Treasury securities transactions of that Segregated Indirect Participant.

Required Fund Deposit and Segregated Customer Margin Requirement components

Government Securities Division

The formula used for calculating a GSD Member's Required Fund Deposit and Segregated Customer Margin Requirement includes the following major component charges:

1. VaR Charge[50]—The VaR Charge is generally the largest component of the Required Fund Deposit and Segregated Customer Margin Requirement. It is based on the potential price volatility of unsettled positions using a sensitivity-based Value-at-Risk ("VaR") methodology. The VaR methodology provides an estimate of the possible losses for a given portfolio based on: (1) confidence level, (2) a time horizon and (3) historical market volatility.

The VaR methodology is calculated at a 99 percent confidence level. FICC assumes a portfolio will take three days to hedge or liquidate in normal market conditions. The end-of-day VaR component of the Required Fund Deposit and Segregated Customer Margin Requirement addresses the risk presented by a Member's/Segregated Indirect Participant's end-of-day positions. GSD also calculates VaR for intraday collection, which reflects the changes in positions and thus risk profile due to the submission of new trades and completed settlement activity since the prior end of day. FICC uses historical simulation to estimate the impact of market volatilities on the Member's/Segregated Indirect Participant's portfolio. A ten-year historical look-back period is used, if it includes sufficient stressed market scenarios. If, however, FICC determines that the historical look-back period does not contain adequate shocks, an additional period of stressed market conditions may be added to the ten-year historical data. A haircut method is applied to securities with insufficient requisite data used to employ the sensitivity approach.

The sum of the VaR value and any haircut applied in lieu of VaR is used to cover the potential market price volatility for the portfolio. This approach assumes no diversification benefit between securities margined with VaR and those with haircuts applied. The sum of the VaR and haircuts applied in lieu of VaR are subject to a minimum amount, referred to as a "VaR Floor," which seeks to address the risk that the VaR model calculates a VaR Charge that is too low because of offsets applied by the model from certain offsetting long and short positions, or "Minimum Margin Amount" complementing the VaR Floor, when securities price changes significantly exceeding those implied by the VaR model risk factors. To account for extreme future risks and to ensure data quality and consistency, VaR results undergo daily backtesting.

[50] FICC can utilize an alternative Margin Proxy method as a back-up VaR Charge calculation to the sensitivity approach in the event that FICC experiences a data disruption with its third-party vendor.

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2. Blackout Period Exposure Adjustment—As outlined in Principle 4 (Credit risk), the Blackout Period Exposure Adjustment addresses exposures during the Blackout Period caused by Sponsored GC/GCF Repo/ACS Triparty Transactions collateralized by mortgage-backed securities. The reduction or addition of the Blackout Period Exposure Adjustment will be applied to all Sponsored GC/GCF/ACS Triparty Counterparties with a portfolio that includes mortgage-backed securities and will only be imposed during the Blackout Period.

3. Other Charges—GSD may assess additional charges to the Required Fund Deposit and Segregated Customer Margin Requirement, including but not limited to the following:

Backtesting Charge: The Backtesting Charge is assessed for those Members and Segregated Indirect Participants whose portfolios experience overall margin backtesting deficiencies over the prior 12-month period. The Backtesting Charge is calculated to mitigate exposures to GSD caused by settlement risks that may not be adequately captured by GSD's portfolio volatility model. GSD may assess this charge on a Netting Member's/Segregated Indirect Participant's start of day portfolio and/or its intraday portfolio. In addition, Sponsored GC/GCF/ACS Triparty Counterparties that experience backtesting deficiencies during the Blackout Period that are attributed to Sponsored GC/GCF Repo/ACS Triparty Transactions collateralized with mortgage-backed securities are subject to a Backtesting Charge.

Margin Liquidity Adjustment Charge: GSD may collect a Margin Liquidity Adjustment ("MLA") Charge to address the risk presented to GSD when a Member's or Segregated Indirect Participant's portfolio contains large net unsettled positions in a particular group of securities with a similar risk profile or in a particular asset type. This charge is designed to address the market impact costs of liquidating a defaulted Member's/Segregated Indirect Participant's portfolio that may increase when that portfolio includes large net unsettled positions in a particular group of securities with a similar risk profile or in a particular asset type. These positions may be more difficult to liquidate because a large number of securities with similar risk profiles could reduce the marketability of those large net unsettled positions, increasing the market impact costs to GSD.

Portfolio Differential Charge: The Portfolio Differential ("PD") Charge addresses the variability of clearing activity submitted to GSD throughout the day by measuring the period-over-period increase in the VaR Charge of Members and Segregated Indirect Participants. Specifically, this component is designed to increase the Required Fund Deposits/Segregated Customer Margin Requirements for Members and Segregated Indirect Participants with more variable clearing activities and mitigate exposures for the period of risk between margin collection times. The PD Charge is calculated twice daily as the exponentially weighted moving average of the historical increases in the Member's/Segregated Indirect Participant's VaR Charge that occur between collections of Required Fund Deposit/Segregated Customer Margin Requirement over a lookback period.

Special Charges: Pursuant to Rule 4, FICC may increase a GSD Member's Required Fund Deposit by assessing additional charges aimed at mitigating perceived risks in excess of those systemically addressed (and discussed above). Examples of special charges include additional requirements due to concerns around market volatility, and/or credit matters, etc.

Volatility Event Charge: The Volatility Event Charge is collected from Members to mitigate FICC's exposures arising from potential adverse market impact due to a scheduled event that has the potential to impact market volatility. The Volatility Event Charge is assessed twice a day at GSD, beginning on the day of the coverage period when one or more of the forward-looking market volatility indicators exceed the threshold(s) specified by FICC and ending on the day of the scheduled event.

Furthermore, GSD can apply Clearing Fund premiums for items such as a Member's late submission of requisite financial information and/or a Member's failure to maintain minimum financial requirements in accordance with the GSD Rules.

In determining the Segregated Customer Margin Requirements for each Segregated Indirect Participant, FICC performs substantially the same calculation it currently performs when determining a Netting

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Member's Required Fund Deposit, except (i) customer activity that is recorded in a Segregated Indirect Participants Account is margined on a customer by customer, or gross, basis, and (ii) the Excess Capital Premium charge will not be applied with respect to segregated customer activity. Segregated Customer Margin Requirements are therefore comprised of the following components, as applicable and as defined in the GSD Rules: (1) VaR Charge; (2) Blackout Period Exposure Adjustment for Segregated Indirect Participants that are GCF Counterparties; (3) Portfolio Differential Charge; (4) Backtesting Charge; (5) Holiday Charge; (6) Margin Liquidity Adjustment Charge; (7) Intraday Supplemental Fund Deposit; and (8) Volatility Event Charge.

Finally, the Required Fund Deposit for GSD Members, and hence the size of the Clearing Fund, may be reduced through the CME arrangement, which is described above in the General Background Section. FICC and CME are able to recognize risk offsets arising from the offsetting value of positions maintained by a common member (or a member and its affiliate) at FICC and CME when calculating the margin requirements for those positions. Each of FICC and CME separately calculate these risk offsets for the entire portfolio of eligible trades and give effect to the more conservative of the two calculations. The arrangement must be elected by the Member and is limited to common members of the two clearing organizations or their affiliates.

Funds-Only Settlement Amounts

As described in Principle 4 (Credit risk), separate and apart from collecting the Required Fund Deposit, marking to market is conducted to reflect the difference between the contract value of a trade and the current market value of the security. Twice each business day, each GSD Member must pay (or is entitled to collect) an aggregate funds-only settlement amount across all CUSIPs in which it has outstanding positions. The main components of this amount include, among other payments, a mark-to-market amount for every net settlement position, a mark-to-market amount for every forward net settlement position, fail marks for obligations that were scheduled to settle and have not yet settled, coupon payments and other adjustments. GSD's funds-only settlement process is a cash pass-through process; i.e., those GSD Members that are in a debit position submit payments that are then used to pay GSD Members in a credit position.

Mortgage-Backed Securities Division

The formula used for calculating a MBSD Member's Required Fund Deposit includes the following major component charges:

1. VaR Charge[51]—The VaR Charge is generally the largest component of the Required Fund Deposit. It is based on the potential price volatility of unsettled positions using a sensitivity-based Value-at-Risk model. The VaR methodology provides an estimate of the possible losses for a given portfolio based on: (1) a confidence level, (2) a time horizon and (3) historical market volatility.

For unregistered investment pool members, the VaR methodology is calculated at a 99.5 percent confidence level; the VaR methodology is calculated at 99 percent for all other Member categories. FICC assumes a portfolio will take three days to liquidate or hedge in normal market conditions. The end-of-day VaR component of the Required Fund Deposit addresses the risk presented by a Member's end-of-day positions. MBSD also calculates VaR for intraday collection, which reflects the changes in positions and thus risk profile due to the submission of new trades and completed settlement activity, since the prior end of day. FICC uses historical simulation to estimate the impact of market volatilities on the Member's portfolio. A ten-year historical look-back period is used, if it includes sufficient stressed market scenarios. If, however, FICC determines that the historical look-back period does not contain adequate shocks, an additional period of stressed market conditions may be added to the ten-year historical data. A haircut method is applied to securities with insufficient requisite data used to employ the sensitivity approach.

[51] FICC can utilize an alternative Margin Proxy method as a back-up VaR Charge calculation to the sensitivity approach in the event that FICC experiences a data disruption with its third-party vendor.

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The sum of the VaR value and any haircut applied in lieu of VaR is used to cover the potential market price volatility for the portfolio. This approach assumes no diversification benefit between securities margined with VaR and those with haircuts applied. The sum of the VaR and haircuts applied in lieu of VaR are subject to a minimum amount, referred to as a "VaR Floor", which seeks to address the risk that the VaR model calculates a VaR Charge that is too low because of offsets applied by the model from certain offsetting long and short positions, or "Minimum Margin Amount" complementing the VaR Floor, when TBA price changes significantly exceeding those implied by the VaR model risk factors. To account for extreme future risks and to ensure data quality and consistency, VaR results undergo daily backtesting.

2. Backtesting Charge—The Backtesting Charge is assessed for those Members whose portfolios experience overall margin backtesting deficiencies over the prior 12-month period. The Backtesting Charge is calculated to mitigate exposures to MBSD caused by settlement risks that may not be adequately captured by MBSD's portfolio volatility model.

3. Margin Liquidity Adjustment Charge—MBSD may collect a MLA Charge to address the risk presented to MBSD when a Member's portfolio contains large net unsettled positions in a particular group of securities with a similar risk profile or in a particular asset type. This charge is designed to address the market impact costs of liquidating a defaulted Member's portfolio that may increase when that portfolio includes large net unsettled positions in a particular group of securities with a similar risk profile or in a particular asset type. These positions may be more difficult to liquidate because a large number of securities with similar risk profiles could reduce the marketability of those large net unsettled positions, increasing the market impact costs to MBSD.

4. Special Charges—Pursuant to Rule 4, FICC may increase a MBSD Member's Required Fund Deposit by assessing additional charges aimed at mitigating perceived risks in excess of those systemically addressed (and discussed above). Examples of special charges include additional requirements due to concerns around market volatility and/or credit matters, etc.

5. Volatility Event Charge—The Volatility Event Charge is collected from Members to mitigate FICC's exposures arising from potential adverse market impact due to a scheduled event that has the potential to impact market volatility. The Volatility Event Charge is assessed once a day at MBSD, beginning on the day of the coverage period when one or more of the forward-looking market volatility indicators exceed the threshold(s) specified by FICC and ending on the day of the scheduled event.

Furthermore, MBSD can apply Clearing Fund premiums for items such as a Member's late submission of requisite financial information and/or a Member's failure to maintain minimum required financial resources, in accordance with the MBSD Rules.

Cash Settlement Amounts

As described in Principle 4 (Credit risk), separate and apart from collecting the Required Fund Deposit, marking to market is conducted to reflect the difference between the contract value of a trade and the current market value of the security. Once each business day, each MBSD Member must pay (or is entitled to collect) an aggregate funds-only settlement amount across all CUSIPs in which it has outstanding positions. This cash settlement process occurs pursuant to MBSD Rule 11 (Cash Settlement) and consists of a mark-to-market amount, principal and interest payments, Treasury Market Practices Group (TMPG) fail charges, billing and other adjustments. MBSD's funds-only settlement process is a cash pass-through process; i.e., those MBSD Members that are in a debit position submit payments that are then used to pay MBSD Members in a credit position.

Intraday Supplemental Fund Deposit, Intraday VaR Charges and Intraday Mark-to-Market Charges

With respect to each Division, FICC may require additional intraday margin should FICC deem it necessary or appropriate. Intraday market moves and positions are tracked and additional Clearing Fund monies are collected, as deemed necessary. Intraday Supplemental Fund Deposits and Intraday Mark-to-Market

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Charges for GSD as well as Intraday VaR Charges and Intraday Mark-to-Market Charges for MBSD are due within a timeframe specified by FICC. Changes in Member portfolios and security pricing are monitored intraday, and additional Clearing Fund charges may be collected to cover the price movement from those Members with a significant exposure in an identified security or net portfolio and the market value of those positions. These intraday routines are run frequently throughout the day and generate reports identifying Members' then-current positions and the estimated impact to the Clearing Fund requirements. These reports are reviewed, and FRM may impose an additional margin call.

The components of each Division's margining methodology are set forth in Rule 4 of the MBSD Rules and in the Margin Component Schedule in the GSD Rules, available on the DTCC website.

Pricing and margin input data sources

As described under Principle 5 (Collateral), the PMID Framework sets forth the manner in which FICC identifies, measures, monitors, and manages the risks related to (i) the pricing of securities processed or otherwise held by FICC and (ii) inputs to FICC's margin methodologies.

FICC utilizes multiple third-party vendors to price its eligible securities and uses a pricing hierarchy to determine a price for each security. FRM monitors each pricing file and performs rigorous exception reporting (large variance, stale, missing, off-the-market reports) to ensure that the prices are accurate, and uses established procedures or, where applicable, valuation models to generate prices for securities when vendor information may not be readily available or reliable.

FICC also uses historical price data that is incorporated into the historical look-back period of the VaR margin model.

In addition, FICC identifies substantive margin input data sources and maintains alternative margin input data sources when substantive margin input data sources are unavailable or unreliable.

Model inputs, assumptions and parameters

As noted above, with respect to each Division, FICC calculates margin requirements assuming a three-day liquidation/hedge period for unsettled positions. The Clearing Fund requirement is generally calculated to achieve a confidence level of 99 percent, and this is evidenced with daily backtesting.

The liquidation/hedge period is conservative based on the types of securities to be liquidated and the availability of options to "hedge" the portfolio to be liquidated essentially in "real time." Hedging strategies may be used promptly following the failure of a Member in order to minimize market risk. This allows sufficient time to liquidate the portfolio in an orderly manner and minimize market impact.

Simulated closeout exercises are performed to understand the profit and loss of a portfolio under various stressed conditions, including market liquidity, limits on average daily volume and significant price changes due to stressed conditions.

As regards to the historical data used in the VaR model, the current look-back period is ten years, plus, to the extent applicable, an additional stressed period. FICC bases its determination of the sample periods for historical data on mathematically sound principles that result in a stable base for providing projected returns, supported by continual backtesting. The appropriate sample period is also discussed with industry experts and FICC's independent model validators.

The Model Risk Framework sets forth the manner in which FICC identifies, measures, monitors, and manages the risks related to the design, development, implementation, use, and validation of quantitative models in accordance with applicable legal requirements, including CCAS 17ad-22(e)(4), (6), and (7), and

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addresses related matters. The Model Risk Framework has been filed with and approved by the SEC,[52] and is reviewed and approved by the Board on an annual basis.

Procyclicality and specific wrong-way risk

In times of stress, market volatility may cause the Clearing Fund to increase, which is inherent in the risk management process of FICC. The choice of the VaR look-back period reflects a balance between stability of the VaR calculation and appropriate responsiveness to changing market conditions. FICC addresses the procyclicality of the formula by adopting a 10-year lookback period that incorporates an additional stress period if FICC determines that the historical look-back period does not contain adequate shocks, utilizing a VaR Floor (or "Minimum Margin Amount" complementing the VaR Floor) and ongoing Member outreach, seeking to increase Member awareness of this impact.

FICC currently mitigates wrong-way risk (defined as the risk that counterparty exposures will increase when the creditworthiness of a counterparty deteriorates) by limiting eligible Clearing Fund collateral to cash and U.S. government and agency securities.

Backtesting, sensitivity analysis and model performance monitoring

FICC assesses the robustness of its margining methodologies through rigorous backtests.

FICC conducts backtesting on a daily basis on individual Member portfolios to determine: (a) the efficacy of the underlying VaR model and (b) the sufficiency of the calculated Required Fund Deposit. For the sufficiency test, coverage is defined as the number of test events without deficiencies divided by the total number of events in the testing period. The backtests compare the Clearing Fund requirements with actual price changes in a portfolio. The portfolio values are calculated by using the actual positions in a Member's portfolio and the observed security price changes over the following three days.

Backtesting results are calculated and reviewed by FRM management on a daily basis. Exceptions identified in backtests are reviewed in order to identify causes and formulate responses, as needed. The results of this review and related metrics are reported to, and discussed with, the MRC monthly and shared with the BRC and FICC's supervisors in accordance with their regularly scheduled meetings. Members with repeated deficiencies, subject to certain criteria, may be contacted as well. In addition, Backtesting Charges may be imposed as discussed above.

FICC employs daily stress-testing to determine whether there will be adequate coverage levels for potential losses in a portfolio incurred under extreme but plausible market conditions. While the VaR component of FICC's Clearing Fund formula is intended to provide coverage at a 99 percent confidence interval (assuming a three-day period to liquidate or hedge the relevant portfolio), it is possible under certain historical or hypothetical stressed scenarios, which may be outside of the 99 percent confidence interval, that the Required Fund Deposit could be insufficient to provide complete coverage for losses realized as a result of a Member's failure. Stress tests facilitate FICC's determination of the possibility where the Required Fund Deposit would be insufficient in this regard.

Additionally, the MRM is responsible for oversight of all risk models. The MRM is responsible for assessing model efficacy. In accordance with the Model Risk Framework, all new models (including material changes to existing models) are subject to independent validation and approval by the MRM prior to their use in production. The margin models are also reviewed at least annually by the MRM. The results of these reviews are provided to the MRC, BRC and FICC's supervisors. If a model issue is detected, deficiencies

[52] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

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would be escalated through the MRM and additional mitigation strategies would be discussed with the MRC and BRC.

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Principle 7: Liquidity risk; CCAS 17ad-22(e)(7)

Principle 7: Liquidity risk	**CCAS 17ad-22(e)(7)**
An FMI should effectively measure, monitor, and manage its liquidity risk. An FMI should maintain sufficient liquid resources in all relevant currencies to effect same-day and, where appropriate, intraday and multiday settlement of payment obligations with a high degree of confidence under a wide range of potential stress scenarios that should include, but not be limited to, the default of the participant and its affiliates that would generate the largest aggregate liquidity obligation for the FMI in extreme but plausible market conditions. ***Key consideration 1***: An FMI should have a robust framework to manage its liquidity risks from its participants, settlement banks, nostro agents, custodian banks, liquidity providers, and other entities. ***Key consideration 2***: An FMI should have effective operational and analytical tools to identify, measure, and monitor its settlement and funding flows on an ongoing and timely basis, including its use of intraday liquidity. ***Key consideration 3***: A payment system or SSS, including one employing a DNS mechanism, should maintain sufficient liquid resources in all relevant currencies to effect same-day settlement, and where appropriate intraday or multiday settlement, of payment obligations with a high degree of confidence under a wide range of potential stress scenarios that should include, but not be limited to, the default of the participant and its affiliates that would generate the largest aggregate payment obligation in extreme but plausible market conditions. [Not applicable, as FICC is not a payment system or SSS.] ***Key consideration 4***: A CCP should maintain sufficient liquid resources in all relevant currencies to settle securities-related payments, make required variation margin payments, and meet other payment obligations on time with a high degree of confidence under a wide range of potential stress scenarios that should include, but not be limited to, the default of the participant and its affiliates that would generate the largest aggregate payment obligation to the CCP in extreme but plausible market conditions. In addition, a CCP that is involved in activities with a more-complex risk profile or that is systemically	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (7) Effectively measure, monitor, and manage the liquidity risk that arises in or is borne by the covered clearing agency, including measuring, monitoring, and managing its settlement and funding flows on an ongoing and timely basis, and its use of intraday liquidity by, at a minimum, doing the following: (i) Maintaining sufficient liquid resources at the minimum in all relevant currencies to effect same-day and, where appropriate, intraday and multiday settlement of payment obligations with a high degree of confidence under a wide range of foreseeable stress scenarios that includes, but is not limited to, the default of the participant family that would generate the largest aggregate payment obligation for the covered clearing agency in extreme but plausible market conditions; (ii) Holding qualifying liquid resources sufficient to meet the minimum liquidity resource requirement under paragraph (e)(7)(i) of this section in each relevant currency for which the covered clearing agency has payment obligations owed to clearing members; (iii) Using the access to accounts and services at a Federal Reserve Bank, pursuant to Section 806(a) of the Payment, Clearing, and Settlement Supervision Act of 2010 (12 U.S.C. 5465(a)), or other relevant central bank, when available and where determined to be practical by the board of directors of the covered clearing agency, to enhance its management of liquidity risk; (iv) Undertaking due diligence to confirm that it has a reasonable basis to believe each of its liquidity providers, whether or not such liquidity provider is a clearing member, has: (A) Sufficient information to understand and manage the liquidity provider's liquidity risks; and (B) The capacity to perform as required under its commitments to provide liquidity to the covered clearing agency;

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important in multiple jurisdictions should consider maintaining additional liquidity resources sufficient to cover a wider range of potential stress scenarios that should include, but not be limited to, the default of the two participants and their affiliates that would generate the largest aggregate payment obligation to the CCP in extreme but plausible market conditions.

Key consideration 5: For the purpose of meeting its minimum liquid resource requirement, an FMI's qualifying liquid resources in each currency include cash at the central bank of issue and at creditworthy commercial banks, committed lines of credit, committed foreign exchange swaps, and committed repos, as well as highly marketable collateral held in custody and investments that are readily available and convertible into cash with prearranged and highly reliable funding arrangements, even in extreme but plausible market conditions. If an FMI has access to routine credit at the central bank of issue, the FMI may count such access as part of the minimum requirement to the extent it has collateral that is eligible for pledging to (or for conducting other appropriate forms of transactions with) the relevant central bank. All such resources should be available when needed.

Key consideration 6: An FMI may supplement its qualifying liquid resources with other forms of liquid resources. If the FMI does so, then these liquid resources should be in the form of assets that are likely to be saleable or acceptable as collateral for lines of credit, swaps, or repos on an ad hoc basis following a default, even if this cannot be reliably prearranged or guaranteed in extreme market conditions. Even if an FMI does not have access to routine central bank credit, it should still take account of what collateral is typically accepted by the relevant central bank, as such assets may be more likely to be liquid in stressed circumstances. An FMI should not assume the availability of emergency central bank credit as a part of its liquidity plan.

Key consideration 7: An FMI should obtain a high degree of confidence, through rigorous due diligence, that each provider of its minimum required qualifying liquid resources, whether a participant of the FMI or an external party, has sufficient information to understand and to manage its associated liquidity risks, and that it has the capacity to perform as required under its commitment. Where relevant to assessing a liquidity provider's performance reliability with respect to a particular currency, a liquidity

(v) Maintaining and testing with each liquidity provider, to the extent practicable, the covered clearing agency's procedures and operational capacity for accessing each type of relevant liquidity resource under paragraph (e)(7)(i) of this section at least annually;

(vi) Determining the amount and regularly testing the sufficiency of the liquid resources held for purposes of meeting the minimum liquid resource requirement under paragraph (e)(7)(i) of this section by, at a minimum:

(A) Conducting stress testing of its liquidity resources at least once each day using standard and predetermined parameters and assumptions;

(B) Conducting a comprehensive analysis on at least a monthly basis of the existing stress testing scenarios, models, and underlying parameters and assumptions used in evaluating liquidity needs and resources, and considering modifications to ensure they are appropriate for determining the clearing agency's identified liquidity needs and resources in light of current and evolving market conditions;

(C) Conducting a comprehensive analysis of the scenarios, models, and underlying parameters and assumptions used in evaluating liquidity needs and resources more frequently than monthly when the products cleared or markets served display high volatility or become less liquid, when the size or concentration of positions held by the clearing agency's participants increases significantly, or in other appropriate circumstances described in such policies and procedures; and

(D) Reporting the results of its analyses under paragraphs (e)(7)(vi)(B) and (C) of this section to appropriate decision makers at the covered clearing agency, including but not limited to, its risk management committee or board of directors, and using these results to evaluate the adequacy of and adjust its liquidity risk management methodology, model parameters, and any other relevant aspects of its liquidity risk management framework;

(vii) Performing a model validation of its liquidity risk models not less than annually or more frequently as may be contemplated by the covered clearing agency's risk management

framework established pursuant to paragraph (e)(3) of this section;

(viii) Addressing foreseeable liquidity shortfalls that would not be covered by the covered

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provider's potential access to credit from the central bank of issue may be taken into account. An FMI should regularly test its procedures for accessing its liquid resources at a liquidity provider. ***Key consideration 8***: An FMI with access to central bank accounts, payment services, or securities services should use these services, where practical, to enhance its management of liquidity risk. ***Key consideration 9***: An FMI should determine the amount and regularly test the sufficiency of its liquid resources through rigorous stress testing. An FMI should have clear procedures to report the results of its stress tests to appropriate decision makers at the FMI and to use these results to evaluate the adequacy of and adjust its liquidity risk-management framework. In conducting stress testing, an FMI should consider a wide range of relevant scenarios. Scenarios should include relevant peak historic price volatilities, shifts in other market factors such as price determinants and yield curves, multiple defaults over various time horizons, simultaneous pressures in funding and asset markets, and a spectrum of forward-looking stress scenarios in a variety of extreme but plausible market conditions. Scenarios should also take into account the design and operation of the FMI, include all entities that might pose material liquidity risks to the FMI (such as settlement banks, nostro agents, custodian banks, liquidity providers, and linked FMIs), and where appropriate, cover a multiday period. In all cases, an FMI should document its supporting rationale for, and should have appropriate governance arrangements relating to, the amount and form of total liquid resources it maintains. ***Key consideration 10***: An FMI should establish explicit rules and procedures that enable the FMI to effect same-day and, where appropriate, intraday and multiday settlement of payment obligations on time following any individual or combined default among its participants. These rules and procedures should address unforeseen and potentially uncovered liquidity shortfalls and should aim to avoid unwinding, revoking, or delaying the same-day settlement of payment obligations. These rules and procedures should also indicate the FMI's process to replenish any liquidity resources it may employ during a stress event, so that it can continue to operate in a safe and sound manner.	clearing agency's liquid resources and seek to avoid unwinding, revoking, or delaying the same-day settlement of payment obligations; (ix) Describing the covered clearing agency's process to replenish any liquid resources that the clearing agency may employ during a stress event; and (x) Undertaking an analysis at least once a year that evaluates the feasibility of maintaining sufficient liquid resources at a minimum in all relevant currencies to effect same-day and, where appropriate, intraday and multiday settlement of payment obligations with a high degree of confidence under a wide range of foreseeable stress scenarios that includes, but is not limited to, the default of the two participant families that would potentially cause the largest aggregate payment obligation for the covered clearing agency in extreme but plausible market conditions if the covered clearing agency provides central counterparty services and is either systemically important in multiple jurisdictions or a clearing agency involved in activities with a more complex risk profile.

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Liquidity risk management framework

FICC, and its affiliates DTC and NSCC, have adopted a Clearing Agency Liquidity Risk Management Framework ("Liquidity Risk Management Framework") which sets forth the manner in which FICC measures, monitors and manages the liquidity risks that arise in or are borne by it, including (i) the manner in which FICC deploys its liquidity tools to meet its settlement obligations on an ongoing and timely basis and (ii) FICC's use of intraday liquidity, in accordance with applicable legal requirements, including CCAS 17ad-22(e)(7)[53], and addresses related matters. The Liquidity Risk Management Framework has been filed with and approved by the SEC,[54] and is reviewed and approved by the Board on an annual basis.

Liquidity risk is the risk that FICC would not have sufficient funding resources to complete settlement obligations of a defaulting Member's unsettled transactions. FICC's liquidity risk management strategy and objectives are designed to ensure that FICC maintains sufficient liquid resources to meet the potential amount of funding required to settle outstanding transactions of a defaulting Member or affiliated family of defaulting Members in a timely manner. FICC's liquidity risk is managed by the Liquidity Risk Management ("LRM") team within FRM, and subject to oversight by the MRC and the BRC.

As a central counterparty, each Division's liquidity needs are driven by the requirement to cover settlement and funds-only settlement, on an ongoing basis, in the event of a failure of a Member. As a cash market CCP, if a Member defaults, FICC will need to complete settlement of guaranteed transactions on the failing Member's behalf from the date of insolvency (referred to as "DOI") through the remainder of the settlement cycle. As such, FICC measures the sufficiency of its qualifying liquid resources through daily liquidity studies across a range of foreseeable scenarios, including amounts needed over the settlement cycle in the event that the Member or Member family with the largest aggregate liquidity exposure becomes insolvent in extreme but plausible market conditions (referred to as the "Cover One Requirement"). FICC settles only in U.S. dollars.

With respect to each Division, FICC seeks to maintain qualifying liquid resources (as defined in Rule 17ad-22(a)) in an amount sufficient to cover this risk. Each Division's liquidity resources include: (1) the cash in the Clearing Fund; and (2) the rules-based committed repo facility, the CCLF facility,[55] that each Division separately maintains.

FICC may have access to additional non-qualifying resources in the form of uncommitted master repurchase agreements ("MRAs"). While these are not designated as qualifying liquid resources, these uncommitted MRAs may be a source of funding that FICC would obtain by pledging securities in the Clearing Fund (U.S. Government Treasury securities, Agency securities guaranteed by the U.S. Government and certain U.S. Agency/Government-Sponsored Enterprise pass-through securities) and securities underlying the transactions that would have been delivered to the defaulting Member had it not defaulted. FRM performs a weekly assessment of the repo market activity relative to observed FICC liquidity needs.

CCLF

Regarding CCLF, once FICC declares a CCLF event, Members will be required to hold and fund their deliveries to the insolvent Member up to a predetermined cap by entering into repo transactions with FICC

[53] Compliance with Rule 17ad-22(e)(7)(vi) is described in the Clearing Agency Stress Testing Framework and compliance with Rule 17ad-22(e)(7)(vii) is described in the Clearing Agency Model Risk Framework.

[54] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

[55] Participation in the CCLF facility is a membership requirement for all full-service FICC Members. Members must attest to their ability to participate in the CCLF facility. Daily reports provide Members with information on their current and potential future commitments. FICC may also seek to obtain a loan from its clearing bank(s) at the discretion of such bank(s).

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until they complete the associated closeout. These facilities allow Members to effectively manage their potential financing requirements with predetermined caps.

Once FICC issues an Important Notice announcing it is invoking CCLF obligations with respect to a Division (or both Divisions), each solvent Member will receive a report that identifies their delivery obligations owed to the Defaulting Member (i.e., positions that would then be liquidated). FICC will then instruct solvent Members to hold and fund those deliveries (entirely or partially) up to their maximum contribution amount. Solvent Members will be responsible for suppressing securities deliveries to MBSD or GSD, as appropriate. FICC will then return securities delivered by solvent Members to the extent that they should have suppressed such deliveries. Operationally, Members will then pair off fails with FICC and set up reverse repos against FICC as their counterparty for the amount they are funding until FICC has liquidated the underlying securities. This will allow solvent Members to achieve finality of settlement while closing out any obligations against FICC and adequately reflecting their financing transactions. Each Member certifies that the CCLF requirement has been incorporated into its liquidity planning and related operational plans, including in the event of any changes to such Member's CCLF requirement.

For MBSD CCLF, these repo transactions will remain open until it completes the liquidation of the underlying securities by selling them to a new buyer. For GSD CCLF, the repo transactions will remain open until 30 calendar days after entry into the CCLF Transaction for U.S. government bonds or 60 calendar days after entry into the CCLF Transaction for mortgage-backed securities. At that point, FICC will instruct the funding Member to close the repo trade and deliver the securities to FICC at the original obligation amount to complete settlement on the contractual settlement date of the liquidating trade. By receiving and delivering securities on the same day, FICC is able to liquidate positions without any need for liquidity.

FICC will reimburse Members for the financing costs they incur when funding their own positions. These costs will become part of the liquidation P&L, as they would if FICC had done the financing. If there are insufficient resources to pay for these costs, they will become part of the loss allocation process.

Allocation of CCLF caps for MBSD and GSD Members

MBSD uses two algorithms to establish the maximum CCLF cap (or Defined Capped Liquidity Amount) for each Member—one for Members that are banks or have bank affiliates, and a second model for all other Members. These maximum contributions are re-calculated on a regular basis coinciding with a semi-annual reset for Members that are banks or have bank affiliates, and a monthly reset period for all other members. Both algorithms determine each Member's largest potential liquidity need for a given period by reviewing its associated sell positions versus other clearing Members at the family level on a bilateral basis within each CUSIP. By establishing a maximum contribution amount for each firm, MBSD effectively establishes a predetermined amount that firms can utilize to size their potential liquidity needs in case a major Member defaults. That, in turn, will help Members assess their potential exposure to liquidity risks in various scenarios.

For GSD CCLF, there is no distinction between banks or non-banks. Members are subject to the Individual Regular Amount and Individual Supplemental Amount based on the Aggregate Total Amount which is a sum of the Cover 1 liquidity requirement plus a liquidity buffer within the look back-period. The Individual Regular Amount is allocated to all Members and the Individual Supplemental Amount is applicable only to Members generating liquidity needs in excess of $15 billion. The Individual Supplemental Amount has a Liquidity Tier structure and uses frequency of liquidity created within those Liquidity Tiers. GSD believes that this tiered approach reflects a reasonable, fair and transparent balance between the need for sufficient liquidity resources and the burdens of the funding obligations on Members' management of their liquidity. This increment would appropriately distinguish Members that present the highest liquidity needs on a frequent basis and allocate more of the Individual Supplement Amount to the Members in the top Liquidity Tiers.

In the context of a CCLF event, FICC will accept securities delivered by a solvent Member to the insolvent Member in excess of the solvent Member's maximum contribution on a delivery-versus-payment (DVP)

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basis for FICC. FICC will seek to fund these deliveries via its Clearing Fund cash deposits. If these cash deposits are insufficient, FICC will seek to fund the requisite amount by routing these securities to other Members. This redistribution of securities will occur in the form of repo transactions. For MBSD, this redistribution of securities will be limited to Members that are banks or that have bank affiliates. For GSD, the redistribution of securities will be applicable to all Members. In addition, the combined amount of suppressed and redistributed deliveries will not exceed a Member's maximum contribution amount. All positions funded by Members will continue to be guaranteed by MBSD or GSD.

As all securities cleared and settled through FICC settle in U.S. dollars, FICC has no cross-currency exposure. As noted above under the discussion of Credit Risk (Principle 4), FICC is not systemically important in multiple jurisdictions, and as a U.S. CCP, its activities do not have a complex risk profile. FICC does not have access to routine credit at the central bank for liquidity; as such, it does not account for this option in its liquidity planning.[56]

Measurement and monitoring of liquidity risk and needs

On a daily basis, FRM and each Division's Settlement Operations groups monitor settlement flows and projected debit obligations. Each Member's incoming credits and debits are reviewed to estimate the size of cash outflow required to satisfy settlement needs. In addition, automated risk systems are utilized to measure and monitor liquidity demands. On a daily basis, FRM contacts those Members that have large projected settlement obligations to confirm their settlement capabilities.

FRM performs daily liquidity studies to determine if each Division's current available qualifying liquid resources are sufficient to satisfy the single-largest family default under extreme but plausible conditions.

In the event that a GSD Member or MBSD Member, as applicable, defaults on its open obligations, the applicable Division's liquidity needs are driven by the Member's settlement obligations on the date of its insolvency (DOI), plus the failed Member's security purchases on each day of the settlement cycle. To the extent that a Member's open portfolio to be closed out is "net-long," FICC is responsible for the receipt of securities and payment of cash. As such, "buy" settlement positions drive the potential liquidity risk that is posed to FICC, since FICC would be responsible for the payment of cash required to settle those purchases.

Accordingly, FICC calculates its liquidity needs per individual Member at a legal entity level and further aggregates amounts at a family level (that is, including all affiliated entities[57] that are also FICC Members, under the assumption that all such affiliates fail simultaneously). Members' total liquidity needs are based on the sum of their aggregate daily debit positions on a per CUSIP basis, less any applicable offsets in the portfolio (that is, if the Member has transacted a securities purchase and sale in the same CUSIP, a settlement credit can be generated). These positions represent the securities that FICC would have an obligation to receive and pay for in the event of a Member default.

Based on the results of the calculation, for each Division, FICC determines its aggregate financial resources that would be available to satisfy the largest peak aggregate settlement obligation of a Member or Member family on the day of the simulated default.

[56] However, FICC does maintain an account with the FRBNY, which is one of the options available to it for cash management under the Clearing Agency Investment Policy and which facilitates the management of, and access to, FICC's qualifying liquid resources.

[57] "Affiliated entities" means collectively, each GSD Member or MBSD Member, as applicable, which controls or is controlled by another Member and each Member that is under the common control of any organization, entity or individual. "Control" for these purposes means the direct or indirect ownership of more than 50 percent of the voting securities or other voting interests of any organization, entity or person.

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In order to assess its liquidity obligations under a wide range of scenarios, each Division performs daily liquidity sufficiency testing with respect to normal market scenarios, as a baseline reference point to assess other stress assumptions, and two types of stress scenarios:

- Regulatory stress scenarios, which are designed to meet the requirements set forth in Rule 17ad-22(e)(7)(i) and (7)(vi)(A), and a wide range of stress scenarios that include, but are not limited to, extreme but plausible scenarios, including the default of the affiliated family that would generate the largest aggregate payment obligation for each Division in extreme but plausible market conditions.
- Informational stress scenarios, which are executed as part of internal analysis. These stress scenarios are not designed to meet the requirements of Rules 17ad-22(e)(4)(vi)(A) and (7)(vi)(A) and, as such, are utilized for informational, analytical and/or monitoring purposes only.

Regulatory stress scenarios are subject to review on at least a monthly basis and approval by the ESTC. On a monthly basis, liquidity risk metrics are reported, and these results are used to evaluate the adequacy of FICC's liquidity resources.

Size and composition of qualifying liquid resources; reporting

Each Division's Rules specifically permit borrowing from the Clearing Fund to facilitate settlement. Additionally, Clearing Fund collateral may be pledged for the purpose of securing loans to facilitate settlement. The investment of Clearing Fund cash is managed in accordance with the Clearing Agency Investment Policy (as described under Principle 16 Custody and investment risks). These amounts are available, as needed, for each Division to draw upon to complete settlement.

Reliability of liquidity providers and others

FICC undertakes due diligence to reasonably validate that each liquidity provider has sufficient liquidity resources, understands its liquidity obligations, and has the capacity to perform its liquidity obligations, including the need to be able to provide funding within a short time window. The basis of this analysis includes a credit analysis of each liquidity provider, which includes review of credit ratings assigned by credit rating agencies, and reviews of key metrics including capital adequacy, liquidity, asset quality, and profitability.

With respect to a liquidity provider understanding its liquidity obligations, automated risk systems are utilized to measure and monitor each Division's liquidity demands. In addition, Members receive daily reporting on the settlement obligations and the CCLF reporting is made available to both MBSD and GSD Members. These CCLF reports enable Members to monitor their potential maximum liquidity obligation in a scenario in which CCLF is invoked.

Moreover, FICC's qualifying liquid resources are tested at least annually to confirm the providers are operationally able to perform their commitments and are familiar with the execution and operational arrangements with respect to FICC's CCLF process.

Stress scenarios, review and validation

LRM conducts a daily liquidity study which includes a number of stress scenarios and assumptions used to evaluate each Division's liquidity needs on a conservative basis and to ensure the maintenance of sufficient qualifying liquid resources to timely meet each Division's settlement obligations with a high degree of confidence.[58] In addition, LRM may perform daily liquidity studies for informational and monitoring purposes using stress scenarios that exceed regulatory minimums.

[58] For GSD, one such assumption excludes all cash and Eligible Clearing Fund Securities deposited by a Sponsoring Member with respect to their Segregated Indirect Participants Accounts.

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Scenario stress test results are reviewed by the MRC on at least a monthly basis and escalated to the BRC as necessary. These results are used to evaluate the adequacy of each Division's qualifying liquid resources, for example, the impact of extreme market moves and the potential changes in resource availability that may follow. To the extent that stress tests indicate a potential impact on the sufficiency of each Division's liquidity resources, management may consider options available to supplement resources.

The results of these analyses are reviewed by the ESTC at its monthly meeting as well as FICC's MRC and the BRC. The ESTC uses such results to evaluate the adequacy of the stress testing methodology for the purpose of achieving compliance with the minimum prefunded financial resource requirements set forth in the Liquidity Risk Management Framework.

FICC also performs an annual liquidity assessment that evaluates key liquidity metrics to support recommendations on the appropriate sizing of qualifying liquid resources. This assessment is presented to the MRC and the BRC.

FICC's liquidity stress-testing methodology is reviewed and validated in accordance with the Model Risk Framework, including validation on an annual basis and monthly model performance monitoring.

Replenishment of liquidity resources; uncovered liquidity shortfalls

As the liquidation of a defaulting Member's portfolio completes (including the sale of collateral used to secure a borrowing), the proceeds from the closeout are used to repay liquidity borrowings, thus replenishing each Division's liquidity resources.

If closeout proceeds are insufficient to fully repay such liquidity usage/borrowings, then the amount of any such shortfall would constitute a credit loss that would first be satisfied by recourse to the defaulting Member's available resources pursuant to the provisions of GSD Rule 4 (Clearing Fund and Loss Allocation) or MBSD Rule 4 (Clearing Fund and Loss Allocation) and the applicable terms of any cross-guaranty agreements. Any loss remaining thereafter would be addressed by the loss allocation waterfall, which is described in Principle 13 (Participant-default rules and procedures).

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Principle 8: Settlement finality; CCAS 17ad-22(e)(8)

Principle 8: Settlement finality	**CCAS 17ad-22(e)(8)**
An FMI should provide clear and certain final settlement, at a minimum by the end of the value date. Where necessary or preferable, an FMI should provide final settlement intraday or in real time. ***Key consideration 1***: An FMI's rules and procedures should clearly define the point at which settlement is final. ***Key consideration 2***: An FMI should complete final settlement no later than the end of the value date, and preferably intraday or in real time, to reduce settlement risk. An LVPS or SSS should consider adopting RTGS or multiple-batch processing during the settlement day. ***Key consideration 3***: An FMI should clearly define the point after which unsettled payments, transfer instructions, or other obligations may not be revoked by a participant.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (8) Define the point at which settlement is final to be no later than the end of the day on which the payment or obligation is due and, where necessary or appropriate, intraday or in real time.

Funds-Only Settlement—GSD and MBSD

The GSD Rules and the MBSD Rules require that the transfer of cash with regard to funds-only settlement (*see* GSD Rule 13 and MBSD Rule 11) be processed through the Federal Reserve's NSS. FICC has indirect access to the NSS service via DTC. All entries processed via NSS are final and irrevocable when posted pursuant to the Federal Reserve's Operating Circular 12.[59] Each Division's funds-only settlement process is further described in Principle 9.

Settlement of Securities and Related Payment Obligations—GSD

For all GSD CCP services (other than the GCF Repo service, the CCIT service, the Sponsored GC service, and the ACS Tri-party service described below), securities settlement occurs on a DVP basis via the Fedwire service or on the books of FICC's clearing bank, BNY Mellon. Securities settlement that occurs on the Fedwire service is final and irrevocable when the transfer of securities and related funds is made pursuant to the Federal Reserve's Operating Circular 7.[60] Similarly, securities settlement that occurs on the books of FICC's clearing bank is final when the transfer of securities and related funds occurs pursuant to the agreement between FICC and its clearing bank.

Settlement in connection with the GCF Repo service and the CCIT service occurs on the books of the clearing bank and is final when the transfer of securities and related funds occurs pursuant to the agreement between FICC and its clearing bank. Settlement in connection with the Sponsored GC service and ACS Triparty service occurs on the books of the relevant Clearing Agent Bank and is final when the transfer of

[59] Operating Circular No. 12 is available at www.frbservices.org.

[60] Operating Circular No. 7 is available at www.frbservices.org.

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securities and related funds occurs as specified by the pre-novation counterparty in accordance with such procedures the Clearing Agent Bank may specify.

Settlement of Securities and Related Payment Obligations—MBSD

Securities settlement at MBSD occurs on a DVP basis via the Fedwire service or on the books of FICC's clearing bank, and finality is achieved in the same manner as for the GSD DVP settlement services, as described above.

Settlement instructions

Pursuant to each Division's Rules,[61] Members cannot revoke their securities settlement obligations to FICC. Specifically, each Division's Rules state that "reversal codes shall not be used to identify any delivery of securities to [FICC] without the express prior permission of [FICC]."[62] FICC may accept a revocation of an instruction or obligation under limited circumstances. For a revocation to be accepted, FICC must receive revocation instructions from both Members, and FICC must be certain that removing the affected positions from the settlement process will not affect other Members.

[61] GSD Rule 12, and MBSD Rule 9.

[62] GSD Rule 12, Sections 1 and 11 and MBSD Rule 9, Section 1.

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Principle 9: Money settlements; CCAS 17ad-22(e)(9)

Principle 9: Money settlements	**CCAS 17ad-22(e)(9)**
An FMI should conduct its money settlements in central bank money where practical and available. If central bank money is not used, an FMI should minimise and strictly control the credit and liquidity risk arising from the use of commercial bank money. ***Key consideration 1***: An FMI should conduct its money settlements in central bank money, where practical and available, to avoid credit and liquidity risks. ***Key consideration 2***: If central bank money is not used, an FMI should conduct its money settlements using a settlement asset with little or no credit or liquidity risk. ***Key consideration 3***: If an FMI settles in commercial bank money, it should monitor, manage, and limit its credit and liquidity risks arising from the commercial settlement banks. In particular, an FMI should establish and monitor adherence to strict criteria for its settlement banks that take account of, among other things, their regulation and supervision, creditworthiness, capitalisation, access to liquidity, and operational reliability. An FMI should also monitor and manage the concentration of credit and liquidity exposures to its commercial settlement banks. ***Key consideration 4***: If an FMI conducts money settlements on its own books, it should minimise and strictly control its credit and liquidity risks. ***Key consideration 5***: An FMI's legal agreements with any settlement banks should state clearly when transfers on the books of individual settlement banks are expected to occur, that transfers are to be final when effected, and that funds received should be transferable as soon as possible, at a minimum by the end of the day and ideally intraday, in order to enable the FMI and its participants to manage credit and liquidity risks.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (9) Conduct its money settlements in central bank money, where available and determined to be practical by the board of directors of the covered clearing agency, and minimize and manage credit and liquidity risk arising from conducting its money settlements in commercial bank money if central bank money is not used by the covered clearing agency.

Introduction

With respect to GSD, there are three types of money settlements that occur: (i) the funds-only settlement process that occurs pursuant to GSD Rule 13 and consists of a mark-to-market amount for every net settlement position, a mark-to-market amount for every forward net settlement position, fail marks for obligations that were scheduled to settle and have not yet settled, coupon payments, GCF Repo interest, billing and other adjustments, (ii) the fund payments associated with securities deliveries pursuant to GSD's DVP settlement process and (iii) the fund payments associated with the settlement of GCF Repo, CCIT, Sponsored GC and ACS Triparty trades.

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With respect to MBSD, there are two types of money settlements that occur: (i) the cash settlement process that occurs pursuant to MBSD Rule 11 and consists of a mark-to-market amount, principal and interest payments, TMPG fail charges, billing and other adjustments and (ii) the fund payments associated with securities deliveries pursuant to MBSD's DVP settlement process.

Fund payments in connection with each Division's DVP service and GSD's GCF Repo, CCIT, Sponsored GC and ACS Triparty services are discussed in Principle 8.

As described in further detail in response to Principle 6 (Margin), each FICC Division collects Clearing Fund deposits from its Members. Clearing Fund deposits are collected intraday in central bank money by Fedwire.

Settlement Arrangements

Funds-only settlement

FICC uses the central bank model with a tiered settlement arrangement for its GSD funds-only settlement process (GSD Rule 13) and its MBSD cash settlement process (MBSD Rule 11) and settlement is conducted in central bank funds. Members are required to engage a settling bank that meets FICC's settling bank limited membership criteria to effect money settlement via Federal Reserve's National Settlement Service ("NSS")[63] on behalf of the Members. Funds transfers are final when effected on the NSS. The arrangement with the central bank is supported by applicable laws and rules, including the applicable Federal Reserve circular (Federal Reserve Operating Circular 12). FICC confirms that fund transfers have been effected on each business day with respect to the funds-only settlement process.

Funds-only settlement is a twice-daily process of generating a net credit or debit cash amount for each Member and settling those cash amounts between Members and GSD or MBSD, as applicable.[64] The funds-only settlement process is a cash pass-through process; i.e., those Members that are in a net debit position are obligated to submit payments that are then used to pay Members in a net credit position. Net debits and credits of all Members using the same settling bank are further netted and reported to the settling bank which is required to acknowledge the net-net debits or credits. FICC uses DTC as its settlement agent to instruct the Federal Reserve to debit or credit each settling bank.

Designation of Settling Bank

Members at each Division are obligated to designate a settling bank to effect daily money settlement on their behalf. Members may designate a settling bank of their own choosing (and FICC does not intervene in that relationship), provided the bank meets certain financial and operational criteria, is a Federal Reserve member and agrees to settle through use of NSS. The settling banks are "Funds-Only Settling Bank Members" with respect to GSD and "Cash Settling Bank Members" with respect to MBSD (collectively, referred to herein as "Settling Bank Members" or "Settling Banks") and agree to abide by the requirements applicable to that class of membership; this includes timely adherence to daily settlement protocols. (Settling Bank standards and monitoring are described below.) The Settling Bank may or may not have a

[63] NSS is a multilateral settlement service owned and operated by the Federal Reserve Banks (FRBs). The service is offered to depository institutions that settle for participants in clearinghouses, financial exchanges and other clearing and settlement groups. Settlement agents, acting on behalf of those depository institutions in a settlement arrangement, electronically submit settlement files to the FRBs. Files are processed on receipt, and entries are automatically posted to the depository institutions' Federal Reserve Bank accounts. DTC maintains an account at the FRB and acts as agent for FICC for Funds-Only Settlement.

[64] The GSD Rules refer to this process as "funds-only settlement"; however, the MBSD Rules refer to this process as "cash settlement." For ease of review, the term "funds-only settlement" is used in this principle.

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credit relationship with the Member—it is not expected or required to guarantee the obligations of the Member or to advance funds on the Member's behalf.[65] If a Settling Bank does not settle on behalf of one or more Members, each of those Members nevertheless remains obligated to pay its net settlement obligation through alternative funding arrangements.

DTC acts as settlement agent for FICC

As described above, FICC has a settlement interface with its affiliate, DTC. DTC acts as settlement agent for FICC. In submitting the NSS file, DTC, as FICC's settlement agent, submits instructions to cause the Federal Reserve accounts of the Settling Banks to be charged for their net-net debit balances and credited with their net-net credit balances.

Settling Bank requirements and monitoring

Settling Banks are required to maintain accounts at the Federal Reserve, and meet minimum financial and operational criteria, including minimum capital requirements and a minimum Tier 1 Risk-Based Capital Ratio for regulatory purposes. Settling Banks are subject to supervision and regulation by their federal and state regulators, depending on their form of organization.

FRM monitors the Settling Banks for their compliance with each Division's Rules (in their capacity as Settling Bank Members or Settling Banks).

Quarterly, FRM reviews the capital adequacy of Settling Banks and the ESR monitors FICC Settling Bank exposure and concentration. For Full Service Members financial statements filed with regulatory agencies, and information gathered from various financial publications are analyzed to confirm that each Settling Bank continues to be financially stable.

[65] Each Division's Rules include protections such as allowing the Settling Bank to refuse to settle for a Member.

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Principle 10: Physical deliveries; CCAS 17ad-22(e)(10)

Principle 10: Physical deliveries	**CCAS 17ad-22(e)(10)**
An FMI should clearly state its obligations with respect to the delivery of physical instruments or commodities and should identify, monitor, and manage the risks associated with such physical deliveries. ***Key consideration 1***: An FMI's rules should clearly state its obligations with respect to the delivery of physical instruments or commodities. ***Key consideration 2***: An FMI should identify, monitor, and manage the risks and costs associated with the storage and delivery of physical instruments or commodities.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (10) Establish and maintain transparent written standards that state its obligations with respect to the delivery of physical instruments, and establish and maintain operational practices that identify, monitor, and manage the risks associated with such physical deliveries.

FICC does not have any obligations with respect to physical instruments or commodities. Accordingly, Principle 10 and CCAS 17ad-22(e)(10) do not apply to FICC.

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Principle 11: Central securities depositories; CCAS 17ad-22(e)(11)

Principle 11: Central securities depositories	**CCAS 17ad-22(e)(11)**
A CSD should have appropriate rules and procedures to help ensure the integrity of securities issues and minimise and manage the risks associated with the safekeeping and transfer of securities. A CSD should maintain securities in an immobilised or dematerialised form for their transfer by book entry.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (11) When the covered clearing agency provides central securities depository services: (i) Maintain securities in an immobilized or dematerialized form for their transfer by book entry, ensure the integrity of securities issues, and minimize and manage the risks associated with the safekeeping and transfer of securities; (ii) Implement internal auditing and other controls to safeguard the rights of securities issuers and holders and prevent the unauthorized creation or deletion of securities, and conduct periodic and at least daily reconciliation of securities issues it maintains; and (iii) Protect assets against custody risk through appropriate rules and procedures consistent with relevant laws, rules, and regulations in jurisdictions where it operates.

FICC is not a CSD and does not provide CSD services. Accordingly, Principle 11 and CCAS 17ad-22(e)(11) do not apply to FICC.

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Principle 12: Exchange-of-value settlement systems; CCAS 17ad-22(e)(12)

Principle 12: Exchange-of-value settlement systems	**CCAS 17ad-22(e)(12)**
If an FMI settles transactions that involve the settlement of two linked obligations (for example, securities or foreign exchange transactions), it should eliminate principal risk by conditioning the final settlement of one obligation upon the final settlement of the other. ***Key consideration 1***: An FMI that is an exchange-of-value settlement system should eliminate principal risk by ensuring that the final settlement of one obligation occurs if and only if the final settlement of the linked obligation also occurs, regardless of whether the FMI settles on a gross or net basis and when finality occurs.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (12) Eliminate principal risk by conditioning the final settlement of one obligation upon the final settlement of the other, regardless of whether the covered clearing agency settles on a gross or net basis and when finality occurs if the covered clearing agency settles transactions that involve the settlement of two linked obligations.

GSD Rule 12 and MBSD Rule 9 establish that all deliveries of eligible securities in satisfaction of delivery obligations, and all receipts of eligible securities in satisfaction of receive obligations, must be made against simultaneous payment or receipt in Federal funds. GSD's and MBSD's settlement mechanisms eliminate principal risk by ensuring that the final settlement of one obligation occurs if the final settlement of the linked obligation occurs.

Settlement System—GSD

Settlement of the transactions processed by GSD involves the delivery of securities against payment of funds.

With respect to GSD, FICC's CCP services can be divided into two segments for purposes of this Principle: (1) the GCF Repo service, CCIT service, the Sponsored GC service and the ACS Triparty service, and (2) the DVP service. From a legal and contractual perspective, these services are described in the GSD Rules to which Members are subject by the provisions of their membership agreements with the FICC. There are also relevant contractual provisions between FICC and its clearing bank that govern the clearance and settlement services provided by the clearing bank with respect to each of these services.

The DVP service encompasses both repo and buy-sell activity, including all non-GCF Repo activity, non-CCIT activity, and non-Sponsored GC / ACS Triparty activity (both repo and buy-sell activity). All DVP service activity is done on a DVP basis to help mitigate principal risk. On each business day, FICC makes available to each GSD Member output that provides information (e.g., type of obligation (deliver or receive), name and reference number of the clearing bank, CUSIP number, settlement date, par value, final dollar value and other information descriptive of an eligible netting security) that FICC deems sufficient to enable the Member to settle its net settlement positions and its Same Day Settlement obligations on that business day. Each GSD Member, based on the information provided by FICC, then provides appropriate instructions to its clearing bank to deliver to FICC and/or to receive from FICC, on behalf of the GSD Member, eligible netting securities against payment or receipt of funds at the appropriate settlement value. Obligations are either settled within the clearing bank (that is, on the books of the clearing bank) or using the Fedwire service. All deliveries are made against payment.

The GCF Repo service essentially functions on a tri-party basis. Dealer GSD Members that participate in the service are required to have an account with the participating clearing bank, BNY Mellon. The service operates on an intra-clearing bank basis, and as such, the securities settlement for the GCF Repo service occurs on the books of the clearing bank and is final when the transfer of securities and related funds occurs

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per the agreement between FICC and its clearing bank. The CCIT service utilizes the infrastructure of the GCF Repo service, and, as such, securities settlement also occurs on the books of the clearing bank and is final in the same way as with the GCF Repo service.

The Sponsored GC service allows Sponsoring Members to execute repo transactions with their Sponsored Members, and these repo transactions would settle on the tri-party repo platform of the Sponsored GC clearing bank. Settlement in connection with the Sponsored GC service occurs on the books of the Sponsored GC clearing bank and is final when the transfer of securities and related funds occurs as specified by the pre-novation counterparty in accordance with such procedures the Sponsored GC clearing bank may specify. The CIL offering within the Sponsored GC service operates similarly with the notable exception being that the FICC is granted a lien in the purchased repo securities subject to the Sponsored GC CIL Trade, in lieu of a margin requirement collected on that activity.

The ACS Triparty service operates similarly, in that it allows Agent Clearing Member to execute repo transactions with their Executing Firm Customers, and these repo transactions settle on the tri-party platform of the ACS Triparty Clearing Agent Bank. Settlement similarly occurs on the books of the ACS Triparty Clearing Agent Bank and is final when the transfer of securities and related funds occurs as specified by the pre-novation counterparty in accordance with such procedures the clearing bank may specify.

Settlement System—MBSD

Settlement of transactions processed by MBSD is done on a DVP basis, which helps eliminate principal risk. The system operates as follows: On each business day, FICC makes available to each MBSD Member output that provides information (e.g., type of obligation (deliver or receive), CUSIP number, settlement date, par value, final dollar value and other information descriptive of an eligible security) that FICC deems sufficient to enable the Member to settle its net pool positions on that business day. Each MBSD Member, based on the information provided by FICC, then provides appropriate instructions to its clearing bank to deliver to FICC and/or to receive from FICC, on behalf of the Member, eligible netting securities against payment or receipt at the appropriate settlement value. Obligations may be settled within the clearing bank (that is, on the books of the clearing bank) or using the Fedwire service. All deliveries are made against payment.

Each Division's Reliance on the Fedwire® Service

Each Division relies on another FMI—the Federal Reserve's Fedwire. BNY Mellon, FICC's clearing bank, has the direct relationship with the Fedwire systems. The terms of the Fedwire service are governed by the Federal Reserve's Operating Circular No. 7.[66]

[66] Operating Circular No. 7 available at www.frbservices.org.

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Principle 13: Participant-default rules and procedures; CCAS 17ad-22(e)(13)

Principle 13: Participant-default rules and procedures	**CCAS 17ad-22(e)(13)**
An FMI should have effective and clearly defined rules and procedures to manage a participant default. These rules and procedures should be designed to ensure that the FMI can take timely action to contain losses and liquidity pressures and continue to meet its obligations. ***Key consideration 1***: An FMI should have default rules and procedures that enable the FMI to continue to meet its obligations in the event of a participant default and that address the replenishment of resources following a default. ***Key consideration 2***: An FMI should be well prepared to implement its default rules and procedures, including any appropriate discretionary procedures provided for in its rules. ***Key consideration 3***: An FMI should publicly disclose key aspects of its default rules and procedures. ***Key consideration 4***: An FMI should involve its participants and other stakeholders in the testing and review of the FMI's default procedures, including any close-out procedures. Such testing and review should be conducted at least annually or following material changes to the rules and procedures to ensure that they are practical and effective.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (13) Ensure the covered clearing agency has the authority and operational capacity to take timely action to contain losses and liquidity demands and continue to meet its obligations by, at a minimum, requiring the covered clearing agency's participants and, when practicable, other stakeholders to participate in the testing and review of its default procedures, including any close-out procedures, at least annually and following material changes thereto.

Rules and procedures relating to default management

FICC's Divisions have effective and clearly defined rules and procedures to manage a Member default. Each Division's Rules (with implementing internal procedures) are designed to ensure that FICC has the authority to take timely action to contain losses and liquidity pressures and demands and continue to meet its obligations in the event of a Member default.

FICC has publicly disclosed key aspects of its default rules and procedures. Each Division's Rules clearly state what constitutes a Member default and the consequences of default. Under GSD Rule 21 (Restrictions on Access to Services), and MBSD Rule 14 (Restrictions on Access to Services), the Board of Directors may suspend a Member or prohibit or limit a Member's access to FICC's services in enumerated circumstances. These circumstances include, for example, a Member's expulsion or suspension from a regulatory or self-regulatory organization, default in delivering funds or securities to FICC, and a Member experiencing such financial or operational difficulties that FICC determines, in its discretion, that restriction on access to its services is necessary for its protection and for the protection of its membership. Each Division's Rules provide FICC with some discretion in determining what constitutes adequate cause to cease to act for a Member. GSD Rule 22 (Insolvency of a Member) and MBSD Rule 16 (Insolvency of a Member) enumerate the circumstances under which a Member will be treated as insolvent.

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If any of the enumerated circumstances arise, depending upon the facts and situation, FICC may suspend a Member from any service provided by FICC either with respect to a particular transaction or transactions or with respect to transactions generally, or it may prohibit or limit such Member's access to services offered by FICC. When FICC restricts a Member's access to services generally, FICC is said to have "ceased to act" for the Member.

The determination as to whether or not to cease to act for a Member is not automatic; rather under GSD Rule 21 and MBSD Rule 14 an affirmative determination to do so must be made. The Board of Directors has delegated authority to make such determinations to the BRC.[67] To ensure that timely action may be taken, the BRC Charter also provides for delegated authority to the Chair of the BRC if it is impractical to convene the BRC. Action taken would then be ratified by the BRC at a subsequent meeting.[68]

When FICC ceases to act for a Member, or suspends or limits its access to services, FICC notifies the Member and furnishes it with a written statement of the grounds for the decision, and of the Member's right to request a hearing with respect to that determination.[69] FICC will also notify FICC's own supervisors of any decision to cease to act promptly after such decision is made, and will issue an Important Notice to all Members informing them of the cease to act.[70] Finally, for a defaulting Member that is also a member of other clearing agencies with which FICC has cross-guaranty or other arrangements, FICC will also notify those clearing agencies as required by those arrangements.

Close-out process

Once FICC has ceased to act for a Member, each Division's Rules provide it with the authority to promptly close out and manage the positions of a defaulter and to apply the defaulter's collateral. GSD Rule 22A (Procedures for When the Corporation Ceases to Act) and MBSD Rule 17 (Procedures for When the Corporation Ceases to Act) describe the procedures, including actions FICC may take, when it ceases to act for a Member; this includes provisions for the treatment of core services where Members may have transactions pending with a defaulting Member. The Rules identify which actions are automatic and which are discretionary, and detail how the unsettled transactions of the defaulting Member are to be processed. In this regard, unless the BRC has determined otherwise, FICC will exclude from further processing any trade that, at the time FICC ceased to act for the Member, had not compared upon receipt pursuant to the Rules or that had not been reported by FICC to Members as compared.[71] Any transactions so excluded are to be settled between the parties and not through FICC.

With respect to activity clearing through the Collateral-in-Lieu offering of the Sponsored GC service, FICC is granted a lien on the purchased repo securities that are subject to the Sponsored GC CIL Trade, which allows it to acquire the purchased repo securities and use them to settle with the GC Funds Borrower. Therefore, if FICC ceases to act for a CIL Funds Lender, FICC may, in lieu of closing out the Sponsored GC CIL Trades (or a portion of any such trades), exercise its right as a secured party in relation to the purchased repo securities and instruct the Clearing Agent Bank to remove such purchased repo securities from the account of such CIL Funds Lender.

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[67] This authority is reflected in the Charter of the BRC, which is available on DTCC's website.

[68] In addition, GSD Rule 44 and MBSD Rule 34 also permit action to be taken by senior corporate officers, if so designated.

[69] Notices are provided for in GSD Rule 45 and MBSD Rule 35; hearing procedures are provided pursuant to GSD Rule 37 and MBSD Rule 28. As provided in GSD Rule 21 and MBSD Rule 14, in certain cases FICC may summarily suspend a Member, with the Member having a right to a hearing after the fact; in other cases, the action is subject to the Member's prior right to request a hearing. See also GSD Rule 22A and MBSD Rule 17 regarding notification of a cease to act.

[70] GSD Rule 22A, Section 1 and MBSD Rule 17, Section 1.

[71] GSD Rule 22A, Section 2 and MBSD Rule 17, Section 2.

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FICC communicates all such actions and determinations to its membership (including Members' responsibilities with respect thereto) via Important Notices, which are posted on the DTCC website. Members are responsible for retrieving Important Notices daily.[72]

As discussed in Principle 14 (Segregation and portability), as a cash market CCP in the U.S., the legal regime provides for segregation and protection of customer securities away from direct participant activities and supported by FICC Rules and internal procedures. As a result, FICC maintains separate "house" and "client" accounts for its Members and distinguishes between proprietary and customer activity. Accordingly, the default management of outstanding transactions in "house" will be distinguished from outstanding transactions in "client" accounts. A segregation of "client" positions from "house" positions designed to enable portability of "client" positions, along with the associated margin, where practicable and permitted, to other sponsors with whom the "client" already has established relationship. Any remaining outstanding transactions that are not otherwise exited from FICC's systems are subject to the closeout.

The process of closing out open positions typically involves effecting market purchases and sales; that is, buying in securities the defaulter was obligated to deliver to FICC, and selling out securities the defaulter was obligated to receive from FICC and pay for, or otherwise liquidating the position. FICC utilizes the services of investment advisors and executing brokers to facilitate such transactions promptly following its determination to cease to act. FICC may engage in hedging transactions or otherwise take action to minimize market disruption as a result of such purchases and sales.

Use of financial resources to contain liquidity pressures and demands and losses

As described more fully in Principle 7 (Liquidity risk), FICC maintains liquidity resources to enable it to complete settlement in the ordinary course notwithstanding the default of a Member. Each Division's Rules empower it to draw promptly on these resources as needed, while it proceeds to complete the close-out and liquidation of the defaulter's positions.

Each Division's Rules specifically permit borrowings from the Clearing Fund to facilitate settlement; in addition, Clearing Fund collateral may be pledged for the purpose of securing loans to facilitate settlement.[73]

As the liquidation of a defaulting Member's portfolio completes (including the sale of collateral used to secure a borrowing), the proceeds from the close-out would be used to repay liquidity borrowings, thus replenishing FICC's liquidity resources. Should close-out proceeds be insufficient to fully repay a liquidity borrowing, then FICC would look to its loss waterfall to repay any outstanding amounts.

Resources to cover losses

If, after closing out and liquidating a defaulting Member's positions, FICC were to suffer a loss, such loss would first be satisfied by the amounts on deposit to the Clearing Fund and Eligible Clearing Fund Securities pledged from the defaulting Member (along with any other resources of, or attributable to, the defaulting Member (or certain permitted margin affiliates) that FICC may access under each Division's Rules, such as funds-only settlement amounts and other collateral held by the Division securing the defaulting Member's obligations to the Division). This includes any amounts available to FICC under clearing agency cross-guaranty agreements (agreements with other registered clearing agencies that relate to the guaranty of certain obligations of a common defaulting member, by one or more clearing agency parties to other clearing agency parties).

[72] GSD Rule 45, Section 1, and MBSD Rule 35, Section 1.

[73] GSD Rule 4 and MBSD Rule 4.

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FICC has also entered into a multilateral netting contract and limited cross-guaranty agreement with DTC, NSCC and The Options Clearing Corporation ("OCC"), under which these clearing agencies have agreed to make payments to each other for unsatisfied obligations of a common defaulting participant to the extent they have excess resources of the defaulting Member. Under this arrangement, no party ever needs to pay "out of pocket" and no party can receive more than its loss. Each Division would apply its own respective resources and determine its own liquidation results. In determining FICC's available net resources for purposes of this cross-guaranty agreement, FICC would first offset the available net resources of GSD with those of MBSD.

GSD also has a cross-margining agreement with the CME, which may also provide for additional funds if the defaulting Member was a cross-margining participant.[74]

Loss waterfall

GSD Rule 4 and MBSD Rule 4 contain the loss waterfall provisions for the respective Divisions. Under GSD Rule 4 and MBSD Rule 4, if there is a loss (i) as a result of an obligation of a defaulting Member to FICC that was not fully satisfied by the application of its resources or the proceeds from the liquidation of its portfolio, or (ii) due to an event other than a Member default ("non-default loss," as described in more detail below), FICC has recourse to the loss allocation waterfall.

For purposes of loss allocation, triggering events, which may include both Member defaults and non-default events, would be grouped together chronologically into discrete "event periods" of ten business days. Losses arising from a group of events that occurred within the same event period, whether a default loss or a non-default loss, would be allocated as follows:

1. Before the allocation of losses to Members, FICC would apply 50 percent of its General Business Risk Capital Requirement as of the end of the calendar quarter immediately preceding the applicable event period (referred to as the "Corporate Contribution"), or such greater amount as the Board of Directors may determine to satisfy the losses. The amount of the Corporate Contribution to be applied to any losses arising from events that may occur in subsequent event periods during the next 250 business days would be reduced to the remaining unused portion of contribution, if any. If the losses are attributable to only one Division, the Corporate Contribution to the losses would be up to the amount then available. If the losses occur simultaneously at both Divisions, the Corporate Contribution to the losses would be applied ratably between the Divisions.

2. If a loss is a default loss, FICC will allocate the loss between the Tier One Members and the Tier Two Members, and the loss allocation process applicable to Tier One Members and Tier Two Members will proceed via two separate, parallel and simultaneous streams. If a loss is a non-default loss, FICC will allocate the loss to Tier One Members.

3. With respect to Tier One Members, if a loss remains after applying the Corporate Contribution, FICC will allocate the remaining amount among Tier One Members that were Tier One Members on the first day of the applicable event period, ratably in accordance with their average daily required deposit to the Clearing Fund over the prior 70 business days or such shorter period of time that the Tier One Member has been a Tier One Member, divided by the sum of the average required fund deposit amounts of all Tier One Members subject to loss allocation in such round. Each Tier One Member must pay its allocation amount within two business days of receiving notice of the amount.

4. With respect to Tier Two Members, if a loss remains after applying the Corporate Contribution, FICC will allocate the remaining amount to Tier Two Members to the extent they traded with the defaulting Member and their trades resulted in the loss. FICC will assess Tier Two Members ratably

[74] If the defaulting Member participates in the CME arrangement, GSD will determine any loss or liability incurred in accordance with that agreement which is part of the GSD's Rules.

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based on their loss as a percentage of the entire amount of the remaining loss attributable to Tier Two Members. Tier Two Members are required to pay their loss allocation obligation in full.

5. Losses relating to the events within the same event period may be allocated iteratively.

Non-default losses are generally expected to be absorbed by FICC's capital, as described in more detail in response to Principle 15 (General business risks). However, to the extent that a loss incident to the clearance and settlement business of FICC is determined by the Board of Directors to be a significant and substantial loss that may materially impair the ability of FICC to provide clearance and settlement services in an orderly manner, FICC has recourse to the loss waterfall, as described above.

A Tier One Member is able to limit its liability for loss allocation, up to a loss allocation cap, by withdrawing from participation within the specified timeframe. However, it will remain obligated for any loss allocations for which a withdrawal election was not timely made, and, subject to the specified cap, would remain obligated for its pro rata share of losses and liabilities with respect to any event period to which it is otherwise subject. To date, FICC has never invoked its loss allocation process.

Sponsored Members are not obligated for loss allocation. If the loss arises in connection with Sponsored Member trades (that is, in connection with the default of the Sponsoring Member), the loss shall be allocated to Tier One Members. If the loss arises in connection with a default other than of the Sponsoring Member, the Sponsoring Member shall be responsible for any loss allocated to the Sponsoring Member Omnibus Account.

Implementation of Member default rules and procedures

FICC is well-prepared to implement its Member default rules and procedures, including any appropriate discretionary procedures provided for in the Divisions' Rules. FICC maintains internal plans outlining the roles and responsibilities for addressing a Member default. These include internal procedures of the involved departments (for example, Settlement Operations, FRM and the General Counsel's Office) that outline the actions that may be taken in the event of a default of a Member and the various roles and responsibilities associated with these actions. Each instance where FICC would determine to cease to act is unique, and the internal procedures take this into consideration. FICC also maintains arrangements with one or more investment advisors and executing brokers to facilitate an actual close-out and liquidation.

As noted above, FICC will notify the defaulting Member of any decision to cease to act promptly after such decision is made. FICC will also issue an Important Notice to all Members informing them of the cease to act. If a trustee (or other insolvency official) is appointed for the defaulting Member, FICC will notify them as well. Finally, FICC will also notify other FMIs that are party to clearing agency cross-guaranty or other link arrangements, as applicable.

The FICC close-out procedures are subject to ongoing review and development, including incorporating knowledge gained from actual close-outs and internal tests, which occur at least once per year. Material changes to FICC's internal plans are presented to the MRC and the BRC for approval.

GSD Rules and MBSD Rules are publicly available

The key aspects of the close-out procedures are in each Division's Rules, which are available to Members and to the public generally on DTCC's website. In addition, if FICC were to make a determination to cease to act on behalf of a Member, FICC would provide notice of the action to Members via Important Notice and also communicate relevant details as to the resolution of open transactions to Members in the same manner. Important Notices are posted by FICC on the DTCC website, and are available to the public. Members are responsible for retrieving notices daily from the website.[75]

[75] GSD Rule 45, Section 1, and MBSD Rule 35, Section 1.

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Testing and engagement with participants and other stakeholders

FICC conducts a simulated close-out at least annually. The close-out simulations cover a wide range of hypothetical fact patterns that may represent extreme, but plausible, situations, and include testing arrangements with FICC's investment advisor(s). In addition, FRM conducts internal "table-top" close-out exercises to enable relevant staff across the DTCC enterprise to become familiar with processes and procedures that would be utilized in the event of an actual cease to act. Both the simulation results and the table-top reviews are used to assess—and, where appropriate, improve—default management processes and procedures. Results are shared with the Board of Directors, the BRC, FICC's supervisors and, as appropriate, relevant stakeholders.

As noted above, as a cash market CCP, the process by which FICC liquidates and closes out Member positions is typically through purchases and sales into the market; accordingly, the successful implementation of FICC's default management processes does not generally require the active participation of Members (through, for example, an auction process). As such, Members are not generally included as part of FICC's testing processes. However, FICC does provide Members with information—including instructions—as to how transactions will be handled in the event of an actual close-out via Important Notices. With regard to recovery and wind-down testing, FICC Rules (GSD Rule and MBSD Rule 3 (Ongoing Membership Requirements)) provide, among other things, that FICC/GSD and FICC/MBSD have established standards for designating those Members, Limited Members and Settling Banks, as such terms are defined in GSD Rule 22D and MBSD Rule 17B (Wind-down of the Corporation), who shall be required to participate in annual recovery and wind-down testing that FICC reasonably determines are, taken as a whole, the minimum necessary to confirm that FICC's testing procedures will be practical and effective in the face of a recovery or orderly wind-down.

The successful implementation of FICC's default management process does require a robust default liquidity process. As such, FICC tests the mechanics of each Division's CCLF with each Division's Members so that FICC staff and the Members of each Division may be prepared for the facility to be invoked and utilized under stress conditions. In addition, FICC also periodically tests the mechanics of its uncommitted master repurchase agreements with its counterparties.

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Principle 14: Segregation and portability; CCAS 17ad-22(e)(14)

Principle 14: Segregation and portability	**CCAS 17ad-22(e)(14)**
A CCP should have rules and procedures that enable the segregation and portability of positions of a participant's customers and the collateral provided to the CCP with respect to those positions. ***Key Consideration 1***: A CCP should, at a minimum, have segregation and portability arrangements that effectively protect a participant's customers' positions and related collateral from the default or insolvency of that participant. If the CCP additionally offers protection of such customer positions and collateral against the concurrent default of the participant and a fellow customer, the CCP should take steps to ensure that such protection is effective. ***Key Consideration 2***: A CCP should employ an account structure that enables it readily to identify positions of a participant's customers and to segregate related collateral. A CCP should maintain customer positions and collateral in individual customer accounts or in omnibus customer accounts. ***Key Consideration 3***: A CCP should structure its portability arrangements in a way that makes it highly likely that the positions and collateral of a defaulting participant's customers will be transferred to one or more other participants. ***Key Consideration 4***: A CCP should disclose its rules, policies, and procedures relating to the segregation and portability of a participant's customers' positions and related collateral. In particular, the CCP should disclose whether customer collateral is protected on an individual or omnibus basis. In addition, a CCP should disclose any constraints, such as legal or operational constraints, that may impair its ability to segregate or port a participant's customers' positions and related collateral.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (14) Enable, when the covered clearing agency provides central counterparty services for security-based swaps or engages in activities that the Commission has determined to have a more complex risk profile, the segregation and portability of positions of a participant's customers and the collateral provided to the covered clearing agency with respect to those positions and effectively protect such positions and related collateral from the default or insolvency of that participant.

As explained in the explanatory notes and footnotes to Principle 14 in the FMI Principles report,[76] certain cash market CCPs, like FICC, operate in legal regimes that facilitate segregation and portability in order to achieve the protection of customer assets by alternate means that offer the same degree of protection as the approach recommended by this Principle. These regimes do not directly apply to FICC, but to its

[76] FMI Principles report, Principle 14, explanatory notes 3.14.5, 3.14.6 and footnote 122.

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members. For example, a number of FICC's members are broker-dealers subject to the SEC's customer protection rules, promulgated by the SEC under the Exchange Act in 1972.[77]

FICC does not clear security-based swaps or engage in activities that the Commission has determined to have a more complex risk profile. As such, CCAS 17ad-22(e)(14) does not apply to FICC.

[77] Exchange Act Rule 15c3-3.

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Principle 15: General business risk; CCAS 17ad-22(e)(15)

Principle 15: General business risk	**CCAS 17ad-22(e)(15)**
An FMI should identify, monitor, and manage its general business risk and hold sufficient liquid net assets funded by equity to cover potential general business losses so that it can continue operations and services as a going concern if those losses materialise. Further, liquid net assets should at all times be sufficient to ensure a recovery or orderly wind-down of critical operations and services. ***Key consideration 1***: An FMI should have robust management and control systems to identify, monitor, and manage general business risks, including losses from poor execution of business strategy, negative cash flows, or unexpected and excessively large operating expenses. ***Key consideration 2***: An FMI should hold liquid net assets funded by equity (such as common stock, disclosed reserves, or other retained earnings) so that it can continue operations and services as a going concern if it incurs general business losses. The amount of liquid net assets funded by equity an FMI should hold should be determined by its general business risk profile and the length of time required to achieve a recovery or orderly wind-down, as appropriate, of its critical operations and services if such action is taken. ***Key consideration 3***: An FMI should maintain a viable recovery or orderly wind-down plan and should hold sufficient liquid net assets funded by equity to implement this plan. At a minimum, an FMI should hold liquid net assets funded by equity equal to at least six months of current operating expenses. These assets are in addition to resources held to cover participant defaults or other risks covered under the financial resources principles. However, equity held under international risk-based capital standards can be included where relevant and appropriate to avoid duplicate capital requirements. ***Key consideration 4***: Assets held to cover general business risk should be of high quality and sufficiently liquid in order to allow the FMI to meet its current and projected operating expenses under a range of scenarios, including in adverse market conditions. ***Key consideration 5***: An FMI should maintain a viable plan for raising additional equity should its equity fall close to or below the amount needed.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (15) Identify, monitor, and manage the covered clearing agency's general business risk and hold sufficient liquid net assets funded by equity to cover potential general business losses so that the covered clearing agency can continue operations and services as a going concern if those losses materialize, including by: (i) Determining the amount of liquid net assets funded by equity based upon its general business risk profile and the length of time required to achieve a recovery or orderly wind-down, as appropriate, of its critical operations and services if such action is taken; (ii) Holding liquid net assets funded by equity equal to the greater of either (x) six months of the covered clearing agency's current operating expenses, or (y) the amount determined by the board of directors to be sufficient to ensure a recovery or orderly wind-down of critical operations and services of the covered clearing agency, as contemplated by the plans established under paragraph (e)(3)(ii) of this section, and which: (A) Shall be in addition to resources held to cover participant defaults or other risks covered under the credit risk standard in paragraph (b)(3) or paragraphs (e)(4)(i) through (iii) of this section, as applicable, and the liquidity risk standard in paragraphs (e)(7)(i) and (ii) of this section; and (B) Shall be of high quality and sufficiently liquid to allow the covered clearing agency to meet its current and projected operating expenses under a range of scenarios, including in adverse market conditions; and (iii) Maintaining a viable plan, approved by the board of directors and updated at least annually, for raising additional equity should its equity fall close to or below the amount required under paragraph (e)(15)(ii) of this section.

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This plan should be approved by the board of directors and updated regularly.	

General business risk and capital planning overview

The capital management strategy of FICC's parent company, DTCC, allows it to effectively identify, monitor, and manage the general business risks for each of its subsidiaries, including FICC, as well as for the DTCC group as a whole. DTCC considers those general business risks, as applicable to FICC, to include potential impairment to FICC's financial position that could result in a loss that consequently is charged against capital. The potential for financial impairment could be affected by a variety of factors, including, but not limited to, an unexpected downturn in business volumes or in the economic cycle; external market events with adverse systemic consequences; competitive forces, such as new market participants in the clearance and settlement space; changes in regulatory requirements that may adversely impact FICC and/or FICC's participants; unexpectedly large operating expenses; and operational or cyber risk events.

Capital management is a critical function that supports FICC and other DTCC subsidiaries. To serve our clients and the financial markets, DTCC, the parent company of FICC, holds capital to meet the following critical objectives:

- Loss absorption from general business risks – capital acts as a financial cushion and allows institutions to continue operating during periods when operating losses or other adverse financial results are experienced; It can also reduce the possibility that losses may need to be allocated to members or imposed on investors of DTCC's securities.

- Market confidence – capital provides a measure of assurance to the public and other market participants that an institution will continue to provide financial services even when losses have been incurred, thereby helping to maintain confidence in the financial system.

- Operations and growth – capital supports business operations and strategic growth.

Management seeks to maintain enough capital for FICC to achieve its operating and strategic objectives by monitoring the impact of financial performance on equity, managing leverage so that FICC has a robust balance sheet and maintaining strong credit ratings.

Management also seeks to hold capital for loss absorption so that DTCC can continue operations and services if general business risk related losses were to materialize. DTCC determines how much capital is needed for loss absorption for FICC by:

(1) identifying risks that can threaten FICC's ability to operate that a) are not related to a member default, and b) not covered by financial resources designated to mitigate credit or liquidity risk,

(2) quantifying the potential losses that could arise from each identified risk, and

(3) aggregating the various potential loss estimates into a single loss absorbing capital requirement.

FICC meets the requirements under CCAS 17ad-22(e)(15) by holding enough capital to protect against general business risk related losses so that FICC can continue providing services as an ongoing concern if general business risk related losses were to materialize. CCAS 17ad-22(e)(15) also requires FICC to maintain a viable plan to replenish financial resources, including equity, should those resources fall close to or below the amount being held in compliance with regulatory requirements.

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Monitoring and Managing Capital

DTCC comprehensively monitors and manages FICC's capital adequacy through a disciplined approach to financial planning and management. This approach focuses on the ability of FICC to generate the financial resources needed to support its business, meet its regulatory capital requirements, and maintain its financial resiliency. The process uses a forward-looking view of expected cash flow and net income, and projects FICC's ability to generate the required level of earnings and cash flows to maintain sufficient financial resources and support overall business strategies.

Key aspects of this approach include the annual budget process, business line performance reviews, and the development of a three-year long-range financial plan, for the overall DTCC enterprise and each operating subsidiary, including FICC. The detailed and comprehensive nature of the annual budget process and related business plans, coupled with the monthly frequency of the ongoing business reviews, allows FICC to monitor profitability and quickly identify relevant events and emerging trends, and to assess their potential financial impact. Based on this information, management can take appropriate tactical and strategic measures to minimize business risk. These measures may include, among others, making changes to existing products and services, introducing new products or services, reprioritizing planned or ongoing projects and reallocating resources accordingly, taking cost-reduction measures, modifying fee structures, and adjusting the level of capital and liquid net assets that DTCC maintains at FICC.

This robust assessment and monitoring of capital needs occurs under the oversight of DTCC's Board of Directors. FICC's capital adequacy and financial condition are reviewed on an ongoing basis by DTCC's Chief Financial Office and reported to the Board regularly. The Chief Financial Office may also develop an annual capital plan, as needed, to specifically assess the impact of any future events or trends on FICC's capital position for CCAS 17ad-22(e)(15).

Liquid net assets funded by equity

The Clearing Agency Policy on Capital Requirements ("Capital Policy")[78] and the Clearing Agency Capital Replenishment Plan ("Replenishment Plan")[79] are maintained by the DTCC Treasury Group and approved by the SEC. The Capital Policy sets forth the manner in which FICC identifies, monitors, and manages its general business risk with respect to the requirement to hold sufficient LNA funded by equity to cover potential general business losses, so FICC can continue operations and services if such losses materialize. The Replenishment Plan provides a viable plan for the replenishment of capital for FICC should its equity fall close to or below the amount being held as its "Total Capital Requirement" pursuant to the Capital Policy.

FICC determines its Total Capital Requirement, or the amount of liquid net assets funded by equity that it maintains, as the sum of two components:

(i) The general business risk capital requirement, which is the greatest of three separate calculations:

a) An amount based on FICC's general business risk profile ("Risk-Based Capital Requirement"), where FICC's general business risk profile consists of the specific business risks of potential impairment of FICC's financial position that it may face (including, for example, business and profit/loss risks, operational risks and investment risks,), and that would be sufficient to cover losses so FICC can continue operations and services if those losses materialize.

[78] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

[79] FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

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b) An amount based on the estimated time to execute a recovery or orderly wind-down of the critical operations of FICC ("Recovery/Wind-Down Capital Requirement"), which is determined annually by the Board as the greater of (i) the estimated amount sufficient to ensure a recovery of critical operations and services of FICC; and (ii) the projected amount to facilitate an orderly wind-down of FICC's critical operations.

c) An amount based on an analysis of FICC's estimated operating expenses ("Operating Expense Capital Requirement"), which is determined as the greater of (i) a retrospective estimate of operating expenses based on six times the average monthly operating expense over the prior twelve month period; and (ii) a prospective estimate of six (6) months of operating expenses based on forecasted expense data.

(ii) A "Corporate Contribution" amount which is defined in and may be applied pursuant to, FICC Rule 4. This amount is equal to 50 percent of FICC's General Business Risk capital requirement. According to FICC Rule 4, FICC may voluntarily allocate an amount greater than 50 percent of FICC's General Business Risk capital requirement to address an unsatisfied loss or liability if the Board believes it to be prudent due to existing circumstances. The use and application of the Corporate Contribution, as part of FICC's loss waterfall to address unsatisfied losses, is discussed under Principle 13 (Participant-default Rules and Procedures) above. These resources are held in addition to the LNA funded by equity held by FICC as its General Business Risk capital requirement.

In addition to the Total Capital Requirement, a buffer amount of capital is held to protect resources available to address general business risks. The amount of buffer that FICC holds is periodically reassessed and determined according to various factors, including the volatility of FICC's net income, liquid cash resources, liquid net assets and estimates of potential losses from general business risks.

In addition to the calculation of the Total Capital Requirement described above, DTCC quantifies its own internal view of capital requirements. This view estimates capital requirements for each major business, including FICC, on a consolidated basis across core risk categories such as: business risk, market risk, investment risk, credit risk, and operational risk. Management believes that developing its own risk-based capital view for DTCC's total business portfolio is an important assessment tool to complement regulatory mandates. This expanded approach enables DTCC to more fully assess its risk-based capital needs, and the capital needs of FICC, under adverse economic scenarios resulting from potential market downturns or contractions.

FICC's liquid net assets are conservatively invested pursuant to the Clearing Agency Investment Policy to maximize liquidity and mitigate investment, credit, and market risks. The Clearing Agency Investment Policy is described in more detail under Principle 16 (Custody and Investment Risks) below. The Clearing Agency Investment Policy is reviewed by FICC's Board annually.

Plans to raise additional capital

FICC's comprehensive capital plan, as described above, is designed, in part, to project FICC's ability to generate and maintain the required level of capital needed to manage and protect against business risks. FICC has developed a viable plan to replenish its financial resources through various replenishment tools, as set forth in that plan.

The replenishment plan is triggered when (i) equity held by FICC is at or below an amount equal to its Total Capital Requirement (as described above) plus the equivalent of one-month of FICC's operating expenses and (ii) DTCC's Board of Directors determines that the replenishment plan should be implemented.

Each of the replenishment tools may serve different functions, as follows: (1) bridge financing, which would provide immediate financing, and would be considered an initial step in implementation of the replenishment plan; or (2) capital replenishment, which would provide FICC with the required additional equity on a longer-

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term basis. The replenishment tools would include either actions taken by DTCC to raise capital, which would be contributed to FICC, or actions taken by FICC to increase capital.

Actions that may be taken by DTCC include contributing resources from existing cash, borrowing under an existing line of credit, or making a claim for insurance proceeds, if applicable. DTCC could also seek to fund a capital contribution to FICC by authorizing, issuing and selling shares of common stock of DTCC to its participant shareholders, or via a capital markets transaction, such as the issuance or sale of preferred stock by DTCC, or through strategic actions, such as the sale or divesture of assets or businesses. The actions FICC could take to increase capital include increasing fees for services, or when appropriate, reducing expenses.

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Principle 16: Custody and investment risks; CCAS 17ad-22(e)(16)

Principle 16: Custody and investment risks	**CCAS 17ad-22(e)(16)**
An FMI should safeguard its own and its participants' assets and minimise the risk of loss on and delay in access to these assets. An FMI's investments should be in instruments with minimal credit, market, and liquidity risks. ***Key consideration 1***: An FMI should hold its own and its participants' assets at supervised and regulated entities that have robust accounting practices, safekeeping procedures, and internal controls that fully protect these assets. ***Key consideration 2***: An FMI should have prompt access to its assets and the assets provided by participants, when required. ***Key consideration 3***: An FMI should evaluate and understand its exposures to its custodian banks, taking into account the full scope of its relationships with each. ***Key consideration 4***: An FMI's investment strategy should be consistent with its overall risk-management strategy and fully disclosed to its participants, and investments should be secured by, or be claims on, high-quality obligors. These investments should allow for quick liquidation with little, if any, adverse price effect.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (16) Safeguard the covered clearing agency's own and its participants' assets, minimize the risk of loss and delay in access to these assets, and invest such assets in instruments with minimal credit, market, and liquidity risks.

Introduction

FICC safeguards its own assets (cash, cash equivalents, and marketable investment securities), and Clearing Fund securities and cash, in a well-established structure designed to minimize the risk of loss of and delay in access to these assets. FICC investments are subject to the Clearing Agency Investment Policy (which has been filed with and approved by the SEC),[80] as it applies to FICC. Under these requirements, FICC may only invest in instruments with minimal credit, market and liquidity risks, as further described below. Clearing Fund securities and cash, and proprietary assets are maintained and accounted for separately on FICC's books and records.

Maintenance of Eligible Clearing Fund Securities

As described under Principle 5 (Collateral), Members may satisfy a portion of their required Clearing Fund deposit through the pledge of Eligible Clearing Fund Securities. The Eligible Clearing Fund Securities are held at FICC's clearing bank and Eligible Clearing Fund Securities that are deposited to satisfy Segregated

[80] The Clearing Agency Investment Policy addresses the investment of funds of DTCC's clearing agency subsidiaries, including NSCC, FICC and DTC. FICC rule filings and Advance Notices are available on the SEC website and the DTCC website and are published in the Federal Register.

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Customer Margin Requirements of GSD are held in a separate account at FICC's clearing bank to ensure the legal segregation of these deposits, as provided for under Section 1a of GSD Rule 4.

FICC invests cash safely under the Clearing Agency Investment Policy

FICC holds proprietary general corporate funds cash (which includes cash LNA funded by equity) as well as Clearing Fund cash for each Division's Clearing Fund.

As a key liquidity resource, Clearing Fund cash must be available to complete daily settlement as needed. It may be invested in readily available liquid investments in accordance with the Clearing Agency Investment Policy as described below. As provided for in the Clearing Agency Investment Policy and in Section 1a of GSD Rule 4, Segregated Customer Margin cash deposits are only held in an account of FICC at a bank within the meaning of the Securities Exchange Act of 1934 that is insured by the Federal Deposit Insurance Corporation, or in a separate deposit account at the Federal Reserve Bank of New York. In addition to Clearing Fund cash, FICC's proprietary general operating funds are invested in accordance with the Clearing Agency Investment Policy.

Investment criteria and monitoring

The Clearing Agency Investment Policy outlines the parameters for FICC's investments including, among other things, permitted investments and limitations on investments. The Clearing Agency Investment Policy is reviewed and approved annually by the FICC Board. DTCC's Treasury group is responsible for the investment of available firm-wide funds, including FICC funds.

Pursuant to the Clearing Agency Investment Policy, Clearing Fund cash and FICC proprietary funds are invested conservatively in order to maximize liquidity and mitigate both credit and market risk. Under the Clearing Agency Investment Policy, assets are held at regulated and creditworthy financial institutions. Clearing Fund cash may be invested in commercial bank deposits and FICC's FRBNY account. As noted above, Segregated Customer Margin cash deposits are invested in a manner to ensure the legal segregation of these deposits as provided for under Section 1a of GSD Rule 4. FICC's proprietary cash may be invested in the following types of financial instruments:

- Reverse repurchase agreements;
- Money market mutual funds;
- Commercial bank deposits;
- FICC's FRBNY account; and
- Direct obligations of the U.S. Government.

Investments in reverse repurchase agreements are secured by debt obligations of the U.S. Government or those U.S. Government Agencies guaranteed by the U.S. Government. U.S. Government or Agency securities posted as collateral must have a market value equal to 102 percent or greater of the cash invested. Money market mutual funds must be regulated under Rule 2a-7 of the Investment Company Act of 1940, have a stable Net Asset Value, and a credit rating of AAA/Aaa from recognized rating agencies. FICC's investments are short term and can be readily accessed for liquidity, should the need arise; as such, market risk is minimal.

As part of its risk management process, evaluations are performed at least quarterly on the relative credit standing of the financial institutions with which FICC places funds. Investments are generally placed with financial institutions having a credit rating of BBB+/Baa1 or better from recognized rating agencies, as set forth in the Clearing Agency Investment Policy.

Approved financial institutions are monitored by Counterparty Credit Risk within DTCC's Group Chief Risk Office on a quarterly basis for review of their financial condition, and approved counterparties and

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counterparty credit limits may be revised based on these periodic reviews. Additionally, and based on its ongoing monitoring process, to the extent that CCR determines that a change in a counterparty's financial condition warrants a change in FICC's investment limits with the counterparty, CCR would promptly notify Treasury of any change, so that Treasury can implement any required adjustment to the investment strategy.

An element of CCR's ongoing investment counterparty monitoring process includes the identification of credit risk, a general analysis of the counterparty which includes the review of the audited financial statements (includes available audit opinions) and/or FFEIC call report (assess liquidity resources, earnings trends, capital levels and other relevant information), and review of the counterparty's external credit ratings (if applicable). Beyond the scope of its internal review practices, FICC takes additional comfort from the regulatory framework that governs its investment counterparties (which includes oversight and examination by U.S. banking regulators and the SEC), in evaluating the safety and accessibility of its investments. This includes regulation regarding the protection of customer securities from theft, loss, or misuse, and in the event of insolvency.

Exposure and concentration risk

The acceptable levels of exposure to financial institution counterparties are governed by the Clearing Agency Investment Policy, which establishes the overall framework for the investment of funds, as well as CCR's criteria for determining credit limits to appropriately facilitate diversification and mitigate concentration across counterparties. On a daily basis, DTCC's Treasury group adheres to the policy guidelines and reports investment activity compared to established credit limits for each of the respective counterparties. This reporting is distributed to management, CCR, and Market and Liquidity Risk.

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Principle 17: Operational risk; CCAS 17ad-22(e)(17)

Principle 17: Operational risk	**CCAS 17ad-22(e)(17)**
An FMI should identify the plausible sources of operational risk, both internal and external, and mitigate their impact through the use of appropriate systems, policies, procedures, and controls. Systems should be designed to ensure a high degree of security and operational reliability and should have adequate, scalable capacity. Business continuity management should aim for timely recovery of operations and fulfilment of the FMI's obligations, including in the event of a wide-scale or major disruption. ***Key consideration 1***: An FMI should establish a robust operational risk management framework with appropriate systems, policies, procedures, and controls to identify, monitor, and manage operational risks. ***Key consideration 2***: An FMI's board of directors should clearly define the roles and responsibilities for addressing operational risk and should endorse the FMI's operational risk management framework. Systems, operational policies, procedures, and controls should be reviewed, audited, and tested periodically and after significant changes. ***Key consideration 3***: An FMI should have clearly defined operational reliability objectives and should have policies in place that are designed to achieve those objectives. ***Key consideration 4***: An FMI should ensure that it has scalable capacity adequate to handle increasing stress volumes and to achieve its service-level objectives. ***Key consideration 5***: An FMI should have comprehensive physical and information security policies that address all potential vulnerabilities and threats. ***Key consideration 6***: An FMI should have a business continuity plan that addresses events posing a significant risk of disrupting operations, including events that could cause a wide-scale or major disruption. The plan should incorporate the use of a secondary site and should be designed to ensure that critical information technology (IT) systems can resume operations within two hours following disruptive events. The plan should be designed to enable the FMI to complete settlement by the end of the day of the disruption, even in case	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (17) Manage the covered clearing agency's operational risks by: (i) Identifying the plausible sources of operational risk, both internal and external, and mitigating their impact through the use of appropriate systems, policies, procedures, and controls; (ii) Ensuring that systems have a high degree of security, resiliency, operational reliability, and adequate, scalable capacity; and (iii) Establishing and maintaining a business continuity plan that addresses events posing a significant risk of disrupting operations.

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of extreme circumstances. The FMI should regularly test these arrangements. ***Key consideration 7***: An FMI should identify, monitor, and manage the risks that key participants, other FMIs, and service and utility providers might pose to its operations. In addition, an FMI should identify, monitor, and manage the risks its operations might pose to other FMIs.	

Operational risk management framework and identification of risks

The Clearing Agency Operational Risk Management Framework and DTCC Corporate Risk Management Policy describe the framework employed by FICC to identify, analyze, mitigate, report and escalate operational risks. This is done through internal incident data collection, operational risk assessments, scenario analysis, testing by internal and external parties, metrics and key risk indicators, issue tracking and reporting.

Operational Risk ("Op Risk") maintains an operational risk management program that enables the identification, assessment, management, monitoring and reporting of the risks encountered in the day-to-day business of DTCC's clearing agency subsidiaries, including FICC. The program establishes DTCC's overall approach for identifying internal and external sources of risks, assessing the implications, prioritizing and developing plans to address such risks so that they can be remediated to the extent practicable.

Each business unit implements an operational risk management program and related elements in accordance with the Clearing Agency Operational Risk Management Framework and the related supporting procedure documents. At least annually, the policies and procedures are reviewed and updated as needed; the updated policies and procedures are then reviewed and approved by the MRC and the BRC.

The operational risks that DTCC has identified in its Corporate Risk Management Policy are:

- Human Capital Risk
- Processing & Operations Risk
- Cybersecurity and Information Protection Risk
- Information Technology Risk
- Financial Risk
- Legal & Regulatory Compliance Risk
- Business Continuity Risk
- Model Risk
- Third Party Risk

To determine how to address these risks, the firm regularly conducts operational risk assessments, which include a thorough analysis of FICC's business functions and how each of these risk families may be implicated in the business operations. The data collected informs the organization's business planning and helps guide decision making with respect to the need for additional investments that may further reduce risk or the readjustment of risk tolerance. Furthermore, to enhance the current risk framework, DTCC has processes to enable the enterprise to identify and consider future scenarios that may materially impact an FICC business to an extent that threatens everyday viability of the business/organization. These activities allow the organization to identify possible vulnerabilities in critical functions and critical external

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dependencies across the enterprise and understand more fully the risks and the potential impact of stressful events and circumstances on FICC's financial condition.

Management of operational risk

DTCC has several ways in which it identifies and manages the operational risks of its Clearing Agencies, including FICC. Each such risk is underscored by the "three lines of defense" strategy:

- The first line of defense is the business and functional units, including Product Management, Operations Management, Information Technology and other areas critical to daily operation and functioning. Their mandate is to proactively manage risk.

- The second line of defense is comprised of DTCC's control functions, which support the Clearing Agencies, including FICC. This includes those areas that fall under the Group Chief Risk Officer's purview, Legal, Privacy and Compliance. Their mandate is to provide advice and guidance to the first line of defense in adhering to established risk standards and to monitor compliance with those standards.

- The third line of defense is Internal Audit. Internal Audit's mission is to assess the overall control environment, risk management, and the control framework and in doing so, raise awareness of control risk, and promote changes for improving governance processes.

Op Risk is an independent risk management function that operates as part of the "second line of defense." It is primarily responsible for:

- the company-wide establishment and implementation of the operational risk framework;

- ongoing monitoring of company-wide adherence to the operational risk framework by the business units;

- analysis and reporting to senior management and the Board, on a company-wide basis, the operational risk exposures, including the timely escalation of risk exposures identified under the framework; and

- supporting the business units with respect to the ongoing roll-out and application of the framework.

Policies, processes and controls

DTCC's operational risk framework is, in part, designed in accordance with industry standards and best practices, including the operational risk measurement rules under Basel II[81] and the standards under CCAS, and it has evolved based on the changing needs of FICC businesses and applicable regulatory guidance. These standards are incorporated into the methodology and tools used to identify, assess, manage and report on operational risks.

FICC also complies with the requirements of Regulation Systems Compliance and Integrity ("Reg SCI"), adopted by the SEC under the Exchange Act. These requirements include establishing, maintaining and enforcing written policies and procedures reasonably designed to ensure that FICC's critical systems have levels of capacity, integrity, resiliency, availability and security adequate to maintain FICC's operational capability and promote the maintenance of fair and orderly markets.

With respect to human capital and people risk, DTCC defines job responsibilities to recruit qualified talent and compensate them competitively based on market data and internal equity. Personnel have access to

[81] http://www.bis.org/publ/bcbs195.htm.

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a range of in-house, online and external learning offerings and programs to support risk management capabilities, professional/leadership development and business/functional knowledge. DTCC regularly tracks voluntary attrition, conducts exit interviews, and takes appropriate action to mitigate the impact of turnover. Succession and replacement plans are in place to address key-person risk for Managing Directors and other key roles.

Additionally, DTCC sets parameters of acceptable conduct that includes, but is not limited to, fraud prevention.

The Enterprise Program Management Office ("EPMO") is a centralized function that plays a strategic and governance oversight role in managing enterprise wide programs ("Enterprise Programs"), standard programs ("Standard Programs"), and initiatives ("Initiatives") ("collectively, EPMO Portfolio"), in partnership with Portfolio Management. Part of the governance process involves an assessment of various risks, including operational risk, by subject matter experts and informs decision making.

Roles, responsibilities and framework; review and audit

As described under Principle 3 (Framework for the comprehensive management of risks), the Board has delegated to management the responsibility for the day-to-day management of risk, as articulated through individual risk tolerance statements. The risk tolerance statements are provided to the BRC for its approval at least annually. The Board requires management to distribute regular risk tolerance reporting to the Board to provide assurance of management's effective oversight of such risks.

To fulfill the Board and management's risk oversight responsibilities, a risk management framework has been designed to provide information on DTCC's key risks (including those applicable to FICC) and the processes used to identify, assess, measure, monitor, mitigate and report those risks. The risk management framework utilizes two primary risk management processes to consolidate reporting: (1) operational risk assessments created for each business and functional area, and (2) the risk tolerance reporting for each risk family. In addition, Op Risk periodically reports on the status of operational risk across the enterprise, providing an overview of the key risks and associated mitigation plans, if applicable.

Internal Audit performs independent testing and reviews the effective implementation of the operational risk framework by both Op Risk and the business units. The audit frequencies range from 12 to 48 months, based on periodic assessments of inherent risk and control environments. In addition, as a registered clearing agency and designated SIFMU, FICC is subject to continuous monitoring by its regulators, which includes review and monitoring of the effectiveness of FICC's operational risk management policies and procedures.

Operational reliability

FICC monitors operational reliability through multiple processes, including an annual risk assessment, periodic risk and controls assessment, incident reporting, monitoring for lifecycle processing, system availability and data reporting; a metrics review of incident-related key risk indicators that have been identified for event processing, settlement processing, data reporting, system availability and other key processes. These monitoring activities are performed regularly and evaluated by management as required, with appropriate procedures and controls applied as needed.

DTCC's System Delivery Process ("SDP") combines both agile and waterfall methodologies and includes processes, toolsets, guidance and practices that all technology teams follow and adhere to. The process is based on key controls that are required to deliver systems to DTCC's clients with high efficiency and reduced risk. A Continuous Improvement Review Team looks for opportunities to improve the processes in terms of control coverage, further simplification, and user experience. Delivery process assurance provides the delivery teams with a monthly view of control adherence for each project following the system delivery process.

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With respect to the management of releasing new or updated technology, DTCC uses the Information Technology Infrastructure Library ("ITIL") best practice framework for IT service management and operations of its enterprise infrastructure. The ITIL Release Management processes plan, schedule and control the movement of releases to test and production environments.

Further, FICC uses internal Service Level Agreements ("SLAs") across the enterprise between the service provider (IT) and their customers (Product Management). SLAs identify the critical business services and underlying IT services provided to fulfill them, their service performance targets, and the metrics associated with these services. Any SLA breaches are noted, escalated as necessary, and timely reports, with appropriate detail, are distributed to various levels within the organization including senior management. These measures and metrics are re-evaluated on an annual basis as part of FICC's continuous improvement process.

Capacity management

DTCC's capacity management oversight is provided through the Enterprise Testing Engineering Performance ("ETE") and Capacity Planning functions. These teams provide a governance structure to ensure the requirements of ordinary course business are met, and for mitigating events that may arise as a result of extreme market events. With respect to certain systems that are designated as "SCI systems", FICC must comply with the requirements of Regulation Systems Compliance and Integrity ("Reg SCI"), including certain system capacity testing.

The ETE is responsible for Performance Testing and Reg SCI capacity stress testing. Among the area's responsibilities are:

- performance testing that happens in the QA test environments at the request of the business;
- capacity stress testing in test environments for SCI systems only;
- reporting defects and recommendations discovered during Performance Testing and Reg SCI capacity stress testing; and
- reviewing and recommending tools to analyze application and system performance in the QA Environment.

The ETE performs capacity stress tests periodically to demonstrate that the processing of SCI systems meets or exceeds the applicable business line's extreme market condition requirement.

The Capacity Planning team utilizes trend analyses, augmented by annual reviews of business application changes, to forecast the upcoming year's capacity demand for both processor and storage capacity for all data centers. To arrive at projected resource usage, the annual reviews encompass

- business input (i.e., growth, new product lines and new applications, etc.);
- application throughput;
- current capacity, scalability, and SLAs; and
- capacity utilization, capacity forecasts.

The function's design gives priority to and addresses the largest business areas and the associated production processing environments.

Further, the Capacity Planning team is specifically responsible for forecasting and monitoring infrastructure capacity and utilization, on DTCC mainframe, distributed systems, and network. This work results in daily, weekly and monthly usage reports for review by Information Technology Management, as well as capacity management information and metrics provided to the Information Technology Governance Committee quarterly. These updates are presented for discussion, no less than two times per year.

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Cloud capacity management

For cloud instances, consumption and user services are separately monitored and adjusted as needed, and reporting is provided to the IMC on a quarterly basis.

Physical security

Global Security Management utilizes a comprehensive security assessment approach as part of an overall program aimed at developing and maintaining a consistent, structured and integrated methodology for identifying, monitoring, managing and reporting on security risks across physical sites and locations throughout the organization.

The process consists of several components, which include: (a) a Security Vulnerability Assessment checklist that is "risk-specific" and facilitates the analysis and reporting of risk information using a common language, and (b) quantitative information, including internal theft events and security breaches, area threat analysis (from a Federal and local perspective) and local area crime statistics to ascertain the effectiveness of current security control structures.

There is no single relevant international, national or industry-level standard for physical security by which guidance can be solely taken. Accordingly, select guidance is taken from a number of applicable resources.

Information security

The Cyber Security Risk Office (CSRO) governs information security across DTCC by establishing and maintaining the enterprise cyber security management framework, including policies, standards, governance structures, and oversight of controls designed to protect the confidentiality, integrity, and availability of DTCC information assets.

- CSRO defines and oversees cyber security roles, responsibilities, and accountabilities using the Three Lines of Defense model and provides strategic direction through the Cyber Security Steering Committee (CSSC), which reviews and approves updates to cyber security policies and control standards. Annual policy reviews are conducted by the Management Risk Committee (MRC) and the Board Risk Committee (BRC) to ensure continued alignment with regulatory and industry requirements.

- CSRO requires all employees and contingent workers to complete annual and periodic cyber security training, including compliance tracking, simulated phishing exercises, and education on applicable policies and regulations. CSRO also monitors compliance with legal and regulatory security requirements, engages with regulators and industry peers, and oversees a comprehensive threat and vulnerability management program that leverages internal and external intelligence to support risk based remediation and informed cyber security investment decisions.

The DTCC Cyber Security Policy establishes an enterprise wide framework for identifying, assessing, and managing cyber security risks and aligns with internationally recognized and regulatory standards, including ISO/IEC 27001, FFIEC guidance, the NIST Cybersecurity Framework, and the Cyber Risk Institute Profile. Supporting standards and procedures operationalize the Policy by enabling consistent execution, accountability, and repeatability across key cyber security domains, including but not limited to risk management, security operations, third party oversight, incident response, resilience, and metrics reporting.

Business continuity

DTCC is concerned with the governance and implementation of proactive and reactive measures which ensure that enterprise and business functions have resilience and recovery capabilities to continue, should a serious event occur. This is done through the: (1) integration and alignment with the various risk functions throughout the organization and sector, (2) development of guidance and standards relating to business continuity (BC), crisis management, and location risk, (3) monitoring compliance and (4) promotion of

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awareness and education. DTCC Global Business Continuity and Resilience ("BCR") Policy and Control Standards ("BCR Policy") establishes requirements for how DTCC as a whole, including FICC, will effect and maintain controls that address defined threats which, if not otherwise implemented, could result in a high level of risk to the continuity of enterprise operations. This policy defines the governance structure, high-level roles and responsibilities and framework for DTCC's BC process.

Given the nature and breadth of significant business disruptive events, BC controls are aligned to the global, regional, site, service, business and support levels, including alignment with each of DTCC's clearing agency subsidiaries. DTCC provides a set of core business processes for each clearing agency subsidiary, including FICC. The business processes have a relative importance based on the service they provide to the financial sector. The ability to deploy sensible and balanced controls, as well as to triage recovery efforts, is based on this relative importance. FICC falls into the top rankings with a goal of two hours for resumption, and a recovery point objective of 30 seconds. The technology currently permits asynchronous replication sufficiently timed to ensure that there is no more than a 30-second variance in the data stored across regions. Additionally, pursuant to its rules, FICC has the right to disconnect a participant from its systems upon a threat of significant impact to its systems.

BC plans are updated annually and include amendments, as appropriate, from the business impact analysis ("BIA"), which is conducted annually and ratified by business line managers and senior management. The BIA validates the criticality of business areas to ensure the appropriate level of controls.

In support of FICC's business, DTCC has data centers in multiple regions. Production processing is spread across these data centers; specifically, primary and secondary sites are separated by approximately 1,500 miles. The out-of-region site contains additional asynchronously replicated copies of production data. The asynchronous nature of the replication to this site is due to the limits inherent in current-day technology. The technology currently permits asynchronous replication sufficiently timed to ensure that there is no more than a two-minute variance in the data stored at the in-region sites and the data stored out-of-region.

In addition, all critical clearance and settlement transactions utilize private non-Internet networks such as the Securely Managed and Reliable Technology ("SMART"), BT Radianz and the NYSE Secure Financial Transaction Infrastructure ("SFTI") networks. SMART is the primary network that provides connectivity between DTCC and its customers. SMART is a seamless, end-to-end, securely managed private communications system encompassing a geographically dispersed complex of processing centers, communications networks and control facilities. Each element of SMART is highly secure, engineered with multiple independent levels of redundancy. External traffic is split between these connections, which are always active with each having sufficient capacity for the entire traffic volume.

The BC plans enable DTCC to effectively and efficiently assess the impact of the disruption, organize communication and decision making, and coordinate the company's response effort. The BCR Policy includes definitions of the types of communications necessary, the methodology and template text to execute the plans. Moreover, crisis events are managed through established crisis management procedures, which are implemented to support an effective enterprise response.

In addition, individual business lines, support units, and site assessment teams (SAT) are responsible for working with DTCC Security and resiliency personnel and include a cross-section of individuals from various departments throughout the organization, including senior management (decision making), facilities management (locations and safety), human resources (personnel issues and travel), marketing (media contact), finance/accounting (funds disbursement and financial decisions), operations (settlement), and risk management (collection of Participants Fund), among others. In a crisis, DTCC Security and resiliency personnel will immediately disseminate a message to internal and external audiences, as appropriate, to ensure consistent communication.

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Review and testing

Contingency arrangements are reviewed throughout the year but at a minimum once annually per business line or support unit. DTCC also conducts facility specific work area recovery exercises throughout the year, but at a minimum once annually.

DTCC works with its customers and other industry infrastructure organizations to discuss the industry's BC preparations and DTCC's expectations regarding customers' own BC capabilities. Furthermore, DTCC conducts regular "tabletop" exercises to validate how its command teams would respond in the event of a catastrophic loss of the company's headquarters or other locations. As with real events, these various exercises involve debriefing sessions and checklists that are used to identify opportunities for improvement.

DTCC also engages in the following:

- *Ensuring connectivity*—DTCC has telecommunication networks with major participants to ensure continued connectivity for all of DTCC's data processing. This includes the participant's primary locations as well as the participant's backup locations.

- *Connectivity testing*—Each year, DTCC's clearing agency subsidiaries issue Important Notices that establish telecommunications connectivity requirements for major Members and customers. With respect to each Division, FICC requires its larger-volume Members to test their connectivity with FICC at least once a year. This includes testing connectivity from the Member's primary and backup locations to DTCC data processing locations.

- *Reg SCI testing* – As a designated Reg SCI entity, FICC performs functional and operational testing with clients that are, taken as a whole, the minimum necessary for the maintenance of fair and orderly markets in the event of the activation of FICC's BC and disaster recovery plans.

- *Industry testing*—DTCC participates in tests conducted by industry partners. In addition, DTCC representatives participate in several industry committees that focus on BC issues at the industry level.

Risks to the FMI's operations and risks posed to other FMIs

As discussed more fully in Principle 3 (Framework for the comprehensive management of risks), given its central role in the U.S. securities markets, FICC potentially poses a number of risks to other entities, while it is also exposed to risks that may originate from other entities.

FICC's material service providers are subject to a comprehensive third-party review and vetting process. FICC assesses its material service provider risks periodically, throughout the lifecycle of a third-party service provider relationship. This includes protocols for reporting, escalation and remediation of third-party risk and potential exposures. Mitigants may take the form of contractual protections, or additional or backup providers where deemed appropriate and feasible. For example, FICC utilizes price feeds from multiple third parties as an additional risk mitigant, and further employs price review and data scrubbing procedures to minimize the risk of using inaccurate prices. In addition, in compliance with SEC Rule 17ad-25(i), FICC has established policies and procedures to manage its risks from relationships with service providers for core services (CSPs). These policies and procedures set forth the obligations of senior management and the Board (which the Board has delegated to the Board Risk Committee). Senior management evaluates risks related to agreements with CSPs and would report to the BRC for its evaluation any action taken by senior management to remedy significant deterioration in performance with a CSP. The BRC reviews and approves agreements that establish CSP relationships and management's policies and procedures that govern relationships and manage risks with CSPs. FICC's third party risk management processes are also discussed under Principle 3 (Framework for the comprehensive management of risks).

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With regard to risks that FICC might pose to other FMIs (where applicable), FICC has communication and escalation processes, in the event of system issues, to facilitate prompt resolution.

Along with other financial industry organizations, DTCC is an active participant in the Financial Services Sector Coordinating Council for Critical Infrastructure Protection and Homeland Security ("FSSCC"), a private sector group that interfaces with the U.S. Department of Treasury and the Financial and Banking Information Infrastructure Committee on infrastructure protection issues. The FSSCC works to coordinate the financial services industry's initiatives to protect critical financial services infrastructure. The goal is to ensure that these efforts focus on complementary objectives and contribute to achieving the highest possible level of overall industry resiliency.

As noted above, DTCC staff actively participate in industry-wide business continuity testing. Some of these tests include market disruption, cyber security and backup site testing.

DTCC is a member of the Analysis & Resilience Center ("ARC") for Systemic Risk. ARC is a cross-sector organization designed to mitigate systemic risk to the nation's most critical infrastructure from existing and emerging threats.

DTCC is also a member of Sheltered Harbor. Sheltered Harbor is a voluntary industry initiative undertaken by the US financial services sector to enhance the sector's resiliency, and to provide additional protections for consumer account information. Its goal is to extend the industry's capabilities to securely save and restore account data in the event of a loss of operational capability.

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Principle 18: Access and participation requirements; CCAS 17ad-22(e)(18)

Principle 18: Access and participation requirements	**CCAS 17ad-22(e)(18)**
An FMI should have objective, risk-based, and publicly disclosed criteria for participation, which permit fair and open access. ***Key consideration 1***: An FMI should allow for fair and open access to its services, including by direct and, where relevant, indirect participants and other FMIs, based on reasonable risk-related participation requirements. ***Key consideration 2***: An FMI's participation requirements should be justified in terms of the safety and efficiency of the FMI and the markets it serves, be tailored to and commensurate with the FMI's specific risks, and be publicly disclosed. Subject to maintaining acceptable risk control standards, an FMI should endeavor to set requirements that have the least-restrictive impact on access that circumstances permit. ***Key consideration 3***: An FMI should monitor compliance with its participation requirements on an ongoing basis and have clearly defined and publicly disclosed procedures for facilitating the suspension and orderly exit of a participant that breaches, or no longer meets, the participation requirements.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (18) Establish objective, risk-based, and publicly disclosed criteria for participation, which: (i) Permit fair and open access by direct and, where relevant, indirect participants and other financial market utilities; (ii) Require participants to have sufficient financial resources and robust operational capacity to meet obligations arising from participation in the clearing agency; (iii) Monitor compliance with such participation requirements on an ongoing basis; and (iv) When the covered clearing agency provides central counterparty services for transactions in U.S. Treasury securities, (A) Require that any direct participant of such covered clearing agency submit for clearance and settlement all of the eligible secondary market transactions to which such direct participant is a counterparty; (B) Identify and monitor its direct participants' submission of transactions for clearing as required in paragraph (e)(18)(iv)(A) of this section, including how the covered clearing agency would address a failure to submit transactions in accordance with paragraph (e)(18)(iv)(A) of this section; and (C) Ensure that it has appropriate means to facilitate access to clearance and settlement services of all eligible secondary market transactions in U.S. Treasury securities, including those of indirect participants, which policies and procedures the board of directors of such covered clearing agency reviews annually.

FICC's access and participation requirements

With respect to each Division, FICC has established participation criteria and requirements relating to financial resources, creditworthiness and operational capability. These requirements are designed to limit the risks a Member may present to FICC or to each Division's respective membership, while facilitating fair

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and open access by market participants; they are objective, risk-based, and are set forth in each Division's Rules, which are publicly available on the DTCC website.

In general, FICC membership requirements are set forth in GSD Rules 2A (Initial Membership Requirements), 3A (Sponsoring Members and Sponsored Members) and 8 (Agent Clearing Service), and MBSD Rule 2A (Initial Membership Requirements) as well as GSD Rule 3 (Ongoing Membership Requirements) and MBSD Rule 3 (Ongoing Membership Requirements). As indicated in the General Background of FICC in part III of this Disclosure Framework, each Division has several categories of membership.

The participation criteria and requirements that are applicable to GSD Netting Members and MBSD Clearing Members are described below.

Membership application requirements

All applicants for full-service membership must provide FICC with certain information, which is reviewed by FRM to ensure that the applicant has (1) sufficient financial ability to make anticipated contributions to the Clearing Fund and to meet obligations to FICC; (2) either an established business history of a minimum of six months or personnel with sufficient operational background and experience to ensure the ability of the firm to conduct its business with FICC; and (3) appropriate settling and clearing bank arrangements.

1. *Type of Legal Entity*. As set forth in GSD Rule 2A and MBSD Rule 2A, an applicant must be a specified type of legal entity (such as a broker-dealer, bank or other specified entity that, under its regulatory regime, as applicable, may engage in the transactions that will be processed through FICC).

   With respect to GSD, a netting membership applicant must either be (i) a bank or trust company chartered as such under the laws of the United States, or a State thereof, or a bank or trust company established or chartered under the laws of a non-U.S. jurisdiction and participates in GSD through its U.S. branch or agency; (ii) a registered government securities dealer; (iii) a futures commission merchant; (iv) an inter-dealer broker; (v) a government securities issuer; (vi) an insurance company, (vii) a registered clearing agency; (viii) a registered investment company or (ix) an entity may apply to be a foreign member in one of the existing categories if FICC, in its sole discretion, has determined that the entity: (a) has a home country regulator that has entered into a memorandum of understanding with the SEC regarding the sharing or exchange of information and (b) maintains a presence in the United States, either directly or through a suitable agent.

   With respect to MBSD, a clearing membership applicant must either be (i) a bank or trust company chartered as such under the laws of the United States, or a State thereof, or a bank or trust company established or chartered under the laws of a non-U.S. jurisdiction and participates in MBSD through its U.S. branch or agency; (ii) a registered securities dealer; (iii) an inter-dealer broker; (iv) an unregistered investment pool; (v) a government securities issuer; (vi) an insurance company;[82] (vii) a registered clearing agency; (viii) an insured credit union, (ix) a registered investment company or (x) an entity may apply to be a foreign member in one of the existing categories if FICC, in its sole discretion, has determined that the entity: (a) has a home country regulator that has entered into a memorandum of understanding with the SEC regarding the sharing or exchange of information and (b) maintains a presence in the United States, either directly or through a suitable agent.

   Each Division may make its services available to persons in such other categories as FICC may from time to time determine, subject to any requisite regulatory approval.

2. *Financial Responsibility*. FICC sets financial requirements for establishing and continuing membership that are based on the type of legal entity (such as broker/dealers, banks, trust companies, etc.). In addition, GSD Rule 4 (Clearing Fund and Loss Allocation) and MBSD Rule 4 (Clearing Fund and Loss

[82] Specific requirements for insurance companies would need to be defined through a rule filing.

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Allocation) each provide that all Members are required to make a deposit to the Division's Clearing Fund, as applicable, with the amount of each Member's required deposit being determined by FICC in accordance with the formula in Rule 4 of each Division's Rules.

3. *Operational Capability*. FICC will assess its Members' operational capability both in connection with the application process as well as during periodic reviews. FICC confirms that the applicant will be able to satisfactorily communicate with FICC and fulfill anticipated commitments to and to meet the operational requirements of FICC with necessary promptness and accuracy and is able to establish appropriate arrangements to effect obligations to FICC. This includes designating a settling bank that meets FICC's settling bank requirements and designating a clearing bank. Prior to activating membership, FICC applicants are required to conduct testing with FICC (i.e., connectivity testing) to confirm that they are able to transmit files to, and receive files from, FICC.

Finally, all applicants must execute a membership agreement, and submit other required membership application documents, under which they agree, among other things, to be bound by FICC's Rules and meet all anticipated financial obligations to FICC.

Before denying a membership application, FICC must provide the applicant with a concise written statement setting forth the specific grounds under consideration upon which any such denial may be based, and notify the applicant of its right to request a hearing to determine whether the application should be denied.

Ongoing compliance with membership requirements

FICC monitors its Members' ongoing compliance with access criteria by reviewing and analyzing financial and operational information and through ongoing market, news and regulatory monitoring. Pursuant to Rule 3 (Ongoing Membership Requirements) of each Division's Rules, Members are required to provide FICC with regulatory reports and other required financial information in a timely manner. This includes annual audited financial statements and monthly, quarterly, semi-annual, or annual financial reports, depending on the reporting frequency of the Member, and notification of changes in condition. Members are informed by Important Notice as to the due dates for providing this information.[83]

Pursuant to its Rules, FICC may fine Members that fail to timely provide such information, and those fines may be reported to the SEC.

FICC collects information periodically from Members and applicants in order to understand their overall business condition and risk management practices, which may cover operational, credit, legal and other risks, and financial standing. Due diligence topics may include but are not limited to capital management, liquidity management, risk controls, climate risk exposure, etc.

Through FRM, FICC reviews this information to confirm that applicants and Members meet their applicable minimum financial requirements. FRM monitors Clearing Fund requirements throughout the day, and Operations staff monitors the operational capability of FICC Members, as applicable.

FRM utilizes an internal credit risk rating matrix (referred to as the "CRRM") to risk rate U.S. bank, U.S. broker/dealer and foreign bank Members as part of the ongoing monitoring of Member firms. The CRRM rating is produced systematically from financial data contained in the Members' regulatory or financial reports and from qualitative assessments from Counterparty Credit Risk analysts of FRM. The rating is expressed on a defined scale that categorizes Members according to their relative creditworthiness. The key elements used to set a Member's internal credit rating focus on certain quantitative and qualitative factors, as defined by the CRRM. The CCR analyst considers whether other factors, not captured by the CRRM, warrant an override to downgrade the model-generated rating. The resulting rating determines the

[83] FRM may receive FOCUS reports automatically from the Member's designated examining authority, and CALL reports automatically from a third-party vendor.

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level of financial review that will be performed on that Member and may impact the Member's Clearing Fund requirement.

Members with a weaker internal credit rating are automatically placed on FICC's Watch List. Members on the Watch List may be monitored more closely than those with a stronger credit rating.

Members can be added to the Watch List for a variety of reasons, including concerns about legal and/or regulatory issues, changes to organizational structure, a Member's liquidity arrangements, news reports relating to the Member, concern regarding a Member's ability to maintain membership standards, etc. A Member can be added to the Watch List regardless of its CRRM rating. Members not subject to monitoring through the CRRM may be manually placed on the Watch List, as FICC deems necessary and appropriate. FICC also has the authority under each Division's Rule 3 (Ongoing Membership Requirements) to seek adequate assurances from a Member if FICC determines that such adequate assurances are warranted based on its review of such Member's ability to meet its financial obligations.

Suspension and orderly exit of Members

Each Division's Rules contain provisions that facilitate the suspension and orderly exit of Members that no longer meet the participation requirements:

- A Member may voluntarily retire from membership by notifying FICC in writing, as provided in GSD Rule 3 and MBSD Rule 3 as applicable. Notwithstanding any such notification, the Member remains obligated to satisfy any open obligations and liabilities arising out of its membership existing at the time of its notification. A retiring Member will typically designate a retirement date, by which all open activity will be closed out and new activity will no longer be submitted or accepted. FICC issues an Important Notice to Members when a Member voluntarily retires from membership.

- Under GSD Rule 21A (Wind-Down of a Netting Member) and MBSD Rule 15 (Wind-Down of a Member), FICC can facilitate the orderly wind-down of a Member's business when FICC is notified by the Member that it intends to wind-down its business. Under this Rule, FICC may impose conditions on, or take actions with respect to, the "Wind-Down Member" in order to mitigate risk, including (but not limited to) permitting the Wind-Down Member to submit only transactions that serve to support the wind-down, restricting or modifying the Wind-Down Member's use of any or all of FICC's services and requiring the Wind-Down Member to post increased Clearing Fund deposits. FICC will notify the Member, all other Members (via Important Notice) and the SEC when it implements this Rule and determines that the affected Member is a "Wind-Down Member."

- Under GSD Rule 21 (Restrictions on Access to Services) and MBSD Rule 14 (Restrictions on Access to Services), as applicable, FICC has the right to restrict a Member's access to services, by limiting or excluding the Member's participation in one or more classes of transactions or services, or by ceasing to act for the Member generally. Under GSD Rule 22 (Insolvency of a Member) and MBSD Rule 16 (Insolvency of a Member), FICC may cease to act for a Member who has become insolvent, as defined in these Rules. The circumstances under which FICC may cease to act are clearly set forth in the Rules, and include a determination by the Board of Directors or appropriate Board Committee that the Member is no longer in compliance with the membership requirements. If FICC ceases to act for a Member, the Rules provide that FICC must notify the affected Member, other Members (via Important Notice) and the SEC of FICC's action. The Member may request a hearing to consider the decision, which, depending upon the circumstances on which the action was based (such as insolvency or settlement default), may be after the cease to act or suspension takes effect.

FICC's Rules contain provisions that facilitate the disconnection or suspension of activities of a participant with a systems disruption that could have an adverse impact on DTCC's systems:

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- Under GSD Rule 50A (Participant System Disruption) and MBSD Rule 40A (Participant System Disruption), FICC has the ability to disconnect a participant from the DTCC system, suspend the receipt and/or transmission of files or communications to/from a participant to the DTCC system or take other actions that FICC considers appropriate to help mitigate risk when there is a systems disruption of a participant that could have a significant impact on DTCC's systems. The circumstances under which FICC may do so are set forth in the Rules, and include a determination by appropriate executive officers that there is a major event that is likely to have a significant impact on the DTCC systems which determination is then ratified, modified or rescinded by the Board.

GSD Sponsored Service

GSD offers a program under Rule 3A that allows Netting Members to act as "Sponsoring Members" for purposes of sponsoring certain institutional firms into GSD membership. Netting Members are eligible to become Sponsoring Members if they are Tier One Netting Members, other than Inter-Dealer Broker Netting Members. Under Rule 3A, an entity may become a Sponsored Member if it is sponsored into membership by a Sponsoring Member and has executed and delivered to FICC the necessary onboarding membership agreement.

For operational and administrative purposes, a Sponsoring Member acts as a processing agent on behalf of its Sponsored Members and establishes an omnibus account for all of its Sponsored Members' activity.

The operational support requirements necessary to meet the settlement obligations of Sponsored Members are satisfied through requirements that GSD has placed on the Sponsoring Member. GSD Rule 3A provides that a Sponsored Member shall appoint its Sponsoring Member to act as processing agent with respect to the Sponsored Member's satisfaction of its securities and funds-only settlement obligations. Therefore, the Sponsored Members use their Sponsoring Members' operational capability to satisfy their funds and securities settlement obligations. Sponsoring Members are subject to the same robust operational capability membership requirements as Netting Members.

Similar to the operational support requirements, FICC ultimately relies on the financial resources of the Sponsoring Member to ensure that the funds and securities settlement obligations of its Sponsored Members to FICC will be met if the Sponsored Members, who are principally liable to FICC for such obligations, should fail to meet them for any reason. The Sponsoring Member is unconditionally liable to FICC under the Sponsoring Member Guaranty[84] to satisfy such obligations and failing to satisfy such obligations is grounds for FICC to cease to act for the Sponsoring Member both in its capacity as a Sponsoring Member as well as in its capacity as a Netting Member.

GSD Agent Clearing Service

The Agent Clearing Service is an alternative indirect access model that operates as an agented clearing model and provides indirect participants (referred to as "Executing Firm Customers") with access to central clearing through a Netting Member (referred to as "Agent Clearing Members"). Unlike the Sponsored Service, FICC has no relationship with the Executing Firm Customers and all obligations (i.e., margin and settlement) under the GSD Rules remain with the Agent Clearing Member. Therefore, the Agent Clearing Member acts as processing agent and credit intermediary for their Executing Firm Customers in clearing.

[84] GSD Rules define the "Sponsoring Member Guaranty" as a guaranty, stated in a manner acceptable to FICC, that a Sponsoring Member delivers to FICC whereby the Sponsoring Member guarantees to FICC the payment and performance by its Sponsored Members of their obligations under the GSD Rules, including, without limitation, all of the securities and funds-only settlement obligations of its Sponsored Members under the GSD Rules.

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Under GSD Rule 8, Netting Members, other than Inter-Dealer Broker Netting Members, are eligible to be Agent Clearing Members and must deliver to FICC an application that includes, for example, information about the Netting Member's customers, past and projected volumes of customer activity and the Netting Member's controls for monitoring and mitigating customer risks. Once approved, an Agent Clearing Member can onboard its Executing Firm Customers by providing FICC with certain information and attestations regarding those customers.

Executing Firm Customers do not onboard as limited members of FICC and, therefore, FICC entirely on the Agent Clearing Members to meet the settlement and risk management obligations that arise with respect to activity cleared through this indirect access model.

GSD CCIT Service

The CCIT Service enables tri-party repurchase agreement transactions in GCF Repo Securities between Netting Members that participate in the GCF Repo Service and institutional counterparties (other than investment companies registered under the Investment Company Act of 1940, as amended). Under Rule 3B, the institutional counterparties, called CCIT Members, are cash lenders in the transactions submitted to GSD and therefore participate in GSD on a limited basis. Rule 3B sets forth the eligibility requirements for CCIT Members, which include financial responsibility and operational capability requirements.

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Principle 19: Tiered participation arrangements; CCAS 17ad-22(e)(19)

Principle 19: Tiered participation arrangements	**CCAS 17ad-22(e)(19)**
An FMI should identify, monitor, and manage the material risks to the FMI arising from tiered participation arrangements. ***Key Consideration 1***: An FMI should ensure that its rules, procedures, and agreements allow it to gather basic information about indirect participation in order to identify, monitor, and manage any material risks to the FMI arising from such tiered participation arrangements. ***Key Consideration 2***: An FMI should identify material dependencies between direct and indirect participants that might affect the FMI. ***Key Consideration 3***: An FMI should identify indirect participants responsible for a significant proportion of transactions processed by the FMI and indirect participants whose transaction volumes or values are large relative to the capacity of the direct participants through which they access the FMI in order to manage the risks arising from these transactions. ***Key Consideration 4***: An FMI should regularly review risks arising from tiered participation arrangements and should take mitigating action when appropriate.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (19) Identify, monitor, and manage the material risks to the covered clearing agency arising from arrangements in which firms that are indirect participants in the covered clearing agency rely on the services provided by direct participants to access the covered clearing agency's payment, clearing, or settlement facilities.

Indirect participants of FICC are firms that rely on the services provided by Members in order to have their activity cleared and settled through FICC's facilities.

Identification and monitoring of indirect participant risk

The risks that indirect participants pose to FICC are identified and monitored as part of the application and on-boarding process for FICC Members, and as part of FICC's ongoing member due diligence process. Among the aims of these processes is for FICC to understand the material dependencies between Members and the indirect participants that rely on Members to access FICC's clearing and settlement services, as well as significant Member-indirect participant relationships and the various risk controls and mitigants that Members employ to manage their risks with respect to such relationships.

During the membership application process, FICC requests and reviews, pursuant to its authority under each Division's Rules, information from applicants regarding the types of businesses they conduct, including the anticipated activity to be introduced to FICC by such applicants on behalf of indirect participants.

Thereafter, as part of its ongoing member due diligence process, FICC collects information from the Members periodically, pursuant to its authority under each Division's Rules to understand their overall business condition and risk management practices, which may cover operational, credit, legal and other risks, and financial standing, including information related to the indirect participants that rely on them.

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In particular, FICC requires that Members submit information regarding the risk that significant indirect participants represent to FICC and the Members, the criteria and procedures used by the Members for on-boarding indirect participants, as well as the risk management policies and procedures employed by the Members for ongoing monitoring and risk management of indirect participant activities.

Management of indirect participant risk

FICC employs Rules-based tools including, but not limited to, those described below to manage the risks that both Members and indirect participants may pose to FICC.

Credit Risk Rating. FICC's review of the information received on indirect participants through its member application process and periodic member due diligence described above may factor into the qualitative factors that affect FICC's credit risk rating of the Member. As described in further detail in response to Principles 4 (Credit risk) and 18 (Access and participation requirements), the resulting credit risk rating of a Member helps determine the level of financial review that will be performed on that Member and may impact the Member's Clearing Fund requirement. If FICC believes the CRRM model-generated rating is not sufficiently conservative, or deems such action necessary to protect itself and its Members, FICC may opt to use mitigation actions, such as the override framework, to lower the Member's credit risk rating,[85] and/or place the Member on the Watch List. This would apply if a Member fails to provide sufficient disclosure, as requested during the due diligence process, to allow FICC to assess the risk exposure posed by the Member's activity, including the indirect member's activity.

In addition, FICC Sponsored Members, who are indirect participants in FICC by virtue of accessing FICC's services through one or more Sponsoring Members that are direct Members of FICC, are subject to adverse news monitoring by FICC. Specifically, FICC uses a monitoring tool daily to identify instances of adverse news regarding any Sponsored Member. If any such news is deemed "severe", according to an internal procedure, it is escalated and reviewed by a cross-function team to determine if any action may be necessary, which could include, for example, outreach to the member, requiring additional reporting from the member, or other appropriate risk management measures as permitted by GSD's Rules.

Executing Firm Customers are also indirect participants in FICC by virtue of accessing FICC's services through one or more Agent Clearing Members that are direct Members of FICC. However, unlike Sponsored Members, Executing Firm Customers do not become limited members of FICC and have no responsibility to FICC with respect to settlement or risk management obligations arising out of their cleared activity. Therefore, because FICC and Executing Firm Customers have no privity with, or liability to, each other, FICC manages the risks presented by activity submitted through the Agent Clearing Service through its ongoing due diligence of the Agent Clearing Members.

Monitoring and Margin Collection. As described in further detail in response to Principle 6 (Margin), FICC collects Required Fund Deposits (and, when applicable, Segregated Customer Margin) from its Members on a daily basis in accordance with the methodology set forth in each Division's Rule 4 (Clearing Fund and Loss Allocation) and the GSD Margin Component Schedule. Such margin deposits are calculated based on, among other risk factors, the trading activity submitted to FICC on Members' own behalf and on behalf of any indirect participants that clear through them. FICC may also require additional ("ad hoc") intraday margin should FICC deem it necessary or appropriate based on intraday market moves and intraday trading activity (whether the activity is that of the Member, or indirect participants who clear through the Member). FICC has established internal thresholds to identify significant changes in Sponsored Members' activity, and monthly reports are provided to the MRC and BRC when activity exceeds those internal thresholds.

In addition, for each Sponsoring Members, FICC monitors the total VaR Charge across all its GSD accounts against such Member's most recently reported regulatory net capital. If the ratio is greater than 1.0, the

[85] This may have the effect of automatically placing the Member on the Watch List as a result of the downgrade. The CRRM is utilized to risk rate U.S. banks, U.S. broker/dealer and foreign bank Members.

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Sponsoring Member is generally not permitted to submit additional activity into its Sponsoring Member Omnibus Account.

Adequate Assurances. FICC also has the authority under each Division's Rule 3 (Ongoing Membership Requirements) and under GSD Rule 3A (Sponsoring Members and Sponsored Members) to seek adequate assurances from a Member if FICC determines that such adequate assurances are warranted based on its review of such Member's indirect participant relationships.

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Principle 20: FMI links; CCAS 17ad-22(e)(20)

Principle 20: FMI links	**CCAS 17ad-22(e)(20)**
An FMI that establishes a link with one or more FMIs should identify, monitor, and manage link-related risks. ***Key Consideration 1***: Before entering into a link arrangement and on an ongoing basis once the link is established, an FMI should identify, monitor, and manage all potential sources of risk arising from the link arrangement. Link arrangements should be designed such that each FMI is able to observe the other principles in this report. ***Key Consideration 2***: A link should have a well-founded legal basis, in all relevant jurisdictions, that supports its design and provides adequate protection to the FMIs involved in the link. ***Key Consideration 3***: Linked CSDs should measure, monitor, and manage the credit and liquidity risks arising from each other. Any credit extensions between CSDs should be covered fully with high-quality collateral and be subject to limits. ***Key Consideration 4***: Provisional transfers of securities between linked CSDs should be prohibited or, at a minimum, the retransfer of provisionally transferred securities should be prohibited prior to the transfer becoming final. ***Key Consideration 5***: An investor CSD should only establish a link with an issuer CSD if the arrangement provides a high level of protection for the rights of the investor CSD's participants. ***Key Consideration 6***: An investor CSD that uses an intermediary to operate a link with an issuer CSD should measure, monitor, and manage the additional risks (including custody, credit, legal, and operational risks) arising from the use of the intermediary. ***Key Consideration 7***: Before entering into a link with another CCP, a CCP should identify and manage the potential spill-over effects from the default of the linked CCP. If a link has three or more CCPs, each CCP should identify, assess, and manage the risks of the collective link arrangement. ***Key Consideration 8***: Each CCP in a CCP link arrangement should be able to cover, at least on a daily basis, its current and potential future exposures to the linked CCP and its participants, if any, fully with a high degree of confidence without	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (20) Identify, monitor, and manage risks related to any link the covered clearing agency establishes with one or more other clearing agencies, financial market utilities, or trading markets.

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reducing the CCP's ability to fulfil its obligations to its own participants at any time. ***Key Consideration 9***: A TR should carefully assess the additional operational risks related to its links to ensure the scalability and reliability of IT and related resources.	

FICC Identifies, Monitors and Manages Risks Related to Links

Link arrangements are subject to FICC's Rules and relevant agreements between the link parties. FICC maintains a risk review framework that sets forth procedures for both the review and approval of prospective link arrangements, and for the identification, monitoring and management of risks related to established links on an ongoing basis.[86] Prospective links are subject to management review and can be subject to regulatory review and approval, as well as a formal risk assessment process designed to identify the risks inherent in the link and the controls and processes designed to mitigate those risks.

A central element of the risk review framework is input from a cross-functional group of FICC businesses, functional units and control functions with the requisite subject matter expertise relevant to the particular link. The framework addresses the relevant risk categories including, but not limited to, operational, market, credit, liquidity, legal and compliance risks.

FICC reviews changes to relevant laws and regulations as they arise, and conducts periodic reviews of the contracts memorializing link arrangements. FICC may update or change its link arrangements as deemed necessary or appropriate. When the contracts memorializing such arrangements have been filed as a rule of FICC, any changes are filed as a Proposed Rule Change or Advance Notice before they can be effective.

FICC has an established link which is between GSD and the CME, and has also entered into a multilateral netting contract and limited cross-guaranty agreement with DTC, NSCC and OCC, as described below.

CME Arrangement

Since 2004, FICC and its predecessors have had in place a cross-margining arrangement with the CME. The FICC-CME cross-margining arrangement permits common members of FICC and CME (or affiliated members of FICC and CME) to reduce the aggregate margin requirements they must post to FICC and CME in respect of a portfolio of eligible FICC and CME cleared positions (e.g., cash Treasury positions and Treasury futures) on the condition that the members provide FICC and CME with the ability to look to the value of the entire portfolio in the event of the members' default. As a result of the cross-margining arrangement, FICC and CME are able to recognize risk offsets arising from the offsetting value of positions maintained by a common member (or a member and its affiliate) at FICC and CME when calculating the margin requirements for those positions. Each of FICC and CME separately calculate these risk offsets for the entire portfolio of eligible trades and give effect to the more conservative of the two calculations. The cross-margining agreement between FICC and CME covers certain cash U.S. Treasury securities cleared by FICC's GSD and various specified interest rate futures products cleared by CME. The cross-margining agreement is incorporated in FICC's rules and available on DTCC's website.[87]

The margin savings provided to GSD Members pursuant to the CME arrangement are monitored by FICC on a daily basis. Any significant change to the CME arrangement would be subject to a risk analysis, including analysis as to whether such change would allow FICC to remain in compliance with the applicable requirements of the CCAS and observant with the applicable FMI Principles. Any change to the cross-

[86] As a separate matter, risk reviews are also performed for certain material relationships for FICC that in and of themselves do not otherwise qualify as a link.

[87] https://www.dtcc.com/~/media/Files/Downloads/legal/rules/ficc_cme_crossmargin_agreement.pdf

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margining agreement would generally require a rule filing and be subject to any requisite regulatory approval.

FICC and the CME have satisfied themselves that the provisions of the cross-margining agreement are enforceable under U.S. law and have designed the arrangement to avail themselves of the special protections that are available in the event of the insolvency of a cross-margining participant. The cross-margining agreement also provides for FICC and CME to cooperate to effect an orderly termination of the agreement in the event of a default of either clearing organization, so as to manage the potential spill-over effects from the default of such clearing organization and limit adverse effects on Members.

Multilateral Netting Contract and Limited Cross-Guaranty Agreement with DTC, NSCC, and OCC

FICC has also entered into a multilateral netting contract and limited cross-guaranty agreement with DTC, NSCC and OCC, under which these clearing agencies have agreed to make payments to each other for unsatisfied obligations of a common defaulting participant to the extent they have excess resources of the defaulting participant. Under this arrangement, no party ever needs to pay “out of pocket” and no party can receive more than its loss. Each Division would apply its own respective resources and determine its own liquidation results. In determining FICC’s available net resources for purposes of this cross-guaranty agreement, FICC would first offset the available net resources of GSD with those of MBSD.

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Principle 21: Efficiency and effectiveness; CCAS 17ad-22(e)(21)

Principle 21: Efficiency and effectiveness	**CCAS 17ad-22(e)(21)**
An FMI should be efficient and effective in meeting the requirements of its participants and the markets it serves. ***Key consideration 1***: An FMI should be designed to meet the needs of its participants and the markets it serves, in particular, with regard to choice of a clearing and settlement arrangement; operating structure; scope of products cleared, settled, or recorded; and use of technology and procedures. ***Key consideration 2***: An FMI should have clearly defined goals and objectives that are measurable and achievable, such as in the areas of minimum service levels, risk-management expectations, and business priorities. ***Key consideration 3***: An FMI should have established mechanisms for the regular review of its efficiency and effectiveness.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (21) Be efficient and effective in meeting the requirements of its participants and the markets it serves, and have the covered clearing agency's management regularly review the efficiency and effectiveness of its: (i) Clearing and settlement arrangements; (ii) Operating structure, including risk management policies, procedures, and systems; (iii) Scope of products cleared or settled; and (iv) Use of technology and communication procedures.

Scope of services responsive to market needs

Through each Division, FICC provides clearing, settlement, risk management and central counterparty services, which are part of its fundamental directive as a registered clearing agency under the Exchange Act. In providing these services, FICC seeks to enhance and expand the scope of services it offers to support an evolving market, often in response to initiatives requested by Members and other industry parties.

FICC engages with the industry and evaluates new initiatives, enhancements, strategic thinking and its performance through the following mechanisms:

- FICC employs a structured approach for the implementation and prioritization of new initiatives. Among other things, the framework ensures new initiatives are fully aligned with the corporate strategy, offer sound financial viability and that risk matters are properly evaluated.

- FICC and its parent company, DTCC, lead strategic thinking across the industry through their development of various White Papers, which are designed to provoke discussions on relevant topics of note in an effort to continually improve the overall market infrastructure. The White Papers are available on the DTCC website.

- FICC actively evaluates whether it is meeting and will continue to meet the requirements and needs of its membership through periodic customer satisfaction surveys conducted by DTCC for all of its business lines, including FICC.

In order to best be able to consider the interests of participants' customers, securities issuers and holders, and other relevant stakeholders of the covered clearing agency including investors, issuers, participants, participants' customers, other persons that access the national system for clearance and settlement indirectly, and regulators of the financial markets in which FICC serves, FICC has established certain

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advisory councils comprised of representatives from FICC participants, their customers, and other stakeholders with appropriate seniority and expertise in the relevant subject matter. Additionally, as required by SEC Rule 17ad-25(j), DTCC solicits council representatives' views on material developments in the clearing agencies' risk management and operations and escalates these viewpoints to the Board or the appropriate Board committee for its consideration.

FICC Members also may have the opportunity to participate in product development through their involvement in ad hoc working groups. FICC establishes ad hoc working groups, as necessary, in order to assist with more discrete programs, products or processes.

FICC's management team additionally interacts with the industry through (a) industry associations and committees (b) industry conferences, (c) forums organized by FICC to inform Members of current and new products and services, and (d) onsite or virtual Member meetings.

Pricing for accessibility of services

FICC provides fair and reasonable pricing to its clients, which is based on factors that include customer demand, client feedback and the cost of providing the service. The goal of the pricing methodology is to recover development costs and operating expenses, and to accumulate capital sufficient to meet regulatory and economic requirements.

FICC pricing is reviewed, at a minimum, annually, and an ad hoc review may be triggered should a market or economic event drive the need. If a review results in a change, the change is proposed to management and the Board for review and approval.

Membership representation in governance

FICC is a wholly owned subsidiary of DTCC, which, as described in Principle 2 (Governance), is owned and governed by the users of DTCC's three clearing agency subsidiaries. The majority of directors on the Board are representatives of financial institutions that are users of these clearing agencies' services. As further described in Principle 2, in addition to management representatives, the Board also includes non-participant directors with industry, regulatory or academic backgrounds important to shaping institutional goals of DTCC and the clearing agency subsidiaries.

Goals and objectives

FICC holistically measures and monitors performance against its goal of achieving certain measurable and achievable targets. Mid-year and year-end reports inform the relevant Board Committees of progress against the firm's corporate and strategic pillars and drivers. These reports include year-to-date operational health metric results, key priority status updates, and relevant table stakes.

Processes and metrics

DTCC uses a variety of metrics to measure FICC's efficiency and effectiveness. In addition to monitoring the annual milestones for strategic goals, DTCC measures technology and infrastructure effectiveness and efficiency based on industry standards adapted to DTCC's specific needs.

FICC evaluates its efficiency and effectiveness on a periodic basis. This information is communicated to senior management and provided to the Board and/or Board Committees as a standing item during their regularly scheduled meetings.

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Principle 22: Communication procedures and standards; CCAS 17ad-22(e)(22)

Principle 22: Communication procedures and standards	**CCAS 17ad-22(e)(22)**
An FMI should use, or at a minimum accommodate, relevant internationally accepted communication procedures and standards in order to facilitate efficient payment, clearing, settlement, and recording. ***Key consideration 1***: An FMI should use, or at a minimum accommodate, internationally accepted communication procedures and standards.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (22) Use, or at a minimum accommodate, relevant internationally accepted communication procedures and standards in order to facilitate efficient payment, clearing, and settlement.

Communication procedures and standards

FICC's policy on communications procedures and standards provides the manner in which FICC will accommodate relevant internationally accepted communication procedures and standards. Specifically, FICC will support relevant internationally accepted communication procedures and standards where there is industry support, and a business case for implementing such standards. In cases where a business relevant to FICC has decided to adopt an industry standard, FICC will work with the appropriate industry organizations that are defining those standards to ensure FICC's unique needs, given its position in the industry, are met. FICC has adopted International Standards Organization (ISO) messaging formats that are used in transactions between financial institutions, for interacting with its Members and other connected parties. FICC also supports secure file transmission protocols.

FICC supports industry-standard communication interfaces through a direct interface (direct end-to-end line) such as DTCC's SMART network, and the SFTI® and BT Radianz networks. FICC also supports direct interface through the Internet. These communication interfaces are available for all entities that interface with FICC systems, including any linked FMIs. Members are eligible to use any one or a combination of the communication interfaces supported by FICC. Members identify their communication interface(s) during the on-boarding process and may modify these elections at any time subsequently, as needed.

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Principle 23: Disclosure of rules, key procedures and market data; CCAS 17ad-22(e)(23)

Principle 23: Disclosure of rules, key procedures, and market data	**CCAS 17ad-22(e)(23)**
An FMI should have clear and comprehensive rules and procedures and should provide sufficient information to enable participants to have an accurate understanding of the risks, fees, and other material costs they incur by participating in the FMI. All relevant rules and key procedures should be publicly disclosed. ***Key consideration 1***: An FMI should adopt clear and comprehensive rules and procedures that are fully disclosed to participants. Relevant rules and key procedures should also be publicly disclosed. ***Key consideration 2***: An FMI should disclose clear descriptions of the system's design and operations, as well as the FMI's and participants' rights and obligations, so that participants can assess the risks they would incur by participating in the FMI. ***Key consideration 3***: An FMI should provide all necessary and appropriate documentation and training to facilitate participants' understanding of the FMI's rules and procedures and the risks they face from participating in the FMI. ***Key consideration 4***: An FMI should publicly disclose its fees at the level of individual services it offers as well as its policies on any available discounts. The FMI should provide clear descriptions of priced services for comparability purposes. ***Key consideration 5***: An FMI should complete regularly and disclose publicly responses to the CPSS-IOSCO disclosure framework for financial market infrastructures. An FMI also should, at a minimum, disclose basic data on transaction volumes and values.	*Each covered clearing agency shall establish, implement, maintain and enforce written policies and procedures reasonably designed to, as applicable:* (23) Provide for the following: (i) Publicly disclosing all relevant rules and material procedures, including key aspects of its default rules and procedures; (ii) Providing sufficient information to enable participants to identify and evaluate the risks, fees, and other material costs they incur by participating in the covered clearing agency; (iii) Publicly disclosing relevant basic data on transaction volume and values; (iv) A comprehensive public disclosure that describes its material rules, policies, and procedures regarding its legal, governance, risk management, and operating framework, accurate in all material respects at the time of publication, that includes: (A) Executive summary. An executive summary of the key points from paragraphs (e)(23)(iv)(B), (C), and (D) of this section; (B) Summary of material changes since the last update of the disclosure. A summary of the material changes since the last update of paragraph (e)(23)(iv)(C) or (D) of this section; (C) General background on the covered clearing agency. A description of: (1) The covered clearing agency's function and the markets it serves; (2) Basic data and performance statistics on the covered clearing agency's services and operations, such as basic volume and value statistics by product type, average aggregate intraday exposures to its participants, and statistics on the covered clearing agency's operational reliability; and (3) The covered clearing agency's general organization, legal and regulatory framework, and system design and operations; and (D) Standard-by-standard summary narrative. A comprehensive narrative disclosure for

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	each applicable standard set forth in paragraphs (e)(1) through (23) of this section with sufficient detail and context to enable a reader to understand the covered clearing agency's approach to controlling the risks and addressing the requirements in each standard; and (v) Updating the public disclosure under paragraph (e)(23)(iv) of this section every two years, or more frequently following changes to its system or the environment in which it operates to the extent necessary to ensure statements previously provided under paragraph (e)(23)(iv) of this section remain accurate in all material respects.

FICC Rules are Public

With respect to each Division, FICC provides GSD Members and MBSD Members with comprehensive rules and procedures, including its fees and key aspects of the default rules. These rules are readily available on DTCC's website for Members to see, as well as supervisors and to the public. Through other publications (including this Disclosure Framework) and Member outreach, FICC also provides further information to enable Members to understand the risks, fees and other material costs they incur through membership at FICC.

GSD Rules and MBSD Rules

Pursuant to the Proposed Rule Change Process, the GSD Rules and MBSD Rules, as applicable, are filed with the SEC prior to or concurrent with their effectiveness and any changes to such documents require a public filing with the SEC.

As discussed under Principle 1 (Legal basis), all Proposed Rule Changes and Advance Notices are posted on DTCC's website, and also published by the SEC on the SEC's website, www.sec.gov, and in the *Federal Register* (which is published online by the U.S. Government Printing Office). Members may be notified of Proposed Rule Changes and Advance Notices via Important Notices published on DTCC's website. Publication of Proposed Rule Changes and Advance Notices allows Participants and other interested parties the opportunity to comment on proposals, and to understand what the proposals would do and how stakeholders may be affected.[88]

All proposed changes to GSD's and MBSD's rules are subject to internal review and governance before they are filed with the SEC. Changes to the GSD Rules and MBSD Rules, as applicable, may be initiated from a number of sources, including internal review, lessons learned from actual events or simulations, process or product enhancements, Member feedback and evolving regulatory requirements.

In addition to its Rules, FICC also provides supplementary information on its services through Important Notices and a series of other documents that are made publicly available on the DTCC website.

[88] Proposed Rule Changes and Advance Notices follow a format prescribed by SEC Form 19b-4. This includes a narrative description of the relevant proposal, with background information including the statutory basis for the proposal and discussion of its impact on Participants and stakeholders.

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Design and operations

As part of FICC's on-boarding process, Members are provided with information about the design and operations of FICC's systems, including record layouts, if applicable, as well as other technical information necessary for Members to establish connectivity with FICC and interact with FICC's systems.

Information about the design and operations of FICC's systems is also included in service descriptions and other forms of user documentation that are available on the DTCC website.

Member rights, obligations and risks

Members' rights, obligations and risks are reflected in the GSD Rules and MBSD Rules, as applicable with respect to each Division, and include, but are not limited to (i) initial and ongoing membership requirements; (ii) FICC's ability to temporarily suspend, or definitively cease to act on a Member's behalf, with respect to some or all services in enumerated circumstances; (iii) the obligation of Members to make deposits to the Clearing Fund; (iv) the requirement for Members to provide additional assurances of financial responsibility and operational capability, as FICC deems necessary; (v) Members' rights and obligations under the loss allocation process; (vi) Member obligations with respect to their use of FICC's services, including with respect to certain indemnifications of FICC; (vii) Members' obligation for compliance with applicable law; (viii) settlement and payment obligations; (ix) settling bank and clearing bank requirements; and (x) disciplinary sanctions (including fines) and a Member's right to appeal a decision adverse to it.

The GSD Rules and MBSD Rules contain information referring to non-routine, though foreseeable, events, including, but not limited to (i) events that could trigger a suspension of services to a Member, (ii) application of the Clearing Fund to complete settlement, (iii) allocation of losses, and (iv) the right of FICC to withhold the return of excess Clearing Fund deposits. The Rules also provide for FICC's right to waive or suspend any Rules, when FICC deems necessary or expedient. These provisions recognize, and provide for, a degree of discretion to management, to enable FICC to address situations as facts and circumstances may require.

Other resources

The initial and ongoing membership requirements are designed to demonstrate that each Division's Members are able to (i) satisfactorily communicate with FICC, (ii) fulfill their anticipated commitments, (iii) meet the operational requirements with necessary promptness and accuracy, and (iv) establish appropriate arrangements to effect payment obligations to FICC. Members must demonstrate operational readiness both during the membership on-boarding process before being provided with access to FICC's systems as well as on an ongoing basis as part of their continuing membership requirements. Relationship Managers and Account Managers are available to FICC participants to provide a number of resources or one-on-one education as needed to ensure participants understand FICC's Rules, procedures and the risks they face from membership in FICC. If a Member is observed to be lacking appropriate understanding of the GSD Rules or MBSD Rules, as applicable, FICC's operations or other requirements, the concerned DTCC department will reach out to Relationship Management staff to discuss remedies with the relevant firm's management. Remedial actions to address an apparent lack of understanding of the GSD Rules or MBSD Rules, as applicable, and the obligations and risks of membership may include one-on-one coaching, webinars or other forms of additional training and/or telephone support. FICC also maintains the right to institute disciplinary proceedings, including fines, for a Member's violation of the GSD Rules or MBSD Rules, as applicable.

FICC provides access to the Real Time Matching Report Center, Clearing Fund Management ("CFM") System, and FICC Customer Reporting ("FIR"), which are client accessible websites for accessing risk reports and other risk disclosures. These websites enable a Member to view and download clearing fund requirement information and component details. The reporting enables a Member to view, for example, a portfolio breakdown by CUSIP, including the amounts attributable to the sensitivity-based VaR model. In addition, Members are able to view and download spreadsheets that contain market amounts for current clearing positions, and the associated VaR charge.

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FICC also provides access to the FICC Risk Client Portal, which is a Member accessible website portal that provides Members the ability, for information purposes, to view and analyze certain risks relating to their portfolio, including calculators to assess the risk and Clearing Fund impact of certain activities. FICC maintains the FICC Client Calculator available on the FICC Risk Client Portal that provides functionality to Members to enter 'what-if' position data and recalculate their VaR charge to determine margin impact pre-trade execution. The FICC Client Calculator allows Members to see the impact to the VaR charge if specific transactions are executed, or to anticipate the impact of an increase or decrease to a current clearing position.

Members, as well as others, also can access the DTCC Application Programming Interface ("API") Marketplace, which provides a single, central location for API specifications developed by DTCC, streamlining access to services and helping to improve the user experience. The API Marketplace is an online, DTCC API "App Store," allowing direct programmatic access to DTCC processing functionality, complete with documentation and training materials to help developers use the APIs.

Fees

With respect to each Division, FICC publicly discloses its fees for services in the GSD Rules and the MBSD Rules, available on the DTCC website.

Generally, a forecast of fee changes is provided to participants during the third quarter of the year. Proposed changes to fees are then finalized by FICC management, approved by the Board and filed as a Proposed Rule Change with the SEC during the fourth quarter of the year. Subject to SEC review, Proposed Rule Changes with respect to fees would then generally have an effective date of the following January 1. FICC may post an Important Notice to the DTCC website to provide further notice of the proposed fee changes. Any proposed changes to fees that happen outside of this annual process are also filed as Proposed Rule Changes with the SEC, and participants are provided notice of the proposed changes both through the Proposed Rule Change Process and through an Important Notice posted to the DTCC website.

In addition, DTCC discloses information on fees charged for operating and maintaining the SMART communications infrastructure that provides access to DTCC's (including FICC's) services. These fees are disclosed in the "SMART Connectivity Services DTCC Fee Guide," which is available on the DTCC website.

Disclosure Framework

This FICC Disclosure Framework is available on the DTCC website. FICC updates the Disclosure Framework every two years, at a minimum, or more frequently for material changes. In addition, FICC discloses quantitative information specified by CPMI-IOSCO for CCPs. This information is also available on the DTCC website, is updated quarterly, and includes transaction volume and value data.

DTCC discloses other quantitative information regarding FICC in the DTCC Annual Report, which is published annually, and in the FICC financial statements, which are published quarterly. This information, as well as basic data on transaction volumes and values, is available to the public on the DTCC website.

In addition to the information described above, FICC's public disclosures include Management's Statements Regarding Internal Controls and Reports of Independent Accountants, and other documents and disclosures available on the DTCC website.

Section VI of this Disclosure Framework includes a listing of publicly available resources, including information available on the DTCC website, and on other public websites. Additional information is available to Members on www.dtcclearning.com or upon request. All such information is available in English.

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Principle 24: Disclosure of market data by trade repositories

Principle 24: Disclosure of market data by trade repositories	**Not applicable for CCAS**
A TR should provide timely and accurate data to relevant authorities and the public in line with their respective needs. ***Key consideration 1***: A TR should provide data in line with regulatory and industry expectations to relevant authorities and the public, respectively, that is comprehensive and at a level of detail sufficient to enhance market transparency and support other public policy objectives. ***Key consideration 2***: A TR should have effective processes and procedures to provide data to relevant authorities in a timely and appropriate manner to enable them to meet their respective regulatory mandates and legal responsibilities. ***Key consideration 3***: A TR should have robust information systems that provide accurate current and historical data. Data should be provided in a timely manner and in a format that permits it to be easily analysed.	

FICC is not a TR. Accordingly, Principle 24 does not apply to FICC.

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V.Definitions of Key Terms and Abbreviations

Bankruptcy Code	The U.S. Bankruptcy Code
BC	Business Continuity
BCR	Business Continuity and Resilience
BRC	Risk Committee of the Board of Directors
CCAS	SEC's Covered Clearing Agency Standards
CCIT	Centrally Cleared Institutional Triparty
CCLF®	Capped Contingency Liquidity Facility
CCP	Central counterparty
CCR	Counterparty Credit Risk
CFM	Clearing Fund Management System
Clearing Agency	A self-regulatory organization that operates either as a clearing corporation or a CSD. A clearing corporation compares and/or records securities transactions, and reports to members the resulting activity; and clears and prepares instructions for the automated settlement of those transactions, often acting as intermediaries or CCPs, in effecting settlement of the transactions. Entities acting as clearing agencies are required to register with the SEC.
DNA	Do Not Allocate
Dodd-Frank	The Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010
DTC	The Depository Trust Company
DTCC	The Depository Trust & Clearing Corporation
DVP/RVP	Delivery versus payment/Receive versus payment
Eligible Clearing Fund Securities	With respect to each Division, certain eligible securities accepted by FICC as Clearing Fund collateral.
EPN	Electronic Pool Notification
ESTC	Enterprise Stress Test Committee
Exchange Act	The Securities Exchange Act of 1934
FDIA	The Federal Deposit Insurance Act
FDIC	Federal Deposit Insurance Corporation

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FDICIA	The Federal Deposit Insurance Corporation Improvement Act of 1991
FICC	Fixed Income Clearing Corporation
FMI	Financial Market Infrastructure
FRBNY	The Federal Reserve Bank of New York
FRM	Financial Risk Management
FRS	Federal Reserve System
GSD	Government Securities Division of FICC
GSD Rules	GSD Rulebook
IAD	Internal Audit Department
LNA	Liquid Net Assets
LRM	Liquidity Risk Management
MBSD	Mortgage-Backed Securities Division of FICC
MBSD Rules	MBSD Clearing Rules
MRC	Management Risk Committee
MRM	Model Risk Management
NSCC	National Securities Clearing Corporation
NSS	The FRB's National Settlement Service
OCC	The Options Clearing Corporation
OLA	Orderly liquidation authority from Title II of Dodd-Frank
Op Risk	Operational Risk
QFC	Qualified Financial Contract
Reg SCI	Regulation Systems Compliance & Integrity
RTTM®	real-time trade matching
SBOD	settlement balance order destined trades
SEC	U.S. Securities and Exchange Commission
SFTI	NYSE's Secure Financial Transaction Infrastructure network

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Shareholders Agreement	The DTCC Shareholders Agreement
SIFMU	Systemically Important Financial Market Utility
SIPA	Securities Investor Protection Act of 1970
SIPC	Securities Investor Protection Corporation
SLA	Service Level Agreement
SMART	DTCC's Securely Managed and Reliable Technology network
TBA	to-be-announced
CSRO	Cyber Security Risk Office
UCC	Uniform Commercial Code; the version of the UCC as adopted in the State of New York is sometimes referred to as the NYUCC
VaR	Value-at-Risk

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VI. Additional Publicly Available Resources

1. Statutes, Laws, Regulations and Reports
CPMI-IOSCO Principles for financial market Infrastructures, available at http://www.bis.org and http://www.iosco.org
Exchange Act Rules, available on the SEC website at http://www.sec.gov
Federal Deposit Insurance Act, available on the Federal Deposit Insurance Corporation website at http://www.fdic.gov
Federal Deposit Insurance Corporation Improvement Act of 1991, available on the Federal Deposit Insurance Corporation website at http://www.fdic.gov
New York Business Corporation Law, available at http://public.leginfo.state.ny.us
New York Uniform Commercial Code, particularly Articles, 8 and 9, available at http://public.leginfo.state.ny.us
Regulation J, available on the Board of Governors of the Federal Reserve System website at http://www.federalreserve.gov; and Operating Circulars 6 and 12, available on the Federal Reserve Bank Services website at http://www.frbservices.org
Securities Exchange Act of 1934, as amended, available on the SEC website at http://www.sec.gov
Securities Investor Protection Act of 1970, available on the SEC website at http://www.sec.gov
Standards for Covered Clearing Agencies, available at http://www.gpo.gov/fdsys/pkg/FR-2016-10-13/pdf/2016-23891.pdf
The Dodd-Frank Wall Street Reform and Consumer Protection Act, available at http://www.gpo.gov/, especially Title II regarding orderly liquidation authority, and Title VIII regarding Clearing and Settlement
Title 11 of the United States Code (The Bankruptcy Code), available at https://uscode.house.gov/
2. FICC and DTCC Documents (available on the DTCC website)
Board Code of Ethics
Board of Directors Committee Charters
Board of Directors Mission Statement and Charter
FICC and NSCC Quantitative Disclosures for Central Counterparties
GSD Rulebook
Identification of Board of Directors
Identification of Senior Management Team

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Important Notices
MBSD Clearing Rules
Principles of DTCC Governance
Procedures for the Annual Nomination and Election of the Board of Directors
Rule Filings, Advance Notices and related Regulatory Approval Orders
White Papers and Industry Reports
3. Financial Statements and Reports (available on the DTCC website)
DTCC Annual Reports, including Consolidated Audited Financial Statements
FICC Audited Financial Statements
FICC Interim (Unaudited) Financial Statements

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